

Solid-State Transformers, End to End

What the technology is, what it cannot yet do, who is actually testing one and at what voltage — and where Megmeet stands in all of it

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BOTTOM LINE UP FRONT

Every claim below is carried at full length, with its tier, in the chapter that owns it; the tags are repeated here so the summary can be checked without turning a page.

The hall-edge contest is not being decided on efficiency. Every serious vendor claims 98.5% and none publishes the measurement boundary. [\[WEB, verified 2026-09-23\]](#) NVIDIA rates the transformer-rectifier unit and the solid-state transformer *identically*. [\[GUIDANCE nvidia-800vdc p21–23\]](#) It is being decided instead on two things that can actually be checked: **the service-voltage class a vendor has specified**, and **whether anything has been sold**. On that test the field inverts. The only solid-state transformer any filing describes as supplying the market is specified 10–13.8 kV and stops there. [\[DOSSIER sungrow v9\]](#) That cannot serve the 34.5 kV class large US campuses take. [\[PRIMER ch.6.1\]](#) The most completely specified 34.5 kV-class design in the ledger is Heron's, and it has not shipped a unit. [\[DOSSIER heron-power v1\]](#) The next is DG Matrix's, whose medium-voltage unit its own CFO deck dates to "2028+" while the product it ships today is a 480 V AC skid. [\[DOSSIER dg-matrix v1\]](#) The only orderable solid-state medium-voltage product from an incumbent is a UPS, not a transformer. [\[DOSSIER abb v7\]](#) The one venture vendor actually shipping hardware ships a 480 V AC skid at 96–97% on its own datasheet, not the medium-voltage unit its marketing rates at 15 MW. [\[DOSSIER dg-matrix v1\]](#) And nobody anywhere holds a UL listing for the 800 V DC bus. [\[WEB, verified 2026-09-23\]](#)

Megmeet sits outside that table on the one axis that defines it. Across thirty-eight sources in its dossier **no kV figure appears anywhere for its solid-state transformer**, and the converter is described only as taking grid high-voltage input to 800 V DC. [\[DOSSIER megmeet v7\]](#) A four-filing text scan of its own disclosures returned the same nothing, and the most recent filing narrowed the claim to "under development" with an "expected" efficiency. [\[WEB, verified 2026-09-23\]](#) What Megmeet does hold is the layer where money moves first: a published, shipping rack-and-sidecar chain that grew 60.92% in the first half of 2026 at the only expanding gross margin among the three largest segments. [\[DOSSIER megmeet v7\]](#)

Analysis. The job is to sell that while the converter above it acquires a class, a listing and a first article — and the risk is that the block above the rack consolidates before it does.

Internal learning document. Not investment advice, and not a Megmeet position on anything. Every factual sentence carries exactly one source tier; every block of inference is labelled **analysis**. Built unattended overnight on 23 September 2026 — the colophon records every assumption made without the reader in the room.

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How to read this document

This briefing **extends** the repository's 20,000-word SST primer of 12 September 2026; it does not replace it. Chapters 1–5 summarise the primer's corresponding chapter in a page, cite its figure numbers, and then add only what the primer does not carry. Where a figure exists in the primer it is referenced by number rather than redrawn; the fourteen new figures are numbered **M1–M14** and are built from a single data file that also feeds the study companion, so the two cannot drift.

THE CITATION CONTRACT

Every factual sentence in this document carries exactly **one** source tier, and no sentence blends two. Where a sentence is inference it carries **[ANALYSIS]** instead, which is a label rather than a source. Where a claim is inference rather than record it sits inside a block labelled **analysis**. The five source tiers, and the analysis label:

[PRIMER ch.x / fig.n]	The repository's SST primer of 12 September 2026, which carries its own numbered first-party sources.
[DOSSIER slug vn]	A Profiler company dossier at the stated version, which cites its own sources and records their provenance tier.
[GUIDANCE nvidia-800vdc pn]	The page-referenced claims ledger for NVIDIA's August 2026 execution paper, held in the repository.
[REPORT 2026-09-08]	The Profiler competitive report on AIDC power conversion.
[WEB, verified 2026-09-23]	Research performed for this document on 23 September 2026. The full URL is in Appendix C.
[ANALYSIS]	A reading of the evidence rather than a source. Used inline where a single sentence is inference — most often a statement that nobody publishes something — and as a block label where a whole passage is.

Two disciplines are applied throughout. No estimate is stated as a fact: where a figure comes from a vendor, a brokerage, a research house or a trade group, the source and its incentive are named in the same sentence. And where a vendor has not disclosed something, the cell reads **undisclosed** — **including Megmeet's own SST**.

Where the tag sits in a wide table. In the reference tables of chapters 6, 7, 10 and 11 a row is one claim assembled from one source, so the tag is written once, at the end of the cell whose sentence it governs, and it covers the rest of that row. Where a row draws on two tiers both tags appear, each against its own sentence, and neither covers the other.

Two fixed vocabularies are used and never varied: limitation status is **unsolved** **lab** **pilot** **product** **standardised**, and evidence tier is **announced** **lab demo** **factory test** **field pilot** **supplying** **undisclosed**.

ONE UNVERIFIED REPORT, BOXED ONCE AND NEVER USED AGAIN

The reader has heard, **unverified**, that Megmeet is in active communication with NVIDIA's and Oracle's engineering teams. That report is hearsay. It has steered the emphasis of chapter 8 — which is longer and more specific than it would otherwise be — and it is stated here once, as hearsay, and **nothing anywhere in this document is sourced to it**. No claim in chapter 8, or anywhere else, rests on it being true. This is the same rule the repository's report engine applies to field notes: they may steer scope, they never appear in content.

PART I

Read this first

The four pages that matter if you read nothing else — the argument in one page, the numbers to memorise, what moved in the last eleven days, and the calendar.

I.1 The one-page cheat sheet

The technology, in five sentences

A solid-state transformer is a converter, not a transformer: it rectifies medium-voltage AC through a stack of series-connected silicon-carbide H-bridge cells, chops each cell's DC into a 5–50 kHz square wave, isolates it through a suitcase-sized medium-frequency transformer, and parallels every cell's output into one regulated 800 V DC bus.

[PRIMER ch.3] Because the transformer core shrinks with frequency — raise f from 60 Hz to 10 kHz and core area falls by roughly 170× — a 4 MW unit weighs about a fortieth of its 60 Hz equivalent, which Infineon puts at 20 tonnes down to about 500 kg. [PRIMER ch.3.1, ch.5.5] The price of that is that it has the overload manners of a computer rather than of a block of iron: an oil transformer carries 200% load for 30 minutes on thermal mass alone, and an SST's continuous rating sits near 1.0–1.2× nameplate with no vendor publishing a curve. [PRIMER ch.5.1] **800 V DC and the SST are two separate decisions** — a customer can build an 800 V DC hall in 2027 with a transformer-rectifier unit and never buy an SST.

[PRIMER ch.2.2] NVIDIA's own execution paper does exactly that: it builds the first power block from a TRU and reserves the MV-direct SST for the generation “toward 2029”. [GUIDANCE nvidia-800vdc p21–23]

The market, in five sentences

Only one covered vendor's filing describes an MV-to-800 V DC solid-state transformer as launched and supplying the market — Sungrow's EnerNeo, specified 10 kV to 13.8 kV, 3 MW standard and 1.5–4.5 MW configurable at 98.5%.

[DOSSIER sungrow v9] The most completely specified product in the public record is Heron Link at 34.5 kV and 4.2 MW with over 98.5% conversion efficiency, and it has shipped nothing, with mass production dated late 2027.

[DOSSIER heron-power v1] The only *orderable* solid-state medium-voltage product from an incumbent is ABB's HiPerGuard, and it is a static MV UPS whose output is medium-voltage AC — not an MV-to-800 V DC converter. [DOSSIER abb v7] The single strongest field evidence anywhere is a 1 MW research unit built at NC State and validated with the New York Power Authority and EPRI on a real distribution feeder, announced in August 2026 — a research unit, not a product, against vendor product claims of 3–7.5 MVA. [WEB, verified 2026-09-23] And no vendor had completed UL certification for a data-centre SST as of the most recent analyst survey, which is a claim by a house that sells the newsletter it appears in.

[WEB, verified 2026-09-23]

Megmeet, in five sentences

Megmeet is the only mainland-China power vendor NVIDIA named for GB200 NVL72 power and appears on both of the rosters NVIDIA published before it stopped naming vendors — the October 2025 ecosystem post and the August 2026 component tier, and its own H1 2026 interim describes it as “one of NVIDIA's designated recommended data-centre power suppliers” — membership of a set, not a rank. [DOSSIER megmeet v7] Its shipping chain begins at three-phase 380–480 V AC: an 800 V HVDC sidecar delivering over 1 MW per rack, an 800 V / 570 kW side rack integrating battery-backup and power-conversion functions, 33 kW and 110 kW shelves, BBU and supercapacitor shelves, CRPS and M-CRPS units to 5.5 kW, and a point-of-load Power Brick. [DOSSIER megmeet v7] The solid-state transformer sits at the top of that published chain and is the one item not built — described only as grid high-voltage input to 800 V DC at an expected efficiency above 98.5%, with **no service-voltage class disclosed anywhere**. [WEB, verified 2026-09-23] The power-products group grew 60.92% to RMB 1.841 billion in the first half of 2026, its gross margin expanding to 25.06% — the

only one of the three largest segments to expand, against smart living and new-energy transport, which both contracted.

[\[DOSSIER megmeet v7\]](#) Against that, headline attributable profit rose 45.21% while ex-non-recurring profit *fell* 16.13% and operating cash flow fell 84.62% — quote both numbers or neither. [\[WEB, verified 2026-09-23\]](#)

ANALYSIS — THE ONE THING TO HOLD IN YOUR HEAD

Everything below the converter in Megmeet's chain is a *socket*: a standardised form factor that gets re-competed every architecture generation, where the company is measurably competitive and growing. The converter is a *block*: something a customer specifies once for a campus and then buys for years. Megmeet is winning at the socket and is not yet in the contest for the block, because a block is specified by its input class and Megmeet has not published one. That is not a capability judgement — it is a disclosure fact, and a specification can be published in a quarter. The risk is only that the block consolidates around two or three certified converters before it is.

1.2 The numbers to know

Twenty-three numbers, each with the tier it comes from. If you memorise one column, memorise this one — most of them are the answer to a question an engineer will actually ask.

What	Number	Why it is the number, and what it is not
Power per mm ² of copper, 800 V DC against 415 V AC	+157%	NVIDIA's conductor argument. The common North American upgrade to 480 V AC buys only +16%; 1500 V DC would buy +382% but has no indoor US component base. [PRIMER ch.2.2, fig.2]
Current for one megawatt at 54 V DC	18.5 kA	The number that ended 54 V distribution. Current, not power, sizes conductors. [PRIMER ch.2.1, fig.1]
Core-area shrink, 60 Hz → 10 kHz	~170×	$V \approx 4.44 \cdot f \cdot N \cdot B \cdot A$. The whole reason an SST exists. [PRIMER ch.3.1]
SST mass against its 60 Hz equivalent	~40× lighter	Infineon's own comparison — about 20 tonnes down to about 500 kg, and 14× smaller. A supplier's figure for a technology it sells silicon into. [PRIMER ch.1, ch.5.5]
Cells per phase, 13.8 kV with 1.7 kV SiC	14–15	42–45 across three phases, each a complete converter with its own capacitor, MFT, gate drivers and sensors. [PRIMER ch.3.2, fig.5]
Cell multiplier, 13.8 kV → 34.5 kV	~2.5×	Cells, magnetics, gate drivers and control channels all scale with it. This is why a 13.8 kV SST is a different product, not a setting. [PRIMER ch.6.1]
SST efficiency band, MV to DC	97–98.5%	Against 99.0–99.5% for a good 60 Hz distribution transformer <i>alone</i> . NVIDIA rates the MV rectifier and the SST identically. [PRIMER ch.3.5]
Oil transformer overload	200% / 30 min	The single biggest engineering advantage the incumbent keeps. No first-party SST vendor publishes an overload curve at all. [PRIMER ch.5.1]
DC fault current rise rate Siemens measured	200 A/μs	At that rate, capacitor-discharge interruption "was only possible with solid-state circuit breakers"; fused clearing took about 20 ms. [PRIMER ch.5.2]
Insulation a 35 kV interface needs	~100 kV	"Three times the insulation", in TDK's assessment. NC State reaches 100 kV BIL at over 99.5% transformer efficiency at 40 kHz. [PRIMER ch.5.3]
Largest independently validated SST on a live US feeder	1 MVA	13.2 kV, NC State with NYPA and EPRI. Vendors quote 3–7.5 MVA products: the gap is the risk a buyer is being asked to carry. [PRIMER ch.5.6]
ERCOT large-load queue against approvals	410 / 9.0 GW	About 410 GW queued as of March 2026 against 9.0 GW approved to energise and a 3.9 GW observed large-load peak. An SST shortens none of it. [PRIMER ch.6.1]
US transformer lead time, standard units	128 weeks	144 for generator step-ups, worst cases near four years. This is what every SST is really sold against. [PRIMER ch.6.2]

What	Number	Why it is the number, and what it is not
Data-centre load dropped, 10 July 2024, Virginia	~1,500 MW	Six faults in 82 seconds; none of it disconnected by utility equipment — all of it on customer-side protection. [PRIMER ch.6.3]
Data-centre load dropped, 22 July 2026, Dominion territory	3.8 GW	A second, larger event on a normally cleared fault. Not the same event as 2024 — do not conflate them. [WEB, verified 2026-09-23]
ERCOT NOGRR 282 ride-through test	2 s @ 0.85 pu	Plus 150 ms below 0.35 pu, no more than 150% current, 90% recovery within 2 s, six sags in 90 s. A converter meets it by design — and must prove it in a submitted model. [WEB, verified 2026-09-23]
NVIDIA's standardised DC power block	4.8 MW	6,000 A-class switchboard, parallel 1,250 A busways, 4+1 catcher through a DC static transfer switch, 20 MW deployment units. Every vendor has to answer how it composes to this. [GUIDANCE nvidia-800vdc p17–21]
Rack power ladder, Gen 1 → Gen 4	145 → 1,000 kW	145 → 330 → 570 → about 1,000 kW. Page-referenced in the repository's copy of the August 2026 paper; not confirmed on any public NVIDIA page. [GUIDANCE nvidia-800vdc p7]
Megmeet H1 2026 power-products growth	+60.92%	To RMB 1.841 billion, the only one of the three largest segments with an expanding gross margin, at 25.06%. [DOSSIER megmeet v7]
Megmeet H1 2026 ex-non-recurring profit	–16.13%	Against a headline attributable net profit of +45.21%. Always quote both; an analyst will have both. [WEB, verified 2026-09-23]
Megmeet FY2025 revenue	RMB 9.403bn	About USD 1.3 billion. Not USD 13 billion — 亿元 is one hundred million, and machine translation renders it as "billion" with depressing regularity. [DOSSIER megmeet v7]
Sungrow EnerNeo class ceiling	13.8 kV	The only SST a filing describes as supplying the market, and it cannot serve a 35 kV campus. [DOSSIER sungrow v9]
Megmeet SST service-voltage class	undisclosed	Zero kV mentions across 38 pinned dossier sources, and zero hits for kV, 千伏 or 中压 in a four-filing text scan. [WEB, verified 2026-09-23]

I.3 Five things that moved since 12 September

The primer closed with a dated watch-list on 12 September 2026 — its §8.2, *What to watch, and when*. [PRIMER ch.8.2] Three company dossiers landed on 19 September and one on 21 September, and three of the five rest on web findings that postdate the primer. [WEB, verified 2026-09-23] *Analysis*: these five are what a reader of the primer does not yet know.

Five things that moved between 12 and 23 September 2026

1	DG Matrix is not shipping the product the primer implies	It is the sharpest single correction in this brief. Anyone placing DG Matrix among the shipping medium-voltage vendors is placing the wrong product.
2	A fifth SST name exists and the primer does not carry it	Not a competitor. The reason to know the name is that a buyer may raise it, and the reason not to price it is that there is nothing to price.
3	NVIDIA opened a qualification programme and SSTs are not in it	The qualification path for an SST is unwritten. Which of the two mechanics a future MV category adopts is the biggest open commercial variable in the layer.
4	The ride-through case got a bigger event and a harder number	These are two different events, two years apart, and should not be conflated. Together they are why the SST's best US argument keeps getting stronger.
5	The policy fence moved twice and the two regimes do not line up	A 34.5 kV behind-the-meter SST is probably outside EO 14421 and inside the FCC list, which has no voltage threshold. A unidirectional rectifier-only architecture may sit outside both. The asymmetry is the finding.

Figure M1 — What moved between 12 and 23 September 2026. Boundary: movement measured against the primer as published on 12 September 2026. Sources per item: dossier revisions of 19 and 21 September, and two web findings verified 23 September. Composed for this briefing from megmeet-sst-briefing-data.json → watchlistDelta.

1 · DG Matrix is not shipping the product the primer implies

The primer places DG Matrix at “pilots” in the hall-edge layer and describes “a 12–34.5 kV AC MV skid to 400–1,500 V DC ... with paid pilots since Q1 2025”. [PRIMER ch.7.3] The dossier that landed on 19 September establishes that the product on the market is the Interport Flex Series: a six-port **low-voltage** skid whose own specification sheet rates the AC port at 400–480 V three-phase, the DC port at 300–850 V, and peak efficiency at 96–97%. [DOSSIER dg-matrix v1] The medium-voltage Interport Cell Series that would compete with Heron and Amperesand at the hall edge is dated “2028+” by the company’s own CFO deck. [DOSSIER dg-matrix v1] This is the sharpest single correction in this document: anyone placing DG Matrix among the shipping medium-voltage vendors is placing the wrong product.

2 · A sixth SST name exists and the primer does not carry it

Novos Power is a San Diego company registered in California on 7 November 2025, with two to ten employees, no disclosed funding round, no SEC Form D, no patent or paper visible in any public database and no named customer. [DOSSIER novos-power v1] Its only named product is the Variable Airgap Solid State Transformer, and “variable airgap” is undefined on the record — no company document, patent, application, paper or talk abstract explains what is varied or why. [DOSSIER novos-power v1] Its input range is stated three ways across its own channels, 13–48 kV on the site and 35 kV

in a July 2026 post, and 48 kV lies above the 38 kV ceiling of UL 2877, the standard that would govern a medium-voltage power supply. [DOSSIER novos-power v1] It is not a competitor; the reason to know the name is that a buyer may raise it.

3 · NVIDIA opened a qualification programme and SSTs are not in it

NVIDIA DSX Ready launched on 21 September 2026 as a qualification programme for partner products meeting the DSX AI-factory reference-design requirements, and at launch it covers exactly two categories: battery energy storage — Hitachi Energy, LG Energy Solution and Tesla — and coolant distribution units — LG Electronics, LiquidStack and Vertiv. [WEB, verified 2026-09-23] The two mechanics differ: BESS partners run the required qualification tests and submit supporting data for NVIDIA review and approval, while CDU partners use a self-qualification suite. [WEB, verified 2026-09-23] NVIDIA says further categories will follow but publishes no timeline and names no transformer, switchgear, rectifier or MV-conversion category. [WEB, verified 2026-09-23]

ANALYSIS

There is no qualification path for a solid-state transformer, and which of the two mechanics a future MV category adopts is the largest open commercial variable in this layer. An NVIDIA-adjudicated test-data gate is a long, expensive, defensible moat; a self-qualification suite is a fast, cheap, crowded one. The useful posture in a first meeting is not to ask when the category opens but to offer to help write it — the category being unwritten is the opportunity, not the obstacle.

4 · The ride-through case got a bigger event and a harder number

The primer's anchor is the Virginia event of 10 July 2024: a lightning arrester failure on a 230 kV line produced six faults in 82 seconds and about 1,500 MW of data-centre load left the grid on customer-side protection, roughly 1,260 MW of it at the third voltage depression and none of it disconnected by utility equipment. [PRIMER ch.6.3, fig.12] A second and larger event is reported on 22 July 2026 in Dominion's territory, in which 3.8 GW of data-centre load dropped on a normally cleared fault. [WEB, verified 2026-09-23] Meanwhile ERCOT's NOGRR 282 has become a numeric acceptance test rather than a principle: ride through two seconds at 0.85 per unit and 150 milliseconds below 0.35 per unit, draw no more than 150% of normal current, recover 90% of normal consumption within two seconds, and withstand six sags in 90 seconds.

[WEB, verified 2026-09-23]

DO NOT CONFLATE THESE TWO EVENTS

Analysis: they are two years apart, with different magnitudes. The 2024 event is the one NERC's incident review documents and the primer reconstructs; the 2026 event is the one the current standards work is being written against. A briefing that merges them into "the Virginia event" will be corrected by anyone who works on large-load interconnection.

5 · The policy fence moved twice, and the two regimes do not line up

The primer records the FCC's 28 July 2026 action adding foreign-produced connected power inverters to the Covered List. [PRIMER ch.7.4] Executive Order 14421, "Declaring a National Emergency to Secure the United States Bulk-Power System", followed on 26 August 2026, covering transformers, inverters, circuit breakers, generation turbines, reactors and battery storage plus associated software, firmware and remote-access capability linked to a covered foreign entity — but applying to transmission facilities rated 69 kV and above, with local distribution explicitly excluded.

[WEB, verified 2026-09-23] An August 2026 FCC update then removed pure AC-to-DC rectifiers and off-grid inverters from the Covered List and added a 45X safe harbour. [WEB, verified 2026-09-23]

ANALYSIS — THE ASYMMETRY IS THE FINDING

A 34.5 kV behind-the-meter hall-edge SST is probably *outside* Executive Order 14421, because it sits on local distribution below 69 kV, and *inside* the FCC Covered List, which has no voltage threshold at all and captures any bidirectional AC ↔ DC device with remote communication. A unidirectional rectifier-only architecture may sit outside both. The two regimes were written for different problems and they do not compose; the binding one for a hall-edge converter is the FCC list.

I.4 The calendar to day one, and the first ninety days

7 October 2026 is a Wednesday. [WEB, verified 2026-09-23] *Analysis:* the market hands you four dated milestones inside the first ninety days, and none of them waits.

The run-in and the first ninety days — blue is yours, orange is the market's

2026-09-23 to 10-06	day -14	Read this brief and the primer. Get the OCP LVDC SST Specification v0.3 from an unblocked network — it could not be retrieved in this run. Re-read the ten week-one questions and decide which three you ask on day one. [ANALYSIS]
2026-10-07	day one	Day one. Senior Sales Manager, SST Solutions. [ANALYSIS]
by 2026-10-31	day +24	Q3 2026 reports from Megmeet, Zhonhen and Sinexcel at the CSRC statutory deadline. Read Megmeet's for a service-voltage class, an order figure and any AI-data-centre revenue line — the three things the H1 interim did not contain. [REPORT 2026-09-08]
2026-11-01	day +25	EU funding restrictions on high-risk-country inverters begin their phase-in: projects assessed under the previous framework but may have to adopt extra cybersecurity measures or exclude high-risk suppliers. [WEB, verified 2026-09-23]
2026-Q4	day +30	First thirty days: get the answers to week-one questions 1-5 written down and shared. The class, the first-article date, the listing path, the FCC analysis and the lab ceiling are the five facts every subsequent conversation depends on. [ANALYSIS]
2026-12-24	day +78	DOE implementing regulations due under Executive Order 14421. Watch whether the 69 kV threshold and the local-distribution exclusion survive into the regulations — that is what decides whether a 34.5 kV hall-edge unit is inside or outside. [WEB, verified 2026-09-23]
2026-12-31	day +85	FERC deadline for NERC to file the new large-load reliability standards and registry criteria (Docket RD26-7-000; CLO-001/002/003). This is the day ride-through stops being guidance — and it is the SST's strongest US argument becoming enforceable. [PRIMER ch.6.3]
2027-01-05	day +90	Ninety days: a written position on where Megmeet competes in the hall edge and where it does not, a named target account list built on service-voltage class rather than on logo, and a one-page answer to each of the sixteen questions an NVIDIA power engineer and the twelve an OCI electrical engineer would ask. [ANALYSIS]
2027-Q1	day +95	Heron Power pilot units targeted; LITEON GPU-side mass production and first McKinney output; Flex separation of Cloud and Power Infrastructure. The quarter the US shelf capacity moves from announced to producing. [REPORT 2026-09-08] [DOSSIER heron-power v1]
2027-02	day +125	CATL's capital increase into Zhonhen's controlling holdco must close within six months of the 2026-08-14 signing. It decides whether the Panama TRU gets a balance sheet. [DOSSIER zhonhen v8]

Figure M2 — The run-in and the first ninety days. Boundary: external milestones are dated from their own sources, named in the row; internal actions are analysis. Day counts run from 7 October 2026 (Wednesday, verified). Composed for this briefing from megmeet-sst-briefing-data.json → calendar.

THE ONE THING TO DO BEFORE DAY ONE

Obtain the **OCP LVDC Solid-State Transformer Specification v0.3** of July 2026. It is the vocabulary every counterparty in this market will use, more than eighty manufacturers are reported to be building to it, and it could *not* be retrieved for this briefing — `opencompute.org` returned HTTP 403 to every attempt from this environment, by two different tools. [\[WEB, verified 2026-09-23\]](#) Everything this document says about the specification's contents is therefore second-hand, and is flagged as such wherever it appears.

PART II

The technology in its entirety

Each chapter summarises the primer's corresponding chapter in a page, cites its figures by number, and then adds only what the primer does not carry.

1 The term system

Extends the primer's chapter 3 (the conversion stage by stage) and chapter 9 (a forty-term glossary). The primer teaches the mechanism and lists the words. What it does not give is a **system** — the terms grouped into the seven families an engineer actually thinks in, each with the one number that anchors it and the sentence a buyer's engineer would use it in. That is this chapter. Every term here also appears in the printed flashcard deck in Appendix B and in the study companion's spaced-repetition widget.

HOW TO USE THIS CHAPTER

Read the "number to remember" column first, all the way down. Those twenty-five numbers — every term but ISOP, which is a pattern rather than a quantity — are what separates someone who has read about SSTs from someone who can hold a technical conversation for twenty minutes. The definitions you can look up; the numbers you have to have. The first three columns are sourced and each carries its own tier tag; the fourth column, *the sentence to say it in*, is **analysis throughout** — scripted language for a conversation, not a sourced record.

Topology

Term	What it is	The number to remember	The sentence to say it in (analysis)
SST — solid-state transformer	A power-electronic converter providing a transformer's voltage change and galvanic isolation through medium-frequency magnetics, with controllable AC-DC and DC-DC stages either side. [PRIMER ch.3, ch.9]	97-98.5% for MV-to-DC units; 99.0-99.5% for a good 60 Hz distribution transformer alone [PRIMER ch.3.5]	"It is a converter, not a transformer — which is why it can also rectify, regulate and ride through, and why it has the overload manners of a computer rather than of a block of iron."
ISOP — input-series, output-parallel	The connection pattern of an SST: cells stacked in series on the medium-voltage input, tied in parallel on the low-voltage output. [PRIMER ch.3.1, fig.4]	—	"Series on the MV side so each cell sees a fraction of the line; parallel on the LV side so they make one stiff bus."
CHB — cascaded H-bridge	A multilevel converter built from series-connected single-phase full bridges; the standard way to face medium voltage with low-voltage semiconductors. [PRIMER ch.3.2, fig.5]	13.8 kV needs 14-15 cells per phase with 1.7 kV devices; 34.5 kV needs about 2.5x as many [PRIMER ch.3.2, ch.6.1]	"The cell count is the cost driver — each cell is a complete converter with its own capacitor, transformer, gate drivers and sensors."
DAB — dual active bridge	An isolated DC-DC converter with an active full bridge on each side of the transformer; power flow is set by the phase shift between the two square waves, so it is bidirectional by construction. [PRIMER ch.3.3, fig.6]	5-50 kHz switching [PRIMER ch.3.3]	"Both bridges are active, so power can go back to the grid — a passive rectifier can never do that."
CLLC / LLC	Resonant isolated DC-DC topologies whose tank circuit lets the switches turn on and off at zero voltage or current; CLLC is the bidirectional form. [PRIMER ch.3.3]	NVIDIA specifies a 64:1 LLC with a matrix transformer for 800 V straight to 12 V in-rack [PRIMER ch.2.2]	"Resonant variants trade control range for soft switching and lower loss — and tend to lose soft switching at light load."
MMC — modular multilevel converter	A half-bridge-cell multilevel converter used in HVDC; an alternative front end to the CHB at higher voltages. [PRIMER ch.9]	N+1 raises a modular multilevel drive's MTBF by about 1.7x — the one independent redundancy number available [WEB, verified 2026-09-23]	"It is also where the SST's service model comes from: MV drives have been swapping cells in minutes for two decades."

Term	What it is	The number to remember	The sentence to say it in (analysis)
TRU / MV rectifier	A conventional MV/LV transformer feeding a rectifier — diode, thyristor or active — to produce 800 V DC. [PRIMER ch.4.1]	NVIDIA's named near-term path, mature from electrolysis, rail and mining; it rates the MV rectifier and the SST identically at 98.5% or better, as large as 7.5 MVA [GUIDANCE nvidia-800vdc p21–23]	"NVIDIA's own paper calls the TRU a faster and proven path and reserves the SST for the next generation. Do not sell against the TRU on efficiency — they are rated the same."

Magnetics

Term	What it is	The number to remember	The sentence to say it in (analysis)
MFT — medium-frequency transformer	The kilohertz transformer inside each SST cell. Small because volt-seconds per cycle fall with frequency; demanding because it must hold medium-voltage insulation in that small volume under PWM stress. [PRIMER ch.3.1, ch.3.3, ch.5.5]	Raising f from 60 Hz to 10 kHz shrinks core area by roughly 170x; a 4 MW unit weighs about a fortieth of its 60 Hz equivalent — Infineon's comparison is 20 tonnes to about 500 kg [PRIMER ch.3.1, ch.5.5]	"The magnetics are the part with the fewest suppliers — a low-loss nanocrystalline or ferrite core, MV insulation between windings, and a cooling path for losses in a small volume."
BIL — basic insulation level	The lightning or switching impulse a piece of equipment must withstand. Set by the ANSI C84.1 equipment class, not by the vendor. [PRIMER ch.6.1, ch.9]	15 kV class = 95 kV BIL; 25 kV class = 125 kV; 35 kV class = 150 kV [PRIMER ch.6.1]	"The class sets the insulation, the BIL and the cell count — which is why a 13.8 kV SST is a different product from a 34.5 kV one."
PD — partial discharge	Localised breakdown inside insulation that does not bridge the electrodes but erodes them over time. Fast dv/dt from wide-bandgap devices drives it. [PRIMER ch.5.3]	A 35 kV interface needs roughly 100 kV of insulation — "three times", in TDK's assessment [PRIMER ch.5.3]	"ORNL's point is that IEEE C62.82.1 was written for iron: localised overstress can occur even when terminal-level clamping looks adequate, and impulse tests may not reflect internal stress."

Devices

Term	What it is	The number to remember	The sentence to say it in (analysis)
SiC — silicon carbide	The wide-bandgap semiconductor that makes 1.2-3.3 kV MOSFETs practical at kilohertz switching. The enabling device of the SST. [PRIMER ch.3.2, ch.6.6]	Production classes 1.2, 1.7 and (from few suppliers) 3.3 kV; 6.5 and 10 kV exist as samples. SiC fell from over \$10/A in 2018 to \$2-3/A in 2025 [PRIMER ch.3.2, ch.6.6]	"Moving from 1.7 kV to 3.3 kV devices roughly halves the cell count — which is why Amperesand builds on 3.3 kV parts and why Wolfspeed's wafer capacity is a supply question for the whole layer."
Cosmic-ray derating	Atmospheric neutrons cause single-event burnout in high-field power devices, so the cell DC link is set well below the device rating. [PRIMER fig.5]	The customary derating is about 0.55 x device rating — the factor that drives every cell-count figure in this brief [WEB, verified 2026-09-23]	"Burnout can occur at 50% or less of rated breakdown, so the derating is not conservatism, it is the design rule."
Fail-short	The property a series-stacked die needs: when it fails it must keep conducting so the stack stays up until the cell can be bypassed. [PRIMER ch.5.4]	ORNL calls SiC fail-short in series stacks "an open research challenge" [PRIMER ch.5.4]	"Press-pack silicon fails short and keeps conducting for months. SiC's thermal stability, small die area and short short-circuit withstand make that hard — and it is the reliability question a careful buyer will actually ask."

Protection

Term	What it is	The number to remember	The sentence to say it in (analysis)
SSCB — solid-state circuit breaker	A semiconductor switch that interrupts DC fault current in microseconds, because DC has no natural current zero for a mechanical contact to exploit. [PRIMER ch.5.2]	NVIDIA targets 125 A air-cooled and 1250 A liquid-cooled solid-state breakers, with MCCBs as the Day-1 branch device [GUIDANCE nvidia-800vdc p27–30]	"Siemens measured it: at a 200 A/us rise rate, capacitor-discharge interruption was only possible with solid-state breakers. Fuses peaked near 9.4 kA at about 3 ms."
HRMG / HRRG	High-resistance midpoint grounding and high-resistance return grounding — two of the four schemes under evaluation for a two-wire 800 V DC bus, alongside floating and solid. [GUIDANCE nvidia-800vdc p24–26]	Four candidates; the paper leans HRMG [GUIDANCE nvidia-800vdc p24–26]	"HRMG grounds both conductors symmetrically through a midpoint resistor network: balanced conductor-to-ground voltages, uniform fault detection on both conductors, and it keeps operating after the first fault."

Term	What it is	The number to remember	The sentence to say it in (analysis)
IMD / RCM	Insulation monitoring device and residual current monitor. The facility zone uses IMD/IRM and alarms without shutting down; the rack interface zone prefers RCM for fast leakage detection. [GUIDANCE nvidia-800vdc p26–27]	Two protection zones [GUIDANCE nvidia-800vdc p26–27]	"The 125 A whip stays de-energised during install and removal — the mechanical lock must engage before an enable signal closes the contactor. That is EV-charging lineage, and they will expect you to know it."

Control

Term	What it is	The number to remember	The sentence to say it in (analysis)
AFE — active front end	A rectifier built from controlled switches rather than diodes. Draws sinusoidal current at unity power factor, can supply reactive power, and can be programmed to a ride-through curve. [PRIMER ch.5.5, ch.6.3]	IEEE 519 sets 5% total voltage distortion at the point of common coupling with no individual harmonic above 3% [PRIMER ch.6.3]	"This is the SST's real US argument: the converter is the model. A transformer cannot be programmed to a ride-through curve; an active front end is the curve."
Grid-forming	A converter that establishes voltage and frequency rather than following them; includes black start. [DOSSIER heron-power v1]	Heron claims IEEE 2800 and ERCOT NOGRR 282 compliance with under 1% ripple injected at MV [DOSSIER heron-power v1]	"Grid-forming is what turns a gigawatt load into a grid citizen rather than a grid problem."
Ride-through	A load's ability to stay connected through a voltage or frequency disturbance instead of tripping. [PRIMER ch.6.3]	NOGRR 282: 2 s at 0.85 pu, 150 ms below 0.35 pu, no more than 150% current, 90% recovery within 2 s, six sags in 90 s [WEB, verified 2026-09-23]	"The July 2024 Virginia root cause was not the transformer — it was a disturbance-counting scheme and manual retransfer. An SST does not override downstream customer logic, and cooling plant is usually not behind the critical bus."

Grid interface

Term	What it is	The number to remember	The sentence to say it in (analysis)
Service-voltage class	The ANSI C84.1 equipment class of the utility feed. It sets insulation, BIL and cell count — and therefore the product. [PRIMER ch.6.1, fig.11]	12.47/13.2/13.8 kV = 15 kV class; 24.9 kV = 25 kV class; 34.5 kV = 35 kV class; gigawatt campuses tie in at 69-345 kV [PRIMER ch.6.1]	"Large greenfield campuses in Texas and the Southeast trend to 34.5 kV because it moves more power on fewer, smaller feeders. Northern Virginia runs 34.5 kV distribution behind 230 kV ties."
PCC — point of common coupling	Where the customer's installation meets the utility's system, and where IEEE 519 harmonic limits are assessed. [PRIMER ch.6.5]	Limits tighten at higher voltage and lower short-circuit ratio [PRIMER ch.6.5]	"For an SST-fed hall the PCC is usually the utility's MV side — so your harmonic signature is measured at 34.5 kV, not at the rack."
The 4.8 MW block	NVIDIA's standardised data-hall DC power block: MV in, a 6000 A-class DC switchboard, parallel 1250 A busways, 4+1 catcher redundancy through a DC static transfer switch. [GUIDANCE nvidia-800vdc p17–21]	4.8 MW per block, 20 MW deployment units of four scalable units [GUIDANCE nvidia-800vdc p17–21]	"Every vendor now has to answer how it composes to 4.8 MW blocks and 1250 A busways with the tap-can interface — paralleling, busway compatibility, and catcher participation are the three concrete questions."

Standards

Term	What it is	The number to remember	The sentence to say it in (analysis)
UL 1741 / 347A / 2877	The US listing routes. UL 1741 with supplements SA/SB is the dominant inverter/converter path; UL 347A and 2877 cover MV power conversion from 1 kV up to 38 kV. [WEB, verified 2026-09-23]	1 kV to 38 kV is the listed medium-voltage range; 38 kV is the ceiling [WEB, verified 2026-09-23]	"Know your listing status precisely. The most quotable line from PCIM Europe 2026 is that there is no UL certification for the 800 VDC bus and that this is a major gap."
IEC 62477-2	The international envelope for MV converters: 1 kV AC / 1.5 kV DC up to 36 kV AC / 54 kV DC. Edition 2 went to CDV with voting closing 13 March 2026. [WEB, verified 2026-09-23]	36 kV AC ceiling [WEB, verified 2026-09-23]	"The envelope exists internationally. What does not exist anywhere is a listed data-centre SST."

Term	What it is	The number to remember	The sentence to say it in (analysis)
DC arc flash	The hazard NFPA 70E 2027 names and no standard can yet calculate. IEEE 1584-2018 is AC only. [PRIMER ch.6.4]	70E 2027 adds Article 310 with DC thresholds; the NFPA research project aims its model at 70E 2030 — a three-year gap [WEB, verified 2026-09-23]	"Operators will energise 800 V DC busway for years with no validated way to compute incident energy. The interim answer is conservative table-based PPE, de-energised-work rules and per-project AHJ negotiation."
OCP LVDC SST Specification v0.3	The specification the layer is being built to — Microsoft, Google and NVIDIA, July 2026. More than 80 manufacturers reported building to it. [WEB, verified 2026-09-23]	v0.3, July 2026 [WEB, verified 2026-09-23]	"This is the vocabulary your counterparties will use. Read it before the first meeting — and note that it could not be retrieved in this run, because opencompute.org refused every request."

ANALYSIS — THE THREE WORDS THAT DO THE MOST WORK

Class. Not “voltage” — *class*. A buyer’s engineer does not ask what voltage your converter takes, they ask what class, because the ANSI C84.1 class sets the insulation, the BIL, the cell count and therefore the product. Someone who says “we can do medium voltage” has told them nothing; someone who says “35 kV class, 150 kV BIL” has told them everything.

Boundary. Every efficiency figure in this market is meaningless without it. When a vendor says 98.5%, the right response is “across which terminals, at what load point, including auxiliaries and cooling or not?” Nobody publishes the answer, which means the first vendor who does gains an advantage that costs them nothing.

Undisclosed. Used precisely, it is a credibility asset rather than a weakness. It says: I know exactly what my company has and has not published, and I will not guess in front of you. Used loosely — as a way of avoiding a question — it is the opposite.

2 Lineage and adjacent technologies

Extends the primer's chapter 2 (the AC hall), chapter 4 (SST against transformer-rectifier unit), figure 3 (the three chains) and figure 8 (published chain losses). The primer draws the comparison the industry argues about — SST against TRU. What it does not draw is the **full adjacency**: the six devices that can sit at the hall edge, scored on the nine attributes a buyer's engineer actually works down, plus the four lineages an SST inherits from. Confusing any two of these six is the most common error in the market, and three of the four things most often called SSTs are not.

Lineage and adjacency — the SST against what it descends from

	Efficiency	Footprint	Overload / inrush	Fault behaviour	Isolation	Controllability	Cost	Maturity	Standards path
Line-frequency transformer (LFT)	best	poor	best	best	best	poor	best	best	best
LFT + rectifier (TRU / MV rectifier)	strong	fair	strong	strong	best	fair	strong	strong	strong
MV UPS (solid-state)	strong	fair	fair	good	best	strong	fair	good	good
Hybrid LFT + solid-state breaker	strong	fair	best	strong	best	fair	strong	good	good
MVDC collection	good	good	fair	fair	good	strong	fair	fair	fair
SST (ISOP CHB + DAB)	good	best	poor	fair	strong	best	poor	poor	poor

Figure M3 — The SST against what it descends from and competes with. Boundary: cells are qualitative five-step rankings (poor / fair / good / strong / best), which are **analysis** and not measurement; the numbers behind each row are in the notes below and carry their own tiers. Sequential single-hue ramp per the dataviz rule for ordinal data. Composed from megmeet-sst-briefing-data.json → lineage.

2.1 What each row actually is

Device	What it is, and the number that characterises it
Line-frequency transformer	The incumbent. 99.0–99.5% efficient at rated load, 200% overload for 30 minutes on thermal mass alone, and a 128–210 week lead time that is the reason this entire category exists. [PRIMER ch.3.5, ch.5.1, ch.6.2]
LFT + rectifier (TRU)	A conventional MV/LV transformer feeding a multi-pulse or active rectifier. NVIDIA rates it at 98.5% or better, calls it “a faster and proven path for quick 800 VDC deployment”, credits it with “strong fault isolation” and “improved compatibility with existing 34.5 kV utility distribution”, and calls it “highly practical for approximately 5 MW-class power block deployment”. [GUIDANCE nvidia-800vdc p21–22] It inherits the transformer's thermal mass, which is the single biggest engineering advantage it holds over an SST.
MV UPS (solid-state)	Not an SST. ABB's HiPerGuard is 2.5 MW per unit hard-paralleled to 25 MW at 4.16–34.5 kV and 98% efficiency, holds UL 1741 and was the first of its kind to achieve UL 9540, and its installed base includes Applied Digital's 400 MW and 300 MW North Dakota campuses. Its output is medium-voltage AC. [DOSSIER abb v7]
Hybrid LFT + SSCB	The cheapest way to buy most of the SST's fault performance without the converter: keep the iron, add a semiconductor breaker. ABB's SACE Infinitus is the first IEC-certified solid-state circuit breaker. [DOSSIER abb v7]
MVDC collection	The adjacent lineage most SST vendors actually come from, along with MV drives, STATCOMs and traction. ABB's own SST heritage runs to the 2012 PETT railway unit — 15 kV, 1.2 MW, about 96%. [DOSSIER abb v7]
SST (ISOP CHB + DAB)	97–98.5% MV to DC, about one third the weight and volume of an LFT-based chain per ORNL, continuous rating near 1.0–1.2× nameplate with no published overload curve, and priced by trade reporting at roughly twice a conventional transformer. [PRIMER ch.3.5, ch.5.1, ch.5.5, ch.5.6]

2.2 What an SST borrows, and from where

From	What comes across
MV drives	The cascaded-H-bridge topology itself, and the cell-swap service model with two decades of field history behind it. GE Vernova markets its MM7 modular multilevel drive on modularity that "can increase MTBF as well as decrease MTTR", with modules "light enough for one or two people to swap a faulty unit with a spare within a few minutes" — so the swap-in-minutes claim every SST vendor makes is not novel to SSTs. [WEB, verified 2026-09-23]
STATCOMs	The four-quadrant active front end: unity power factor, reactive support, and the grid-forming control that lets a converter behave as a grid citizen rather than as a passive load. [PRIMER ch.5.5]
Traction	The original SST. ABB's PETT railway unit of 2012 ran 15 kV at 1.2 MW and about 96% — the first megawatt-class SST anybody energised, fourteen years before this market existed. [DOSSIER abb v7]
EV charging	The interlocked-connector discipline NVIDIA specifies for the 125 A DC whip: hardwired to the tap can behind a normally-open contactor, mechanical lock engaged before an enable signal closes it, release opening the contactor before unlocking. [GUIDANCE nvidia-800vdc p26–27]
PV and storage inverters	The manufacturing base. Sungrow, SolarEdge, Enphase and Power Electronics are all inverter houses reaching up into the hall edge. [DOSSIER sungrow v9] The FCC's inverter action therefore lands on this layer and not on the rack. [WEB, verified 2026-09-23]

THREE THINGS ROUTINELY CALLED SSTs THAT ARE NOT

Zhonhen's **Panama** is a transformer-rectifier lineup — medium-voltage switchgear, phase-shifting transformers and rectifier cabinets in one prefabricated skid, and NVIDIA's own paper names the Panama Architecture as a TRU implementation, as an *architecture* and not as a company. [\[GUIDANCE nvidia-800vdc p22\]](#) ABB's **HiPerGuard** is a static medium-voltage UPS whose output is MV AC. [\[DOSSIER abb v7\]](#) Vertiv's **PowerDirect** is a rack and sidecar DC product at 400–900 kW, and Vertiv's own release contains no MV or SST reference. [\[DOSSIER vertiv v9\]](#) Getting any of these wrong in front of an engineer costs more credibility than admitting you do not know a number.

3 Where an SST sits in NVIDIA's 800 V DC

Extends the primer's chapter 2.2 (what NVIDIA specified), chapter 4.2 (the staging) and figure 7 (the timeline). The primer covers the October 2025 white paper's argument and the August 2026 execution paper's three options. What this chapter adds is the **engineering detail an SST vendor is actually measured against** — the protection zones, the grounding candidates, the breaker ratings, the block arithmetic — and the two things NVIDIA has done since the primer closed.

3.1 The rack roadmap, which is what forces everything else

NVLink wants more GPUs within copper reach, so rack power climbs: Gen 1 (GB200/GB300) at 145 kW on 54 V DC with 60 A whips in an 8-to-make-6 feed; Gen 2 (Vera Rubin NVL72) at 330 kW with both AC and DC input options — the hinge generation; Gen 3 at 570 kW with 100% liquid-cooled shelves and 800 V DC into the PSUs; Gen 4 at roughly 1 MW-class with native 800 V DC in-rack. [\[GUIDANCE nvidia-800vdc p7\]](#) 54 V DC distribution physically runs out of road, because one megawatt at 54 V is 18.5 kA. [\[PRIMER ch.2.1, fig.1\]](#)

PROVENANCE CAUTION ON THE RACK LADDER

The 145 / 330 / 570 / ~1,000 kW ladder is page-referenced to the repository's copy of NVIDIA's August 2026 execution paper. [\[GUIDANCE nvidia-800vdc p7\]](#) A dedicated search of NVIDIA's public pages on 23 September 2026 could not confirm those four figures on any NVIDIA-hosted page; NVIDIA's public pages bound the same ladder more loosely as “racks ranging from 100 kW to over 1 MW” and “1 MW IT racks starting in 2027”. [\[WEB, verified 2026-09-23\]](#) When quoting NVIDIA publicly, use NVIDIA's public language; the four-step ladder is for your own model.

3.2 The four architectures — and they are not a ladder

NVIDIA's core claim is that the four deployment forms are “not sequential requirements, but flexible deployment options”, chosen by facility, timeline and density. [\[GUIDANCE nvidia-800vdc p8\]](#)

Form	Date	What it is, in the numbers a vendor must answer to
AC baseline	now	480 V AC to the rack, 100 AAC whips, 4-to-make-3 rack feed redundancy, conversion to LVDC inside the rack. Fully supported and not going away. [GUIDANCE nvidia-800vdc p9]
Option A — Power Rack	production Q3 2026	A 19-inch power rack beside the compute racks: twelve 100 AAC whips in at 415–480 V three-phase with no neutral, air-cooled shelves of six 18.3 kW PSUs each carrying 1,200 J for input-current shaping, about 660 kW of 800 V DC out over 125 A interlocked DC whips point-to-point with no busbar; optional BBU of four 20 kW units giving 80 kW for 60 s, N+1; certified under UL 62368-1. Explicitly a bridge. [GUIDANCE nvidia-800vdc p10–14]
Option B — Power Center	deployment as soon as Q3 2027	End-of-row DC power centres adjustable to 2 MW, deployed at about 1.6 MW in four-to-make-three for a 4.8 MW effective cluster, distributing over overhead 800 V DC busway with 1,250 A feeders and 125 A tap cans; MCCB clearing up to about 20 ms at the 125 A branch. Retrofits without modifying upstream AC electrical rooms. [GUIDANCE nvidia-800vdc p14–17]
Option C — DC Power Block	initial implementation now, as a transformer-rectifier block; the August 2026 rollout statement puts the full DC power block at 2027+	About 4.8 MW blocks at the hall edge: MV AC in, step-down transformer, LV rectification, a 6,000 A-class DC switchboard, parallel 1,250 A busways on the same tap-can interface as Option B, 4+1 catcher redundancy through a DC static transfer switch — “DC needs no phase synchronization, so source transfer is simpler than AC”. Scales as 20 MW deployment units of four scalable units, shipped as modularised outdoor containerised packages. [GUIDANCE nvidia-800vdc p17–21]

Form	Date	What it is, in the numbers a vendor must answer to
Option C next-gen	toward 2029	34.5 kV AC converted directly to 800 V DC in blocks scaling toward about 10 MW. The named enabler is the mature SST, “expected to be launched toward 2029”, explicitly “beyond the deployment horizons of current 800 VDC architectures”. [GUIDANCE nvidia-800vdc p22–23]

THE SENTENCE THAT SETTLES THE TRU-VERSUS-SST QUESTION

NVIDIA's own paper credits the transformer-rectifier unit with mature transformer technology, strong fault isolation, improved compatibility with existing 34.5 kV utility distribution, and being “highly practical for approximately 5 MW-class power block deployment”; it names three TRU families, one of which is the Panama Architecture. It describes the SST as “approximately 15 kV-class today” against the 34.5 kV that hyperscale campuses take, and dates the next generation toward 2029. [\[GUIDANCE nvidia-800vdc p21–23\]](#) The industry's own reference document says the near-term block is a TRU. Selling against that is selling against the customer's own architecture paper.

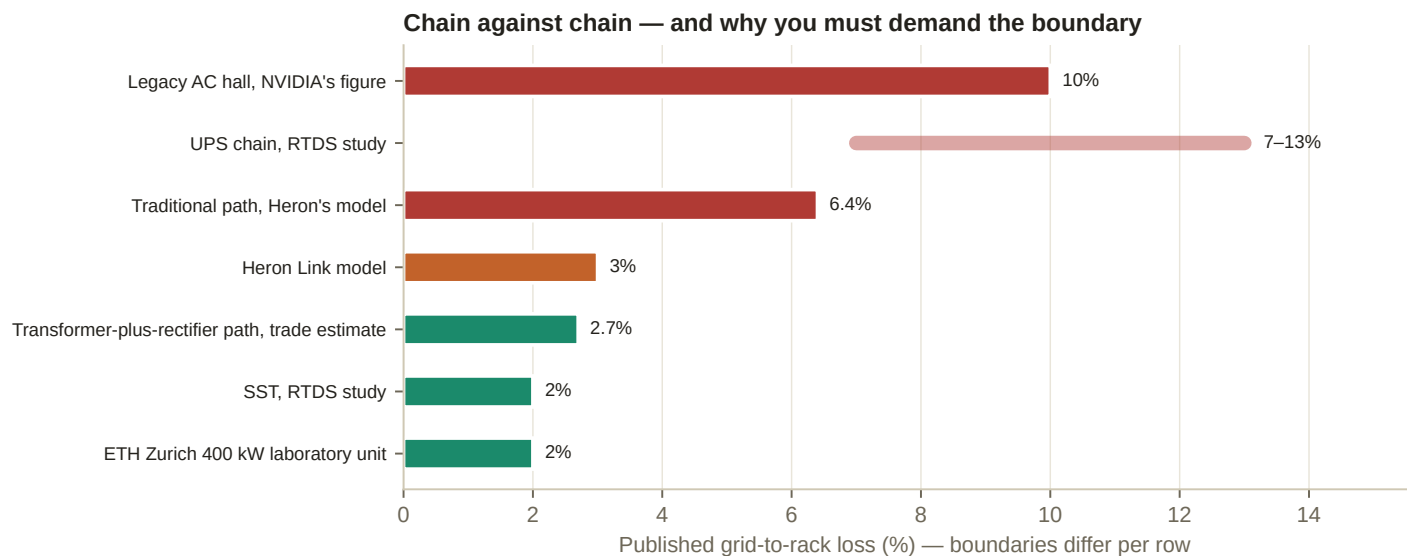


Figure M4 — Chain against chain, and why you must demand the boundary. Boundary: each bar is a *published* grid-to-rack loss figure and the rows are not like-for-like — every author states a different boundary and load point, and several state neither. That incomparability is the finding, not a defect in the chart; a range is drawn where the source publishes one. Composed from `megmeet-sst-briefing-data.json` → `chains.publishedChains`, where every row carries its own source tag. [\[PRIMER ch.4.3, fig.8\]](#)

The practical consequence is the sentence to have ready when a buyer's engineer quotes a competitor's efficiency at you: **ask which boundary and which load point.** [\[PRIMER ch.4.3\]](#) The chart puts the reason in one picture. *Analysis:* the bottom rows are single-converter numbers and the top rows are whole-chain numbers, so they are not the same quantity at all — and the spread between the two kinds is wider than the margin any vendor claims over any other.

3.3 The protection engineering — the genuinely new vocabulary

This is the part of the August 2026 paper that was not in anyone's curriculum, and it is the vocabulary of every engineering conversation about 800 V DC for the next two years.

Topic	What is specified
AC against DC faults	DC has continuous current and no zero crossing, so it needs faster interruption, current limiting and arc-fault detection — but fault energy is “often lower in converter-limited systems”. Series arcing can sustain and is harder to extinguish, making connector design and arc detection critical. The counter-intuitive teaching point: DC's weakness is paired with a structural strength, because a converter-fed system can limit fault current <i>at the source</i> , which a stiff AC utility feed cannot. [GUIDANCE nvidia-800vdc p24]

Topic	What is specified
Grounding	800 V DC distribution is two-wire, positive and return. Four candidates are under evaluation — high-resistance midpoint, high-resistance return, floating and solid — and the paper leans to HRMG, which “provides a more uniform fault monitoring, earlier fault identification, and improved system reliability while maintaining low ground-fault current levels”. The final choice is deferred to system-level studies. [GUIDANCE nvidia-800vdc p24–26]
Protection zones	Two. The Facility Distribution Zone runs from the TRU or SST output to the tap-can input, is qualified-personnel territory, uses continuous insulation monitoring that alarms without shutting down, and coordinates fast source shutdown through the rectifiers on high overcurrent. The Rack Interface Zone runs from tap can to compute rack, has the highest human interaction, prefers residual current monitoring, and keeps the 125 A whip de-energised during install and removal. [GUIDANCE nvidia-800vdc p26–27]
Breakers	Two SSCB target ratings: 125 A air-cooled for the 800 → 54 V shelves, and 1,250 A likely liquid-cooled for future native-800 V DC rack interfaces. The sober counterweight is that MCCBs remain the primary Day-1 branch device on availability, supply chain and certification familiarity. [GUIDANCE nvidia-800vdc p27–29] Siemens data at a 200 A/μs rise rate: a fuse peaks near 9.4 kA at about 3 ms, a thermal-magnetic breaker near 5.7 kA clearing in 6–7 ms, a hybrid about 2.5 ms, and an SSCB holds current near zero within about 1 ms — read from a chart image, so approximate in shape rather than datasheet-exact. [GUIDANCE nvidia-800vdc p29–30]
Power smoothing	It moves rack-side: “localized peak shaving, power smoothing, and slew rate control using minimal energy storage (e.g., electrolytic capacitors) deployed within or near the compute racks”, with facility or behind-the-meter storage as a <i>may</i> . The 1,200 J per PSU is that philosophy in hardware. [GUIDANCE nvidia-800vdc p4, p12] For Vera Rubin, NVIDIA publishes 400 J per GPU of capacitor storage, “6× more rack-level energy storage” than the prior generation, Intelligent Power Smoothing and dynamic power steering. [WEB, verified 2026-09-23]
Certification	System-level requirements first: “equipment certification alone is insufficient”, across eight workstreams from power quality to end-to-end stability criteria, working with UL Solutions and leveraging existing frameworks rather than writing new standards. Plus explicit AHJ enablement — installation practices, inspection guidelines, commissioning procedures and training so local inspectors can approve these builds. [GUIDANCE nvidia-800vdc p30–33]

WHAT THE AUGUST 2026 PAPER DOES NOT SAY — THE GUARDRAIL LIST

No efficiency percentages, no copper-savings figures and no TCO numbers — those live in the October 2025 paper and in vendor decks, and a business-value study is explicitly named as future work. No vendor names for power equipment beyond a Siemens chart credit and UL Solutions: **Panama is named as an architecture, not as a company, and the paper puts nobody on a roster**. No efficiency comparison between TRU and SST, only qualitative attributes. And no retirement of AC — coexistence “for the foreseeable future”.

[\[GUIDANCE nvidia-800vdc p4, p22, p34, p35\]](#)

3.4 What changed after the primer closed

Two things, both material and both dated after 12 September. NVIDIA DSX Ready launched on 21 September 2026 as the first published qualification programme for partner power products, covering battery storage and coolant distribution units only, with no MV-conversion category and no timeline for one. [\[WEB, verified 2026-09-23\]](#) And the first published NVIDIA-aligned end-to-end electrical reference design for Vera Rubin — from Siemens with NVIDIA, Fluence and nVent, dated 1 June 2026 — is built on a nominal 34.5 kV utility connection through medium-voltage distribution and modular low-voltage power blocks, at 136 MW facility and 100 MW IT capacity, delivered as pre-engineered, prefabricated, factory-tested MV and LV skids, UL-aligned and targeting Tier III concurrent maintainability. [\[WEB, verified 2026-09-23\]](#) That release mentions neither 800 V DC nor solid-state transformers anywhere. [\[WEB, verified 2026-09-23\]](#)

ANALYSIS

Read those two together and the near-term shape of the market is clear: the first industrialised NVIDIA-aligned electrical design is a conventional MV-AC-to-LV-AC building, and the first qualification programme does not have a box for your product. Both are commercial openings rather than closures — a factory-tested skid is exactly where an SST wants to land, and Tier III concurrent maintainability is a hard constraint that rewards a converter with a real bypass story. But anyone who walks in claiming NVIDIA's architecture requires an SST today will be corrected by their own reference design.

4 The value case, by perspective

Extends the primer's chapter 2.2 (NVIDIA's copper and staging argument) and chapter 7 (the vendors' positions). The primer gives one perspective properly — NVIDIA's — and describes the vendors' positions. What it does not give is the **buyer-side view from ten different chairs**. Every one of these players wants something different from 800 V DC, and only three of them want an SST specifically. Where a player has said something, it is sourced. Where it has not, the cell is labelled analysis, because inferring a position and then quoting it back is exactly the error this document exists to prevent.

Player	What 800 V DC buys them	What an SST adds over a TRU	What they would need to see
NVIDIA [PRIMER ch.2.2] [GUIDANCE nvidia-800vdc p7, p21-23]	800 V DC carries 157% more power through the same copper than 415 V AC. The rack roadmap is what forces it: NVLink pulls GPUs into copper reach, density climbs, and 54 V distribution runs out of road.	Stage elimination, footprint and control — not efficiency. NVIDIA rates the MV rectifier and the SST identically at 98.5% or better.	ANALYSIS: a qualification category that does not exist yet. DSX Ready launched on 21 September 2026 covering BESS and CDUs only. The first thing an SST vendor can usefully offer NVIDIA is help writing the category.
Oracle / OCI [WEB, verified 2026-09-23] [WEB, verified 2026-09-23]	Nothing published. Oracle has said nothing about 800 V DC, DC distribution, MV conversion or solid-state transformers anywhere in its own material. Its published power content is capacitance, advanced energy storage, hardware GPU ramping and software ramp control.	ANALYSIS: unknown, because the question has never been put publicly. The one place the conversation opens naturally is Project Jupiter — up to 2.45 GW of solid-oxide fuel cells, which are DC-native, consolidated into one microgrid campus.	ANALYSIS: someone else's operating hours. With no internal reference and no published spec, the first question a cautious operator asks is who has one running and at what scale.
The other hyperscalers [WEB, verified 2026-09-23]	Microsoft states that solid-state transformers are "the next step" in its full-stack power transformation after Mt Diablo — no vendor, no site, no kV, no MW, no date. Google, Microsoft and NVIDIA jointly own the OCP specification work.	ANALYSIS: the ability to standardise a block across a fleet. A hyperscaler that specifies one converter for every campus buys schedule certainty, not efficiency.	ANALYSIS: a second source. No hyperscaler will single-source the hall edge, and today no two vendors offer an interchangeable 34.5 kV-class unit.
Neoclouds [WEB, verified 2026-09-23]	NVIDIA names CoreWeave, Lambda, Nebius, Oracle Cloud Infrastructure and Together AI among providers "designing for 800-volt data centers" — NVIDIA's characterisation of them, not their own statements.	ANALYSIS: time to power. A neocloud's constraint is energisation date, and a 24-week lead time against 46-48 for a conventional lineup is the argument that actually lands.	ANALYSIS: a bankable warranty and a spares depot. Neither exists for any MV SST today — only one vendor in the whole research pass publishes a warranty term at all.
Colo developers and landlords [DOSSIER sungrow v9] [DOSSIER heron-power v1]	ANALYSIS: floor area and time to revenue. The SST's published footprint claims are the most concrete thing on offer — Sungrow claims 312 kW/m ² and one-third to one-fifth of a conventional lineup, Heron claims a 70% smaller footprint and an outdoor NEMA 3R siting that removes the electrical room.	ANALYSIS: deletion of the electrical room. For a landlord, converted floor is leasable floor.	ANALYSIS: a tenant who will sign for it. Landlords do not specify converters; anchor tenants do.
Utilities [PRIMER ch.6.3, ch.6.5]	Their published concern is not conversion, it is load behaviour. ERCOT NOGRR 282 turned ride-through into a numeric acceptance test; NERC Project 2026-02 will make interconnection modelling, operational data and protection coordination mandatory, with a FERC filing deadline of 31 December 2026.	ANALYSIS: one authoritative dynamic model of the grid interface instead of a composite of transformer, UPS, VFDs and protection. The converter is the model.	ANALYSIS: an approved-equipment-list entry, which nobody has. This is why every announced US pilot sits behind a customer-owned MV substation.
EPCs [DOSSIER heron-power v1] [WEB, verified 2026-09-23]	ANALYSIS: nothing in the published record. EPCs do not publish architecture positions.	ANALYSIS: fewer trades and a shorter commissioning window. Heron's model puts labour at \$0.08M per MW against \$0.73M for the conventional lineup — a vendor model with vendor-chosen boundaries, and neither figure audited on a delivered US project.	ANALYSIS: a factory-tested skid they can set and terminate. The one published NVIDIA-aligned reference design is delivered as pre-engineered, prefabricated, factory-tested MV and LV skids.
Incumbent hall vendors [REPORT 2026-09-08] [DOSSIER ge-vernova v6]	All six declared an 800 V DC position in October 2025. ABB has the only orderable solid-state MV product and it is a UPS; Eaton published a reference architecture and bought SST intellectual property; Schneider sells the reference design; Vertiv declared engineering readiness; GE Vernova publishes reference designs and holds a hyperscaler commitment tied to a specification milestone; Hitachi Energy holds the transformer calendar.	ANALYSIS: replacement content for a shrinking centralised UPS franchise. The architecture moves conversion toward the rack and takes the UPS with it.	ANALYSIS: evidence that the SST market is larger than the UPS revenue it cannibalises. GE Vernova has priced the prize at 2-3x revenue per gigawatt; Infineon sizes the whole SST slice at just above \$1bn of a \$15bn-plus transformer market by 2030.
Silicon [DOSSIER infineon v1] [DOSSIER vicor v2]	Infineon publishes a dedicated SST application page and is one of fourteen named silicon providers; its AI data-centre revenue went from EUR 250m in FY2024 to above EUR 700m in FY2025 and is guided above EUR 1.6bn for FY2026. Vicor sells no block at all and turned its bus-converter patents into royalties worth a fifth of a quarter's revenue.	ANALYSIS: content per megawatt. Every cell is a bill of materials, and a 34.5 kV unit has about 2.5 times the cells of a 13.8 kV one.	ANALYSIS: volume visibility. Infineon's own market sizing is the most sober in the industry precisely because it has no reason to inflate a slice it already supplies.
Megmeet [DOSSIER megmeet v7]	Its published position is the rack and the sidecar: an 800 V HVDC sidecar converting 380-480 VAC at over 1 MW per rack, a 570 kW side rack, 33 kW and 110 kW shelves, BBU and supercapacitor shelves, CRPS and M-CRPS to 5.5 kW, and a Power Brick. The SST is in the published chain and is marked under development.	ANALYSIS: the move from component vendor to infrastructure vendor. Everything below the converter is a socket someone re-competes; the converter is a block someone specifies.	ANALYSIS: a published service-voltage class, and then a listed unit. Until the first exists, Megmeet is not in the hall-edge comparison at all; until the second exists, nobody in the layer is.

Figure M5 — What 800 V DC buys each player, what an SST adds on top of a transformer-rectifier unit, and what each would need to see. Boundary: the first two columns are sourced where a player has published a position and labelled *analysis* where inferred; the third column is analysis throughout. Composed from megmeet-sst-briefing-data.json → perspectives, where each row carries its own tier tag.

4.1 The four positions worth reading twice

NVIDIA does not buy power. It sets the architecture and lets the ecosystem carry the risk, which is why its published documents give ratings, interfaces and dates but no commercial terms. What it needs from an SST vendor is help writing a qualification category that does not exist. [\[WEB, verified 2026-09-23\]](#)

Oracle has published nothing on conversion at all. Its single engineering post on AI power is a one-minute read whose entire power content is “capacitance, advanced energy storage solutions, hardware-accelerated GPU ramping techniques, and software-based ramp control algorithms” — no voltage, no rack density, no topology.

[\[WEB, verified 2026-09-23\]](#) Chapter 8 takes this apart properly; the point here is that with OCI you are not responding to a specification, you are helping write one.

Utilities are not asking about conversion. Their published concern is load *behaviour*, and it has become numeric: ERCOT’s NOGRR 282 is now an acceptance test, and NERC Project 2026-02 will make interconnection modelling, operational data and protection coordination mandatory against a FERC filing deadline of 31 December 2026.

[\[WEB, verified 2026-09-23\]](#) This is the one place where an SST has a structural advantage that nothing else in the chain can match: the converter *is* the model. [\[PRIMER ch.6.3\]](#)

The incumbents are all running the same hedge. All six declared an 800 V DC position in October 2025 and every one of them has deferred the converter while buying or building the content around it. [\[REPORT 2026-09-08\]](#)

THE DEFENSIBLE PITCH IN THE UNITED STATES

SSTs are marketed against transformer lead times, while the market’s binding constraints are the interconnection queue and the absence of a validated DC arc-flash method — and an SST relieves neither. The argument that actually holds is the regulatory one: **an SST is the only architecture that makes a gigawatt-class load a modelable, dispatchable, ride-through-compliant grid citizen at the exact moment FERC and NERC are making that mandatory.** Everything else in the story is either years out or belongs to the transformer-rectifier unit, which NVIDIA itself calls the faster and proven path. [\[PRIMER ch.6.6\]](#)

5 Limitations → mitigation → who → status

Extends the primer’s chapter 5 (overload, faults, insulation, reliability, cost and standards) and chapter 6 (grid, supply chain, ride-through, codes, sites, economics). The primer describes the limitations. What it does not do is name, for each one, **who is actually working on it and how far they have got** — and the answer for a whole family of them is “nobody”. That is the finding of this chapter and of chapter 7, which takes the same table and reads it as an adoption question rather than an engineering one.

The status word is from the fixed set and is assigned from evidence, not from vendor language: **unsolved** means no named owner or no published method; **lab** means an identified research programme; **pilot** means a named party is doing it at scale but not shipping; **product** means it is purchasable, even if only as a vendor claim; **standardised** means a standards body or regulator owns it.

The full twenty-five-row table is chapter 7. This chapter takes the four that are intrinsic to the converter — the ones a customer’s engineer raises in the first meeting, in the order they raise them.

5.1 Overload — the thermal reservoir an SST does not have

A liquid-filled 60 Hz transformer tolerates 200% load for 30 minutes because tonnes of oil and iron take time to heat, and a data hall exploits that margin every time a row steps from idle to full load or a fault draws current until a breaker clears. [PRIMER ch.5.1] An SST's silicon has microseconds of thermal margin, so surge tolerance collapses to transient events and continuous rating sits near 1.0–1.2× nameplate. [PRIMER ch.5.1] **No first-party SST vendor publishes an overload curve**, and no named body owns the gap as a published rating. [PRIMER ch.5.1] Status: **unsolved** as a rating, **product** as a design property vendors assert. The practical consequence is that an SST is sized for peak rather than average, and DC-bus storage — which NVIDIA already requires for GPU swings — becomes part of the overload strategy rather than an option. [PRIMER ch.5.1]

5.2 Fault handling — the deepest structural difference

ORNL's 2026 review is blunt: MV-SSTs “exhibit limited fault withstand capability ... unlike conventional transformers, which benefit from substantial thermal and magnetic fault tolerance, MV-SSTs must actively detect, limit, and interrupt fault currents within timescales compatible with semiconductor safe-operating limits”. [PRIMER ch.5.2] On the DC side the problem is capacitor energy: Siemens measured that on a common 800 V bus a fault in one load drew discharge from all three healthy loads' capacitors, fused clearing took about 20 ms, and interruption at a 200 A/μs rise rate “was only possible with solid-state circuit breakers; other devices were not able to react fast enough”. [PRIMER ch.5.2] Named owners exist here: ABB claims the first fully IEC-certified solid-state circuit breaker in production, and Eaton has published a 1,500 V / 1,250 A hybrid SSCB. [WEB, verified 2026-09-23] Status: **product** at the device, **pilot** for a complete 800 V DC hall balance of system.

5.3 Insulation and partial discharge — the hardest hardware problem

The medium-frequency transformer must hold medium-voltage insulation in a small volume while being driven by PWM square waves with very fast edges; TDK's assessment is that it is “the biggest challenge”, needing “three times the insulation” — a 35 kV interface requiring roughly 100 kV. [PRIMER ch.5.3] ORNL identifies two distinct stresses: insulation coordination under IEEE C62.82.1, a framework written for iron, where “localized insulation overstress may occur even when terminal-level clamping appears adequate”; and converter-induced stress where fast dv/dt makes voltage sharing “dominated by parasitic capacitances” and drives partial discharge that shortens insulation life. [PRIMER ch.5.3] There is a named academic owner: a Georgia Tech group identifies the mismatch between metal-oxide-varistor protective levels and SST insulation strength, and “the incompatibility of standard impulse test on SST protective structures” — in plain terms, **the standard BIL test does not apply cleanly to an SST**. [WEB, verified 2026-09-23] Status: **lab**.

5.4 Reliability — thousands of devices and one unsolved failure mode

A 4 MW SST contains on the order of a hundred or more SiC modules plus their capacitors, drivers and sensors, against a requirement of 20-year life at 99.9–99.999% availability. [PRIMER ch.5.4] The modular design gives inherent redundancy and vendors design for it — Heron tolerates two cell failures per phase with sub-cycle bypass. [DOSSIER heron-power v1] But ORNL flags a problem specific to wide-bandgap devices: series-stacked designs need a failed die to **fail short and keep conducting for months**, which press-pack silicon achieves and which SiC's thermal stability, small die area and short short-circuit withstand make “an open research challenge”. [PRIMER ch.5.4] Status: **pilot** at the system, **unsolved** at the die. And the arithmetic nobody advertises: the one independent figure available says N+1 raises a modular multilevel drive's MTBF by a factor of about 1.7 — a long way from the step change “five nines” implies. [WEB, verified 2026-09-23]

5.5 What the converter does better than iron

Five genuine advantages, and it is worth being precise about them because three of the five are commonly overstated.

Grid behaviour: the MV-side active front end draws sinusoidal current at unity power factor, can supply reactive power, and can ride through voltage dips to a specified curve — NVIDIA asks for ride-through “surpassing current ITIC curve expectations by referencing ERCOT VRT requirements”, and a transformer can do none of it. [\[PRIMER ch.5.5\]](#) **A native DC**

bus: batteries couple directly with no extra DC-DC stage, as can DC-coupled solar or fuel cells. [\[PRIMER ch.5.5\]](#)

Bidirectionality: two active bridges around the MFT mean stored energy can flow back to the grid for demand

response. [\[PRIMER ch.5.5\]](#) **Footprint and speed:** ORNL cites one third the weight and volume of an LFT-based chain, and Infineon’s own comparison is about 40× lighter and 14× smaller and roughly 50% faster to construct. [\[PRIMER ch.5.5\]](#)

Sungrow claims 312 kW/m² — a vendor figure for its own product. [\[DOSSIER sungrow v9\]](#) Heron claims a 70% smaller footprint and a 24-week lead time against 46–48 weeks for a conventional lineup — its own model, and note that it is not the 128–210-week transformer lead time this document quotes elsewhere, which is for the utility service transformer rather than for the secondary lineup Heron is comparing against. [\[DOSSIER heron-power v1\]](#) **Partial-load efficiency:** idle cells switch off, and a real-time-simulator study put an SST at roughly 2% loss nearly flat across load against 7–13% for a UPS chain. [\[PRIMER ch.4.3\]](#)

THE EFFICIENCY SENTENCE TO GET RIGHT

NVIDIA’s headline numbers — up to 5% better end-to-end efficiency, 45% less copper, 85% more power through the same conductor, up to 70% lower maintenance and up to 30% lower TCO — are claims for the **800 V DC architecture against a legacy AC hall** whose end-to-end efficiency NVIDIA puts at “less than 90%”. They are not claims for the SST against the MV rectifier, which NVIDIA rates at the same 98.5%. [\[PRIMER ch.4.3\]](#) The SST’s advantages in NVIDIA’s own framing are stage elimination, footprint and control. Saying “an SST is 5% more efficient” is the single most common error in this market and it is wrong in two directions at once.

PART III

The market

Who is actually testing one and at what voltage; what is blocking adoption and who owns each blockage; and what the two engineering programmes that matter most have actually committed to.

6 The pilot-and-test ledger, and the 34.5 kV argument in numbers

Extends the primer's chapter 6.1 (the US voltage ladder), chapter 7 (the players) and figure 11. The primer explains why the service class sets the product. This chapter answers the question the primer leaves open: **who has actually shown what, at which class, and how far does the evidence go** — and then makes the 34.5 kV argument in cells, magnetics and BIL rather than in adjectives.

6.1 The ledger

Every vendor with a published medium-voltage position, sorted by the highest class claimed. The evidence tier is assigned from the strongest public evidence located, not from the vendor's own language, and the last column says exactly what the class rests on — because the difference between a datasheet, a product page and a range on a marketing banner is the whole subject of this chapter.

Vendor	MV class	Rating	Output	Evidence	What the class rests on
Novos Power	13–48 kV	10 MW	400–1,500 V DC (claimed)	announced	stated three ways (13-48 kV on the site, 35 kV in a July 2026 post); 48 kV is above the 38 kV ceiling of UL 2877 [DOSSIER novos-power v1]
Siemens + Reinhausen	34.5–36 kV	n/d	800 V DC	announced	"up to 36 kV" — 36 kV is the IEC class containing 34.5 kV; no MW rating, no date [WEB, verified 2026-09-23]
Delta Electronics	6.6–35 kV	n/d	800 V DC	supplying	widest published incumbent range; Delta states "dozens of units in Asia, customer pilots underway in the US" — a vendor claim relayed by trade press [WEB, verified 2026-09-23]
Heron Power	34.5 kV	4.2 MW	800 V DC	announced	34.5 kV is the headline spec, not a range; 4.2 MW is the data-centre configuration (5 MW is the solar/storage configuration of the same platform) [DOSSIER heron-power v1]
SolarEdge	13.8–34.5 kV	5 MW	800–1,500 V DC	announced	2–5 MW modular, Infineon SiC, >99% claimed; customer testing targeted 2027 [WEB, verified 2026-09-23]
Enphase	13.8–34.5 kV	1.25 MW	800 V (±400 V) DC	announced	distributed architecture: 342 x 4 kW GaN modules per 1.25 MW rack; demo late 2026, pilots 2027, volume 2028 [WEB, verified 2026-09-23]
DG Matrix (Cell-MV, roadmap)	12–34.5 kV	15 MW	800 V DC — the MV unit is unshipped	announced	the 15 MW is a marketing rating for an unshipped unit; the CFO deck dates the multi-port MV SST to "2028+" [DOSSIER dg-matrix v1]
ABB (HiPerGuard, MV UPS)	4.16–34.5 kV	2.5 MW	MEDIUM-VOLTAGE AC — a UPS, not a converter	supplying	output is medium-voltage AC — it is a UPS, not an MV-to-800 V DC converter. Plotted because it is the only orderable solid-state MV product [DOSSIER abb v7]
Hyosung	22.9 kV	1.05 MW	undisclosed	lab demo	22.9 kV is a Korean class; the nearest US analogue is 24.9 kV, not 34.5 kV [WEB, verified 2026-09-23]
Eaton (Resilient Power)	12.47–15 kV	2 MW	800 V DC (trade press)	announced	"2 MW, 15 kV class" is trade-press only; eaton.com returned HTTP 503 to the research pass, so no vendor spec sheet was obtained [WEB, verified 2026-09-23]
Sungrow (EnerNeo)	10–13.8 kV	3 MW	800 V DC	supplying	the only unit a filing describes as supplying the market — and it stops at 13.8 kV [DOSSIER sungrow v9]

Vendor	MV class	Rating	Output	Evidence	What the class rests on
ETH Zurich (research)	13.2 kV	0.4 MW	800 V DC	lab demo	400 kW at 98% measured — the best-instrumented efficiency figure in the set, and it is a laboratory unit [WEB, verified 2026-09-23]
NC State / NYPA / EPRI (1 MVA research unit)	13.2 kV	1 MW	13.2 kV AC feeder, 1 MVA	field pilot	the only independently validated megawatt-class SST on a live US distribution feeder: 15 kV SiC MOSFETs, three days supplying an EV load, reactive power injected, energised more than ten times, June 2026 [PRIMER ch.5.6]
Alderbuck Energy	12 kV	n/d	800 V DC	factory test	the only named US AI-compute site with a disclosed class attached to an SST — San Diego Supercomputer Center, host utility SDG&E, CEC-funded. It is 12 kV [WEB, verified 2026-09-23]
Zhonhen (Panama)	10 kV	2.5 MW	240 / 400 / 800 V DC	supplying	a transformer-rectifier architecture, not an SST; plotted because buyers place it against SSTs [DOSSIER zhonhen v8]
JST Power Equipment	10 kV	2.4 MW	800 V DC	factory test	10 kV is a China class [WEB, verified 2026-09-23]
EPFL + Navitas	3.3 kV	0.25 MW	800 V DC → 12 V	lab demo	3.3 kV is not a US service class at all [WEB, verified 2026-09-23]
Megmeet	undisclosed	—	—	undisclosed	The SST is described only as "grid HV input to 800 V DC". Across the 38 sources pinned in dossier v7 the string "kV" does not appear once. [DOSSIER megmeet v7] A four-filing text scan of the FY2025 annual report, the H1 2026 interim and two investor-relations records found zero hits for kV, 千伏 or 中压. [WEB, verified 2026-09-23]
Amperesand	undisclosed	—	—	undisclosed	No kV figure appears anywhere on the company's own site — "configurable MV AC and LV DC" only. The 10-34.5 kV figure in September 2026 reporting is third-party. Job postings cite "medium-voltage (5 kV to 36 kV) systems". [DOSSIER amperesand v2]
GE Vernova	undisclosed	—	—	undisclosed	Reference designs cite 13.8 kV AC to 800 V DC at the perimeter, but no SST product class, rating or date is published. A first delivery to an unnamed hyperscaler was stated for fall 2026. [DOSSIER ge-vernova v6]
Hitachi Energy	undisclosed	—	—	undisclosed	No SST product is named at all. Its engineers are separately quoted cautioning that compact 34.5 kV designs present significant design challenges. [DOSSIER hitachi-energy v5]
Vertiv	undisclosed	—	—	undisclosed	34.5 kV to 800 V DC at 3-5 MW is stated by an engineering director as a goal, not a product. The shipping sidecar is 400-900 kW and is not an SST. [WEB, verified 2026-09-23]
Schneider Electric	undisclosed	—	—	undisclosed	No MV SST programme was found in the pinned sources. The 1.2 MW/rack 800 V DC sidecar was still a prototype at GTC 2026 with no commercial date. [DOSSIER schneider-electric v9]
Sinexcel	undisclosed	—	—	undisclosed	Management states four times that the AIDC critical-power products are samples in development with formal sales not yet realised; the string "800V" does not appear in its 212-page interim. [DOSSIER sinexcel v8]

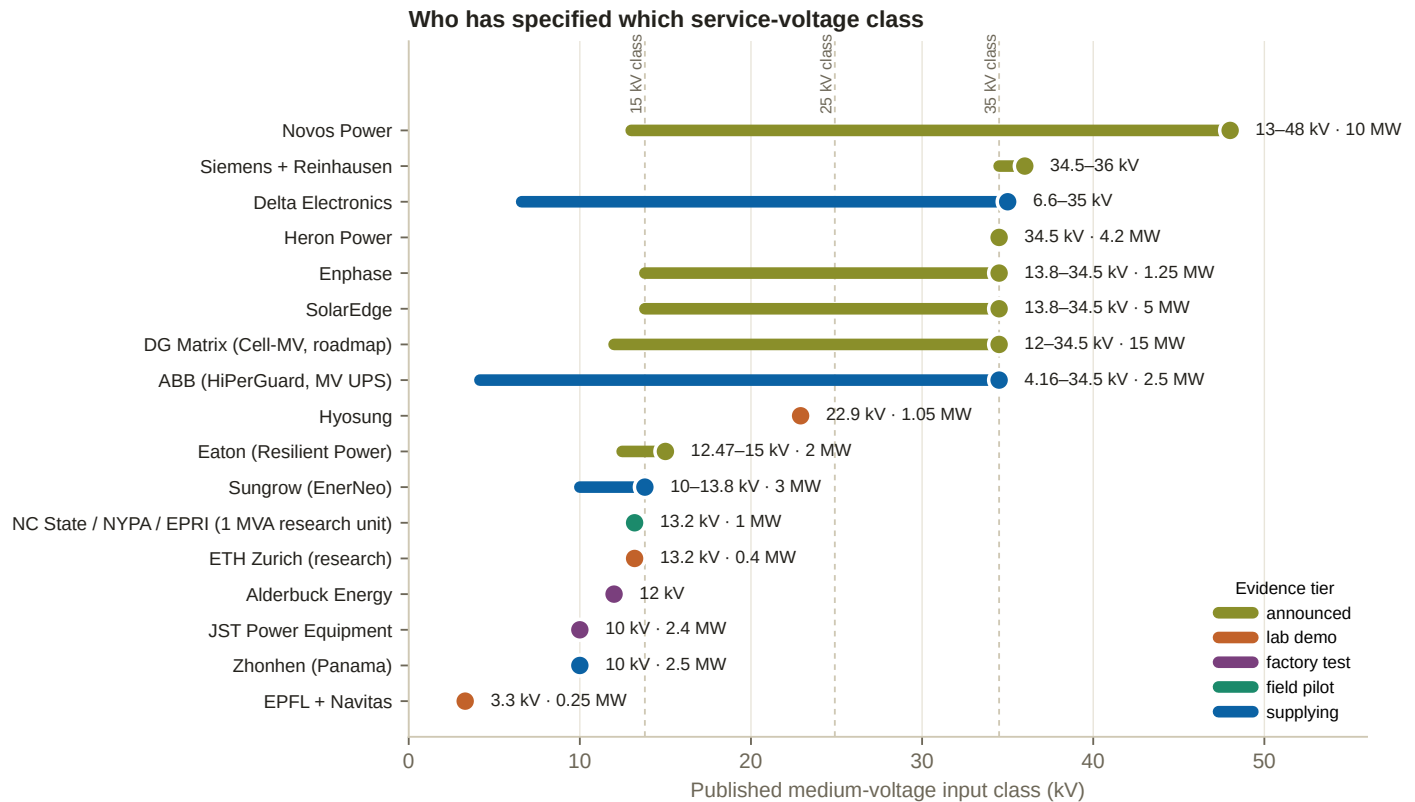


Figure M6 — Who has specified which service-voltage class. Boundary: the bar spans a published range and the marker sits on a single published class; colour is evidence tier, from the fixed six-word vocabulary. Vendors with no published class are not plotted and are listed in the table above. Composed from megmeet - sst - briefing - data . json → classLedger, where every row carries its own source tag.

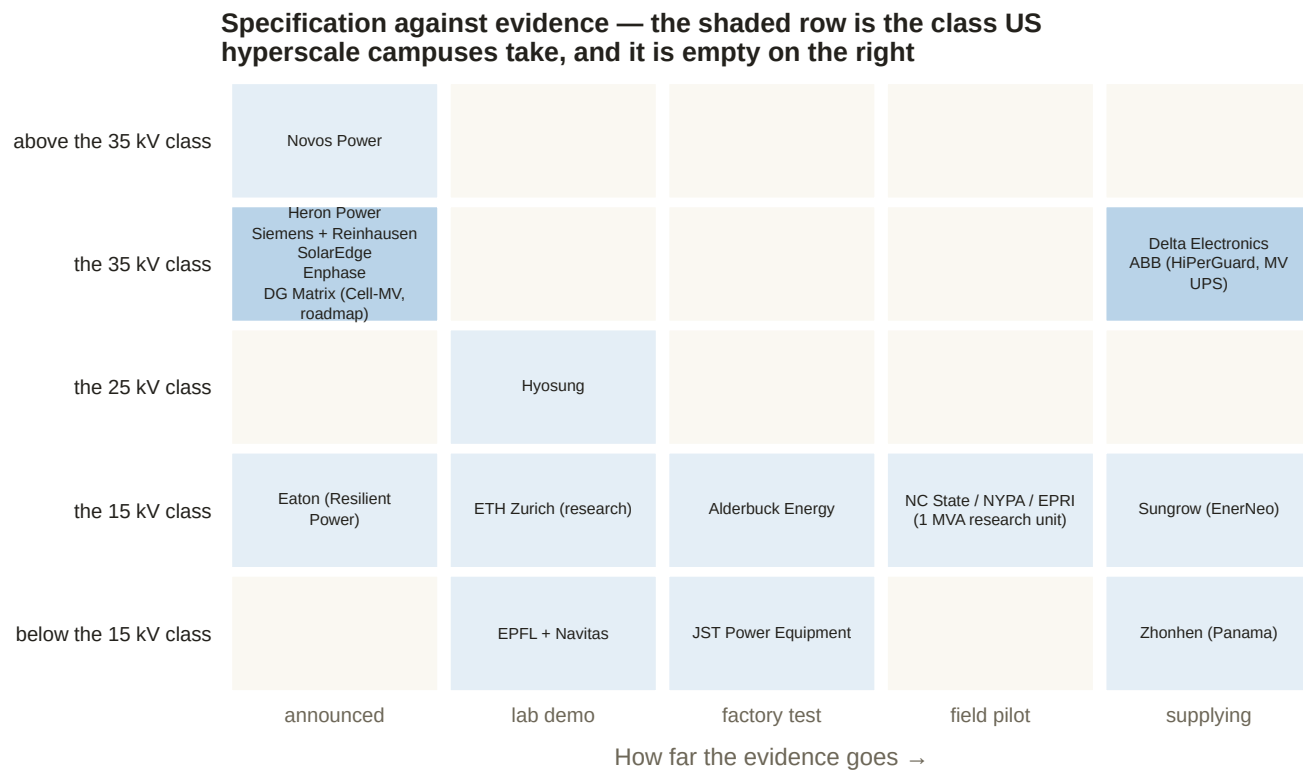


Figure M7 — Specification against evidence. Boundary: rows are ANSI class bands, columns are the evidence tier, and a vendor appears in exactly one cell. Neither axis is a quality score: a vendor high and left has a specification without a product; a vendor low and right has a product that cannot serve a 35 kV campus. A grid rather than a scatter because seven of the seventeen vendors sit in just two cells and overprint at any marker size. Composed from cLassLedger.

READ FIGURE M7 ACROSS THE SHADED ROW

Every count below is read off figure M7, whose rows carry their own tiers in the ledger above. The 35 kV row — the class large US campuses actually take — contains five vendors under “announced” and two under “supplying”, and the two under “supplying” are Delta, whose shipped units are claimed in Asia by the vendor itself, and ABB, whose product is a UPS. The three middle columns of that row are **empty**. There is no 34.5 kV-class solid-state transformer anywhere in the world at lab demo, factory test or field pilot.

6.2 Why 34.5 kV is the class that matters, in numbers

US utilities serve large customers at 12.47, 13.2 or 13.8 kV — the ANSI C84.1 15 kV class — or at 24.9 kV or 34.5 kV, while gigawatt campuses interconnect at 69–345 kV with their own substations. [PRIMER ch.6.1, fig.11] Large greenfield campuses in Texas and the Southeast trend to 34.5 kV because it moves more power on fewer, smaller feeders, and Northern Virginia campuses run 34.5 kV distribution behind 230 kV ties. [PRIMER ch.6.1]

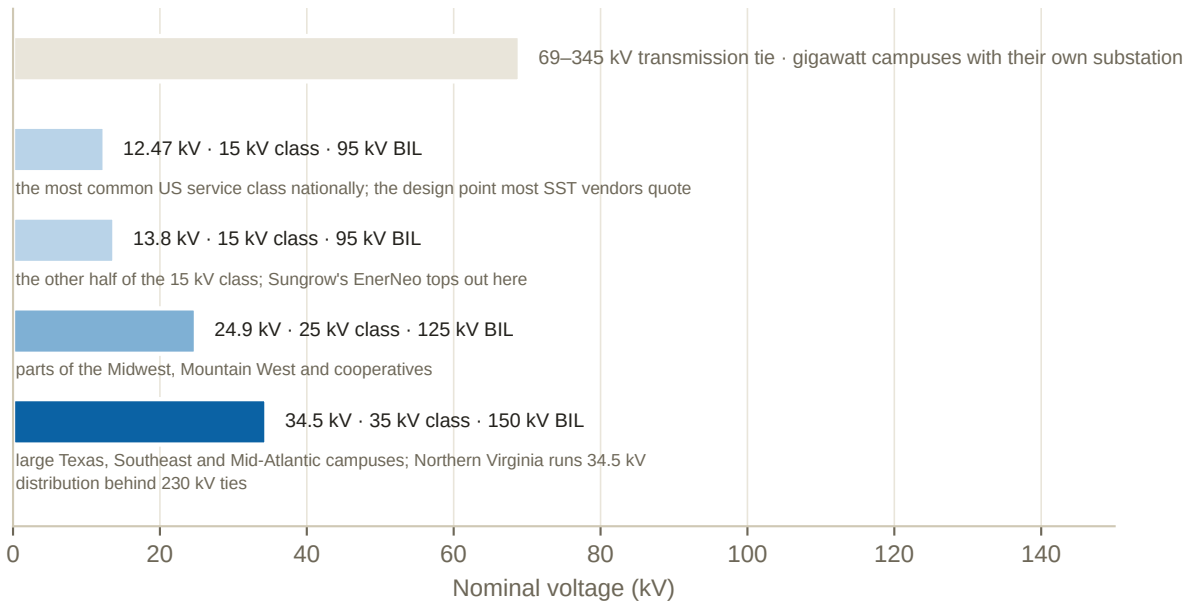
The US ladder — the class sets the product, not a configuration option

Figure M8 — The US ladder. Boundary: nominal voltages with the ANSI C84.1 equipment class and its BIL, plus the transmission tie for comparison. The usage note on each row is from the primer's treatment of the US service mix. Composed from megmeet-sst-briefing-data.json → classCost.classes.

Because a cascaded-H-bridge SST's cell count scales roughly linearly with input voltage, the step from the 15 kV class to the 35 kV class multiplies cells, medium-frequency transformers, gate drivers and control channels by *roughly* 2.5×. [PRIMER ch.6.1] *Analysis:* the counts computed on the primer's own assumptions bracket that figure rather than land on it — 13.8 → 34.5 kV is 15 → 35 cells, a 2.3× step, and 12.47 → 34.5 kV is 13 → 35, a 2.7× step. The primer's 2.5× is the round number for the class step; the two exact ratios are what a bill of materials actually moves by. That is the whole argument, and it is worth seeing as a count rather than a multiplier.

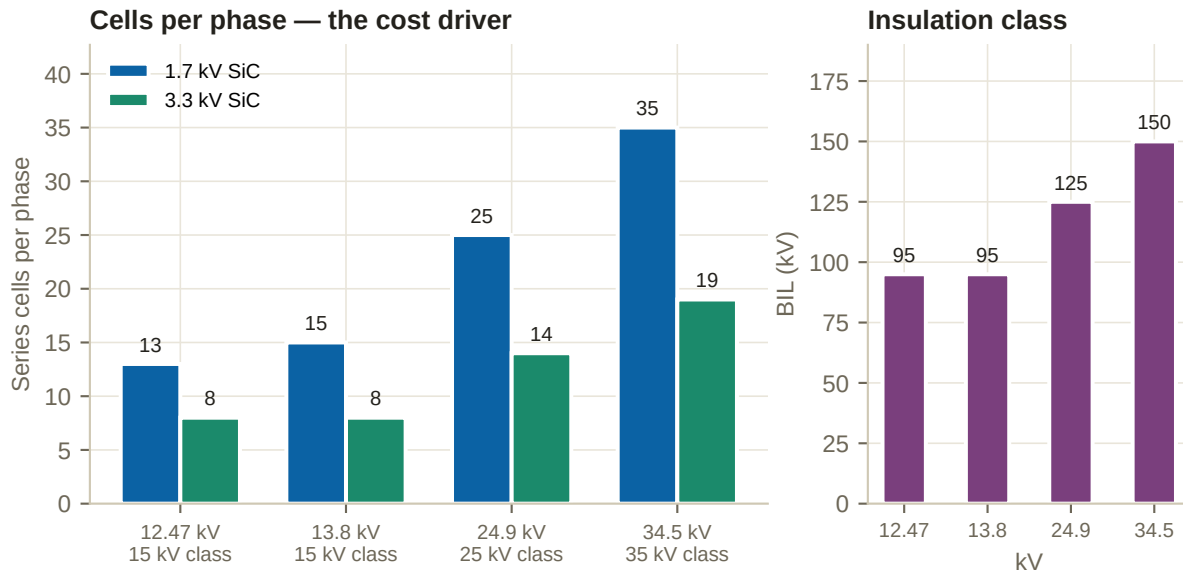


Figure M9 — What the 35 kV class costs inside the converter. Boundary: cell counts are **computed for this briefing** on the primer's stated assumptions — cell DC link = $0.55 \times \text{device rating}$, phase peak = $V_{LL} \cdot \sqrt{2}/\sqrt{3} \times 1.10$ grid overvoltage, plus one redundant cell per phase [PRIMER fig.5]. Real designs vary the derating and the redundancy, so the ratios between classes are what matter and not the absolute counts. BIL is the ANSI C84.1 equipment class, not a vendor figure.

With 1.7 kV devices a 12.47 kV design needs about 13 cells per phase and a 34.5 kV design about 35 — a 2.7× step in cells and therefore in capacitor banks, magnetics, gate drivers, sensors and failure points. Moving to 3.3 kV devices roughly halves both counts, which is precisely why Amperesand builds its front end on 3.3 kV silicon-carbide modules.

[DOSSIER amperesand v2] On top of the count, the insulation class steps from 95 kV BIL to 150 kV, and a 35 kV interface needs roughly 100 kV of insulation inside the medium-frequency transformer itself. [PRIMER ch.5.3]

THREE CAUTIONS THAT WILL OTHERWISE COST YOU

345 kV is not 34.5 kV. Abilene's Oncor interconnection is 345 kV *transmission*, and the campus distribution class is undisclosed. [WEB, verified 2026-09-23] One decimal point turns a transmission fact into a fabricated service-class finding, and it is the single most likely error to introduce into this material.

The only named US AI-compute site with a disclosed class attached to an SST is at 12 kV, not 34.5 kV: Alder buck Energy's unit for the San Diego Supercomputer Center, hosted by San Diego Gas & Electric and funded by the California Energy Commission, announced 23 June 2026 and still to pass factory test and lab validation before installation. [WEB, verified 2026-09-23]

Not everyone agrees 34.5 kV is ready. Hitachi Energy — the incumbent with the deepest MV transformer manufacturing base — is quoted urging caution specifically on compact 34.5 kV designs, on insulation and manufacturability grounds, and saying the technology "is not fully mature for every use case". [WEB, verified 2026-09-23] That is a trade-press paraphrase rather than a Hitachi release, but it is the most interesting single datapoint in the question: the company that would have to build the magnetics is the sceptic.

6.3 Programmes, buyers and what is genuinely energised

The strongest field evidence in the world is a research unit. A 1 MW solid-state transformer designed and built at NC State was tested on a real distribution feeder at EPRI's power delivery laboratory in Lenox, Massachusetts, announced 18 August 2026, funded by the New York Power Authority and a DOE award, and claimed as "the first independently verified, megawatt-class solid state transformer validated on a live utility distribution feeder".

[WEB, verified 2026-09-23] The primer gives its class: a 1 MVA unit on a 13.2 kV feeder — the 15 kV class — using 15 kV SiC

MOSFETs, run over three days supplying an EV load and energised more than ten times in June 2026. [\[PRIMER ch.5.6\]](#)

Analysis: it therefore belongs in the ledger proper rather than in the undisclosed block, and it is the only **field pilot** in the table.

The largest committed volumes are all at the low class. Sungrow's EnerNeo carries framework purchase agreements with HEC Technology for 30 MW across 2026–27 and with ZDATA for 100 MW, with a demonstration expected at the end of 2026 and deliveries in 2027–28 — and the product is specified 10 kV to 13.8 kV.

[\[DOSSIER sungrow v9\]](#)

No SST has been evidenced energised at a named hyperscaler site. Microsoft states that solid-state transformers are “the next step” in its power transformation after Mt Diablo, naming no vendor, no site, no kV, no MW and no date.

[\[WEB, verified 2026-09-23\]](#) GE Vernova stated in April 2026 that a first product would be delivered to an unnamed hyperscaler in the autumn of 2026 followed by about six months of testing, and as of 23 September that delivery has not been reported. [\[WEB, verified 2026-09-23\]](#) Everything else at hyperscaler scale is ecosystem membership or an unnamed counterparty.

ANALYSIS — WHAT THE LEDGER IS ACTUALLY TELLING YOU

Three facts, held together, define the commercial reality of this layer. The most commercially advanced SST in the world cannot serve the class that US hyperscale campuses take. The best-evidenced megawatt-class unit anywhere is a university research project. And the only named US site with a disclosed class is at 12 kV. A vendor arriving with a credible 34.5 kV-class product and a witnessed first article would not be joining a crowded field — it would be the first entrant in an empty one. That is the opportunity and the risk in the same sentence, because being first also means being the vendor who discovers what the standards bodies have not yet written.

7 Obstacles to adoption

Extends the primer's chapter 5 (technical) and chapter 6 (grid, supply chain, ride-through, codes, sites, economics). The primer's thin spot is operations and maintenance — one mention of “maintenance”, none of spares, MTBF or technicians. This chapter fills it, and puts every obstacle in one table with a named owner and a status word. **The finding is the empty column:** of twenty-five obstacles, **ten** sit at **unsolved**, and **eight** of those ten name no owner at all — four of the eight in operations and maintenance, two in utility acceptance, one in policy, and only one technical. (The tenth unsolved row, the interconnection queue, does name an owner: FERC. It is unsolved because no converter can shorten it.)

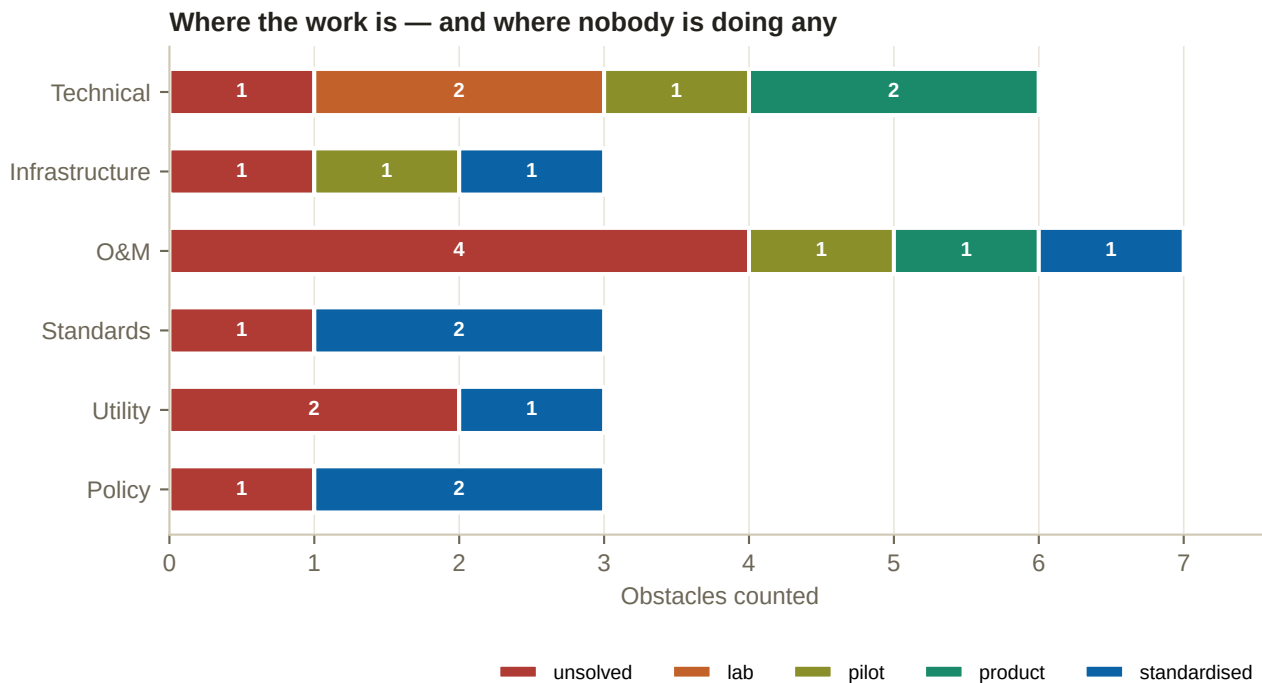


Figure M10 — Where the work is, and where nobody is doing any. Boundary: one row per obstacle in the table below, counted into its family and its status word; the bars are counts and not a severity score. Status vocabulary is fixed and assigned from evidence, not from vendor language. Composed from megmeet-sst-briefing-data.json → obstacles, where every row carries its own source tag.

Obstacle	The mechanism	Who is doing what about it	Status
Technical			
Overload / short-time rating	An oil transformer carries 150-200% of rated kVA for minutes on thermal mass alone; a converter's overload is bounded by junction temperature and must be bought as silicon. No first-party SST vendor publishes an overload curve.	Nobody owns it as a published rating. [PRIMER ch.5.1] Vendors claim thermal margin as a design property. [WEB, verified 2026-09-23]	unsolved
Inrush	No magnetising inrush, so no reactive-power step at close-in.	This one goes the SST's way — energisation is a power-electronics soft start. [WEB, verified 2026-09-23]	product
DC fault current, no natural zero crossing	DC-link capacitors discharge into a fault at 200 A/us; Siemens measured that interruption at that rise rate "was only possible with solid-state circuit breakers". Fused clearing took about 20 ms.	ABB (claims the first fully IEC-certified SSCB in production); Eaton (1500 V / 1250 A hybrid SSCB). [WEB, verified 2026-09-23]	product
SiC cosmic-ray single-event burnout and derating	Atmospheric neutrons cause burnout at 50% or less of rated breakdown, so MV cells must be derated — which is where the 0.55 factor in the cell-count arithmetic comes from.	NASA NEPP on the device physics; device vendors on the FIT budget. [WEB, verified 2026-09-23]	lab

Obstacle	The mechanism	Who is doing what about it	Status
Cell failure, redundancy and bypass	ORNL: series-stacked designs need a failed die to fail short and keep conducting for months. Press-pack silicon does; SiC's thermal stability, small die area and short short-circuit withstand make it "an open research challenge".	Heron (two cell failures per phase, sub-cycle bypass); Amperesand (proprietary redundancy); ORNL flags the open problem. [PRIMER ch.5.4]	pilot
Insulation coordination and partial discharge	Georgia Tech identify the mismatch between MOV protective levels and SST insulation strength, and "the incompatibility of standard impulse test on SST protective structures" — the standard BIL test does not apply cleanly to an SST.	Georgia Tech (Divan, Graber et al.). [WEB, verified 2026-09-23] NC State on BIL; MFT insulation is an active academic area. [PRIMER ch.5.3]	lab
Infrastructure			
Interconnection queue and transmission build	ERCOT: about 410 GW queued against 9.0 GW approved to energise and a 3.9 GW observed large-load peak. 888 miles of 345 kV-and-above built in a year against a benchmark near 5,000.	FERC (RM26-4-000); nobody can shorten it with power electronics. [PRIMER ch.6.1]	unsolved
Transformer and switchgear lead times	Standard units about 128 weeks, generator step-ups about 144, worst cases near four years. One flagship US plant at 57 units a year is about 6% of national demand, arriving 2027.	Hitachi Energy, Siemens Energy, GE Vernova/Prolec — all building capacity, all arriving late. [PRIMER ch.6.2]	pilot
Ride-through and large-load standards	This is the one constraint an SST genuinely relieves: a converter-fronted load is the only kind that can be programmed to a curve and modelled as one device.	NERC Project 2026-02 (CLO-001/002/003); FERC deadline 31 December 2026; ERCOT NOGRR 282 already numeric. [WEB, verified 2026-09-23]	standardised
O&M			
Mean time to repair and hot-swap	The swap-in-minutes claim is not novel to SSTs — GE Vernova markets the same property for its MM7 modular multilevel drive, which has two decades of field service behind it.	Amperesand claims a power stage swaps in under 20 minutes, two people with a lift. The claim is published in a case study by the company that sells it the silicon. [WEB, verified 2026-09-23]	product
MTBF or FIT for a complete MV converter	The one independent number available: N+1 raises the MTBF of a modular multilevel drive by a factor of about 1.7 — a long way from the step change implied by "five nines".	Nobody. No vendor publishes one. [WEB, verified 2026-09-23]	unsolved
Service life against an oil transformer	The published SST design target found is "15+ years". An oil transformer is financed against 30-40. The owner is buying a mid-life silicon replacement into the capital plan, and nobody has costed it.	Nobody. No standards body has set a life class for an MV SST. [WEB, verified 2026-09-23]	unsolved
Spares strategy, spares cost and warranty	An oil transformer's spares strategy is a bushing and a fan. A several-hundred-module SST's spares strategy is an inventory of power electronics with its own obsolescence clock.	Nobody, with one exception: Enphase publishes 10% module redundancy and a 10-year limited warranty. [WEB, verified 2026-09-23]	unsolved
Part-load efficiency	Every efficiency figure in the market — 98%, >98.5%, 98.5-99% — is a peak or nominal number. A data hall does not run at rated load.	Nobody publishes a curve. [ANALYSIS]	unsolved
Firmware update and cyber exposure	A companion FCC waiver preserves qualifying firmware updates on already-authorised covered devices only through at least 1 January 2029 — the right to patch is now a time-limited regulatory grant. Whether NERC CIP attaches to a behind-the-meter hall-edge SST is unaddressed anywhere.	ISA/IEC 62443 as the framework; the FCC as the unexpected regulator of the update channel itself. [WEB, verified 2026-09-23]	standardised
Technicians	A reported shortfall of more than 300,000 electricians against roughly 20,000 retiring a year; an apprentice starting a five-year IBEW programme in 2025 qualifies in 2030.	Google/NECA/Electrical Training Alliance are funding training; the arithmetic does not close on the relevant timescale. [WEB, verified 2026-09-23]	pilot
Standards			
DC arc-flash incident-energy model	NFPA 70E 2027 adds a new Article 310 naming DC thresholds; the calculation method is aimed at 2030. The hazard is named three years before the method exists.	NFPA Fire Protection Research Foundation — project open, seeking \$3-4M, results aimed at NFPA 70E 2030. [WEB, verified 2026-09-23]	unsolved
Listing of an MV power-conversion unit	The envelope exists. What does not exist is a listed data-centre SST: no vendor had completed UL certification for data-centre SST deployment as of May 2026, per an analyst house that sells the newsletter the claim appears in.	UL (1741 +SA/SB, 347A, 2877); IEC SC 22E for 62477-2 Edition 2, CDV vote closed 2026-03-13. [WEB, verified 2026-09-23]	standardised

Obstacle	The mechanism	Who is doing what about it	Status
The 800 V DC specification itself	This is the vocabulary a buyer's engineer will use. The specification document itself could not be retrieved in this run — opencompute.org returned HTTP 403 to every attempt.	OCP — Google, Microsoft and NVIDIA. LVDC Solid-State Transformer Specification v0.3, July 2026; more than 80 manufacturers building to it. [WEB, verified 2026-09-23]	standardised
Utility			
Acceptance of a converter at the service point	NOGRR 282 is now a numeric test: ride through 2 s at 0.85 pu and 150 ms below 0.35 pu, draw no more than 150% current, recover to 90% of consumption within 2 s, survive six sags in 90 s.	ERCOT/PUCT (NOGRR 282; 16 TAC 25.194, Project 58481); NERC Project 2026-02. [WEB, verified 2026-09-23]	standardised
Approved-equipment-list qualification	At 12.47-34.5 kV the service transformer is commonly utility-owned. An SST displacing it must be accepted onto a utility's approved-equipment list — a multi-year qualification with no visible US precedent, which is why every announced US pilot sits behind a customer-owned MV substation.	Nobody visible. No covered vendor discloses a US utility qualification. [PRIMER ch.6.5]	unsolved
MV equipment inside an operating data hall	An SST terminating 34.5 kV at the row turns white space into MV-classified space, with vault, access, ventilation and qualified-person rules. Probably the largest permitting unknown for facility-level SSTs.	Nobody. No US AHJ precedent was located. [PRIMER ch.6.4]	unsolved
Policy			
FCC Covered List	Two prongs, both required: a bidirectional AC-DC converter, and components enabling remote communication. No voltage or power threshold. "Foreign-produced" is the Buy American content test at 48 CFR 25.101(a), not a nationality test — so a US-headquartered vendor building offshore is captured.	FCC PSHSB (DA 26-786, 28 July 2026); DoW/DHS conditional approval, applications due 1 January 2028. [WEB, verified 2026-09-23]	standardised
Executive Order 14421	Signed 26 August 2026. Covers transformers, inverters, breakers, turbines and storage — but applies at 69 kV and above and excludes local distribution. A 34.5 kV hall-edge SST is probably outside it and inside the FCC list.	DOE — implementing regulations due 24 December 2026, FAR recommendations 22 February 2027. [WEB, verified 2026-09-23]	standardised
Insurance and bankability of a novel MV asset	No insurer, reinsurer, broker or owner's-engineer standard naming solid-state transformers was found. For a first-of-a-kind MV asset this is the largest single information gap on the commercial side.	Nobody visible. [ANALYSIS]	unsolved

7.1 The operations-and-maintenance chapter, which does not exist anywhere else

THE FRAMING THAT MATTERS MORE THAN ANY NUMBER BELOW

There is no field-service corpus for medium-voltage solid-state transformers because **there is no installed base**. Every serviceability figure in this section is either a vendor design claim about a product not yet shipped at scale, or borrowed from the nearest analogue with real field history — the cell-based MV drive, serviced by OEM field engineers for two decades. The honest position is that **the O&M model is asserted, not demonstrated**, and holding that distinction in a customer meeting is worth more than any of the individual claims. [\[WEB, verified 2026-09-23\]](#)

Mean time to repair. The most concrete claim available is Amperesand's: a power stage "can be swapped out in less than 20 minutes", with modules "small and light enough for simple two-person replacement in minutes, using readily available lifts". It is published in a case study by Wolfspeed, which supplies the silicon carbide inside the product, so both parties gain from the claim. [\[WEB, verified 2026-09-23\]](#) The comparison that puts it in perspective: GE Vernova markets the identical property for its MM7 modular multilevel drive — modules "light enough for one or two people to swap a faulty unit with a spare within a few minutes" — so swap-in-minutes is established MV-drive practice rather than an SST innovation. [\[WEB, verified 2026-09-23\]](#)

Reliability arithmetic. *No MV SST vendor publishes an MTBF or FIT number for the complete converter.* Every reliability statement located is either qualitative — “five nines”, “excellent cosmic ray performance” — or a manufacturing metric such as Enphase’s 500 defects per million shipped, which is a production-quality figure and must never be relayed as a field reliability rate. [WEB, verified 2026-09-23] The one independent number available says N+1 redundancy raises a modular multilevel drive’s MTBF by a factor of about 1.7, and the same source warns that “extreme care should be taken when using these figures to compare vendors” — a drives consultancy with an interest in arguing vendor numbers are incomparable, which happens to point at a defensible conclusion. [WEB, verified 2026-09-23]

Service life. The published SST design target located is “15+ year lifetime”. A large oil-filled transformer is specified and financed against a service life several times that. No standards body has defined a life class for an MV SST and no vendor publishes an end-of-life or mid-life-refresh plan, so an owner buying an SST is buying a mid-life silicon replacement into the capital plan — and nobody found has costed it. [WEB, verified 2026-09-23]

Warranty and spares. Exactly one warranty term was located across the entire research pass: Enphase’s 10-year limited warranty on the IQ SST. [WEB, verified 2026-09-23] No warranty, service-contract structure, response-time commitment or availability guarantee was published by Eaton, ABB, Hitachi Energy, Amperesand, DG Matrix or Heron Power in anything reachable. [WEB, verified 2026-09-23] Nor does any vendor publish a spares list, a recommended stocking level or a spare-cell price. [WEB, verified 2026-09-23] The structural point is worth carrying into a meeting: an oil transformer’s spares strategy is a bushing and a fan; a several-hundred-module SST’s spares strategy is an inventory of power electronics with its own obsolescence clock.

Firmware and cyber — a transformer that takes software updates. The frameworks exist and an MV-SST-specific profile does not: ISA/IEC 62443 is the product-side OT security standard and vendor platforms already aggregate access, configuration and firmware events for monitoring against NERC CIP and IEC 62443-3-3. [WEB, verified 2026-09-23] The regulatory teeth arrived from an unexpected direction. The stated national-security rationale for the FCC’s July 2026 action is precisely the firmware-update channel, and the companion waiver preserves qualifying Class I and Class II software and firmware updates on already-authorised covered devices only through at least 1 January 2029 — **the right to patch a covered device is now a time-limited regulatory grant, not a vendor decision.** [WEB, verified 2026-09-23] Whether NERC CIP attaches to a behind-the-meter hall-edge SST is addressed by no source located, and is an open question. [WEB, verified 2026-09-23]

Technicians. The United States is reported to need more than 300,000 additional electricians while roughly 20,000 retire every year, with medium-voltage scope and switchgear commissioning named as the specific skills moving into the data-centre footprint — a widely repeated figure from a trade careers source that does not trace to a primary one. [WEB, verified 2026-09-23] Training money is being spent by Google with NECA and the Electrical Training Alliance, and the same source makes the honest point about it: a four-week credential is not a four-year apprenticeship, and an apprentice starting a five-year IBEW programme in early 2025 qualifies in 2030. [WEB, verified 2026-09-23]

7.2 Standards and listing — the three-year gap

The clearest single finding in the whole standards picture is a timing gap. **NFPA 70E 2027 adds a new Article 310 for DC hazards** with explicit thresholds — contact-thermal at power at or above 1,000 W, shock at or above 100 V DC and above 40 mA, arc flash above 150 V DC and above 1.2 cal/cm² — and an 800 V DC hall bus is over every one of them. [WEB, verified 2026-09-23] Meanwhile the NFPA Fire Protection Research Foundation’s project to develop a DC arc-flash hazard model was, in February 2026, still seeking sponsors at an estimated \$3–4 million, with results aimed at the **2030** edition of 70E. [WEB, verified 2026-09-23] So the hazard is named in 2027 and the calculation method arrives, at the earliest, in 2030. In between, operators energise 800 V DC busway with conservative table-based PPE, de-energised-work rules and per-project negotiation with the authority having jurisdiction. [PRIMER ch.6.4]

The listing envelope itself exists. UL 1741 with supplements SA and SB is the dominant US route and MV considerations were brought in by revision; UL 347A and UL 2877 cover MV power conversion from 1 kV up to 38 kV, which puts the 34.5 kV half-edge case inside them. [\[WEB, verified 2026-09-23\]](#) Internationally IEC 62477-2 covers converters from 1 kV AC / 1.5 kV DC up to 36 kV AC / 54 kV DC, and Edition 2 went to a committee-draft-for-vote closing 13 March 2026.

[\[WEB, verified 2026-09-23\]](#) NEC 2026 adds a new Article 270 for medium-voltage grounding and bonding above 1,000 V AC, relevant precisely because SSTs push MV into the building footprint. [\[WEB, verified 2026-09-23\]](#)

THE LINE TO REMEMBER FROM PCIM EUROPE 2026

“Currently, we don’t have a specification, and currently we don’t have a UL certification for the 800-VDC bus. That is a major gap.” [\[WEB, verified 2026-09-23\]](#) Related, and equally load-bearing: the server ecosystem was built around the SELV classification covering circuits below 60 V, so creepage and clearance at 800 V are unresolved in existing designs. [\[WEB, verified 2026-09-23\]](#)

7.3 Utility acceptance — the question has changed

Eighteen months ago the question was whether a utility would accept a converter at the service point. It is now whether any large load can *prove* it rides through — and a converter is the easiest thing to prove it with. The driver is not SST-specific; it is the load-loss events. [\[WEB, verified 2026-09-23\]](#)

ERCOT and the PUCT have turned ride-through into a numeric acceptance test through NOGRR 282, and PUCT rule 16 TAC §25.194 under Project 58481 implements Texas SB 6 at 75 MW and above, with study fees of \$100,000 for 75–249 MW and \$300,000 at 250 MW and above, financial security at \$50,000 per MW of requested peak demand and a non-refundable interconnection fee of \$50,000 per MW of contracted peak demand. [\[WEB, verified 2026-09-23\]](#) NERC Project 2026-02 proposes CLO-001 for interconnection studies and modelling data, CLO-002 for operational data and communications and CLO-003 for protection coordination and disturbance monitoring, against a FERC-directed filing deadline of 31 December 2026. [\[WEB, verified 2026-09-23\]](#) FERC opened a generic rulemaking on large-load interconnection at Docket RM26-4-000 and on 18 June 2026 issued Federal Power Act section 206 show-cause orders to all six RTOs and ISOs on their large-load and co-located-load tariffs. [\[WEB, verified 2026-09-23\]](#) Individual utilities have moved too: Dominion’s new GS-5 rate class for customers above 25 MW was approved in November 2025 for a January 2027 effective date with 14-year contracts [\[WEB, verified 2026-09-23\]](#); AEP Ohio’s data-centre tariff took effect 23 July 2025 requiring large new customers to pay for a minimum 85% of stated monthly energy. [\[WEB, verified 2026-09-23\]](#)

What has *not* changed is the approved-equipment question. At 12.47–34.5 kV the service transformer is commonly utility-owned, and an SST displacing it must be accepted onto a utility’s approved-equipment list — a multi-year qualification with no visible US precedent, which is why every announced US SST pilot sits behind a customer-owned medium-voltage substation. [\[PRIMER ch.6.5\]](#)

7.4 The three obstacles nobody is working on

Worth stating plainly, because each is a question a careful buyer asks and no vendor can currently answer. **Insurance and bankability** of a novel MV asset: no insurer, reinsurer, broker or owner’s-engineer standard naming solid-state transformers was located in the research pass. [\[WEB, verified 2026-09-23\]](#) **Part-load efficiency**: every figure in the market is a peak or nominal number, and a data hall does not run at rated load. [\[WEB, verified 2026-09-23\]](#) **Field-failure and availability data** from an operating installation: none exists to find, because there is no installed base of MV data-centre SSTs to report on. [\[WEB, verified 2026-09-23\]](#)

ANALYSIS — WHERE THE LEVERAGE IS

The technical obstacles all have named owners and most are at *product* or *standardised*. The operational ones mostly have nobody. That inverts the usual assumption about where a challenger has room: the engineering is crowded and the operations manual is blank. **The first vendor to publish an overload curve, a part-load efficiency curve, an MTBF figure for the complete converter and a warranty with a response-time commitment gains a durable advantage at essentially zero engineering cost** — because publishing what you already know is cheaper than inventing anything, and every one of those four is a question a buyer's engineer asks in the first hour.

8 NVIDIA and Oracle: what their programmes specify, and what their engineers will ask

Extends the primer's chapter 2.2 and chapter 4.2 on NVIDIA; the primer does not cover Oracle at all. These are two completely different conversations and the most useful thing this chapter does is say why. NVIDIA has published an architecture with dates, ratings and interfaces, so with NVIDIA you are being checked against a record. Oracle has published almost nothing about the electrical architecture inside its AI data centres, so with OCI you are helping write one.

What each programme has actually committed to, and when

NVIDIA architecture

2025-10-13

800 VDC ecosystem published; about 30 named partners; MV input cited as "e.g. 35 kVAC"; conversion "at the facility level"

2026-03

Joint NVIDIA / Google / Microsoft 800 VDC white paper via OCP

2026-06-01

Siemens + NVIDIA + Fluence + nVent DSX Vera Rubin reference design: nominal 34.5 kV utility, 136 MW facility / 100 MW IT, factory-tested MV and LV skids, UL-aligned, Tier III concurrent maintainability — and no mention of 800 VDC or SSTs at all

2026-07

OCP LVDC Solid-State Transformer Specification v0.3 (Microsoft, Google, NVIDIA)

2026-08-11

Three-tier rollout stated: power rack H2 2026, row power center 2 MW/row 2027, DC power block 2027+; more than 80 manufacturers building to the spec

2026-09-21

NVIDIA DSX Ready qualification programme launches — covering BESS and CDUs only. There is no NVIDIA qualification path for an SST

2026-H2

800 VDC power rack enters production; Vera Rubin NVL72 ships

2027

Row power center, 2 MW per row, overhead 800 VDC busway; full-scale 800 VDC production "coincides with Kyber"

2029

Next-generation SST and MV-direct conversion: 34.5 kV AC to 800 V DC, blocks toward 10 MW — corpus-only, page-referenced to the August 2026 execution paper, not confirmed on a public NVIDIA page

Oracle / OCI

2025-10-14

"First Principles" — OCI's only engineering post on AI power. Its entire power content is capacitance, advanced energy storage, hardware-accelerated GPU ramping and software ramp control. No voltage, no kW/rack, no topology

2026-01-26

Oracle states it is funding onsite transmission lines, battery storage and dedicated substations so costs are not passed to ratepayers

2026-04-27

Project Jupiter (Dona Ana County, NM) to be fully powered by up to 2.45 GW of Bloom fuel cells, replacing planned gas turbines and diesel, as one single microgrid campus — Oracle's boldest published electrical decision

2026-09-15

Investment in more than 1.7 GW of carbon-free electricity supplying ERCOT

2026-09-16

Project Lighthouse: Oracle names American Transmission Co. on the Ozaukee County interconnection — the only host transmission utility Oracle names itself

not published

Rack density, distribution voltage, where AC-to-DC conversion sits, MV-direct conversion, solid-state transformers, and any vendor-qualification programme. The only Oracle-800 VDC link in existence is NVIDIA naming OCI among providers "designing for 800-volt data centers" — NVIDIA's characterisation, not Oracle's statement

Figure M11 — What each programme has actually committed to, and when. Boundary: milestones are dated from the named source only, and the corpus-only NVIDIA item is marked as such in its own text. Two parallel dated lists rather than a shared time axis, because NVIDIA's milestones cluster inside eleven months and overprint at page width. Composed from megmeet-sst-briefing-data.json → timelines.

8.1 NVIDIA — what is on the record

The architecture is published and consistent on its own terms, with one internal inconsistency worth knowing about. NVIDIA's technical blog specifies converting **13.8 kV AC** grid power directly to 800 V DC at the data-centre perimeter using industrial-grade rectifiers, then distributing via row-level 800 V DC busways over two conductors with DC-DC conversion in the compute rack. [\[WEB, verified 2026-09-23\]](#) A different NVIDIA blog of the same date cites medium voltage

“e.g., 35 kVAC” as the utility supply and puts AC-to-DC conversion “at the facility level”. [\[WEB, verified 2026-09-23\]](#) **NVIDIA's own pages name two different MV classes.** Ask which one a given site uses; do not assume, and do not assume they are unaware of the discrepancy.

The load profile is the most important number NVIDIA publishes for a converter vendor: rack utilisation swings from **30% to 100% in milliseconds**. [\[WEB, verified 2026-09-23\]](#) NVIDIA's own answer to it is rack-side — 400 J per GPU of capacitor storage in Vera Rubin, “6× more rack-level energy storage” than the prior generation, Intelligent Power Smoothing and dynamic power steering. [\[WEB, verified 2026-09-23\]](#) The commercial consequence is that the real negotiation is not whether your converter can ride the swing, but *how much of the ride-through is yours and how much is theirs*.

The ecosystem roster is a weaker signal than it looks. NVIDIA's October 2025 blog lists companies “building 800 VDC-compatible infrastructure”, which is a published alignment and not a qualification, a design win or a supply agreement; by August 2026 NVIDIA describes more than eighty manufacturers building to the specification without publishing the list. [\[WEB, verified 2026-09-23\]](#) Critically, **that page does not break the roster into rectifier or SST sub-categories and names no vendor as an SST vendor** — so any claim that a firm is “NVIDIA's named SST partner” is unsupported by it. [\[WEB, verified 2026-09-23\]](#) The three tiers it does name are silicon (fourteen firms including Infineon), power system components (Bizlink, Delta, Flex, Lead Wealth, LITEON and Megmeet) and data-centre power systems (ABB, Eaton, GE Vernova, Heron Power, Hitachi Energy, Mitsubishi Electric, Schneider Electric, Siemens and Vertiv). [\[WEB, verified 2026-09-23\]](#)

The sixteen questions an NVIDIA power engineer would ask

Each is anchored to something in the published record; where the record is silent, the question comes from the absence and is marked as such.

#	The question	Why they ask it, and what you need ready
1	Two-wire 800 V or three-wire ± 400 V — and can your unit do both?	The published architectures genuinely diverge: OCP's Diablo is three-wire ± 400 V, NVIDIA's is two-wire 800 V with higher copper utilisation. A vendor without a side fails the first five minutes. [WEB, verified 2026-09-23]
2	What is your input MV class — 13.8 kV, 34.5 kV, or both?	Their own pages name both, and the Siemens NVIDIA-aligned reference design is built on nominal 34.5 kV. They know the record is inconsistent and will test whether you do. [WEB, verified 2026-09-23]
3	What is your grounding scheme, and do you support continuous insulation monitoring and active arc-fault detection?	The protection posture reported for the specification is high-resistance grounding, continuous insulation monitoring, active arc-fault detection and mechanical connector interlocks. Expect them to ask which of these your product implements and which it assumes the facility provides. [WEB, verified 2026-09-23]
4	What is your available fault current and how fast do you clear?	The published ecosystem expectation is that an SST limits fault current to roughly 1.5× nominal for durations limited to hundreds of milliseconds — a radically different protection philosophy from a magnetic transformer feeding tens of kA, and it changes downstream breaker coordination entirely. Bring the number and the coordination study. [WEB, verified 2026-09-23]
5	What is your isolation rating?	The published expectation is 100 kV insulation for a 35 kV input. [WEB, verified 2026-09-23]
6	When the rack swings 30% to 100% in milliseconds, how much of the ride-through is yours and how much is the rack's?	This is the single most important technical conversation you will have, because it is a negotiation about responsibility rather than a specification check. [WEB, verified 2026-09-23]
7	What is your efficiency, at what load points, and measured across which boundary?	NVIDIA's whole business case is an <i>end-to-end</i> claim, so a point efficiency at full load is not an answer to it. State the boundary and give the curve at the real load profile. [WEB, verified 2026-09-23]
8	What is your unit rating and how do you get to a 2 MW row or a 4.8 MW block?	Their modularity numbers are published: a 2 MW row power centre for 2027 and 4.8 MW blocks with 1,250 A busways and a 4+1 catcher. Your paralleling story has to land on them. [GUIDANCE nvidia-800vdc p14–21]
9	Is it N+1 at the cell level, and are the sub-blocks hot-swappable?	The published expectation is modular sub-blocks with hot-swap capability, against availability targets of 99.9% for AI training and 99.999% for inference. Ask which workload class the site is, because the two numbers imply different products. [WEB, verified 2026-09-23]

#	The question	Why they ask it, and what you need ready
10	What is your failure mode, and what is the bypass?	NVIDIA-aligned reference designs target Tier III concurrent maintainability, and an SST with no bypass path cannot be concurrently maintained. This is a gating question, not a detail. [WEB, verified 2026-09-23]
11	Are you UL-listed for the 800 V DC bus — and if not, what is your path?	Nobody is. Know your listing status precisely and do not overstate it; the honest answer here is worth more than a confident one. [WEB, verified 2026-09-23]
12	How do you get into a DSX Ready category, and will you self-qualify or submit test data?	MV conversion is not yet a category. The right answer shows you know that and want to help define it. [WEB, verified 2026-09-23]
13	What is your telemetry and control interface?	<i>From the absence:</i> NVIDIA publishes no telemetry or management-interface requirement — no PMBus, Redfish, DMTF or Modbus reference on any page located. But the Siemens reference design specifies a centralised management suite with a single pane of glass, so integration into someone else's DCIM is the real requirement. Ask whose. [WEB, verified 2026-09-23]
14	Do you fit in a factory-tested skid, and what is your lead time against the 2027 row date?	The one published NVIDIA-aligned reference design is delivered as pre-engineered, prefabricated, factory-tested MV and LV skids. Lead time is a schedule question against published dates. [WEB, verified 2026-09-23]
15	What is your harmonic signature at the MV terminals?	<i>From the absence:</i> no NVIDIA or OCP harmonic limit could be evidenced. Since an SST is an active front end feeding a utility interconnect they will ask anyway; bring your THD figures and an IEEE 519 assessment before anyone requires one. [WEB, verified 2026-09-23]
16	What is your cyber and firmware posture?	<i>From the absence:</i> not addressed on any NVIDIA or OCP page reachable. An SST is a software-defined, remotely updatable element of the critical power path; expect the review regardless. [WEB, verified 2026-09-23]

8.2 Oracle — the headline finding is an absence

ORACLE HAS PUBLISHED NOTHING ABOUT THE ELECTRICAL ARCHITECTURE INSIDE ITS AI DATA CENTRES

No voltage class, no distribution topology, no rack-power figure, no DC distribution, no medium-voltage conversion strategy, and **no mention of solid-state transformers anywhere in Oracle's own published material** across its blogs, newsroom, data-centre pages and site pages. [WEB, verified 2026-09-23] Oracle's public voice on power is about *sourcing* it — grids, fuel cells, turbines, substations, ratepayers, interconnection — never about converting and distributing it.

Oracle's only substantive engineering post on AI data-centre power is "First Principles: Data Center Innovations to Power Gigawatt Scale Superclusters" of 14 October 2025, by OCI's chief technical architect and its VP of AI infrastructure. It is a one-minute read, and its entire statement on power is that AI gigawatt-scale data centres face "substantial load sizes and significant power fluctuations resulting from synchronous computational workloads", and that to manage them "OCI leverages the current best practices integrating capacitance, advanced energy storage solutions, hardware-accelerated GPU ramping techniques, and software-based ramp control algorithms". [WEB, verified 2026-09-23] Those four levers are OCI's published framing of the problem an SST would have to live inside.

THE ONE THING NOT TO SAY TO AN ORACLE ENGINEER

NVIDIA's blog states that "CoreWeave, Lambda, Nebius, Oracle Cloud Infrastructure and Together AI are among other industry pioneers designing for 800-volt data centers". [WEB, verified 2026-09-23] That is the strongest published Oracle–800 V DC link that exists, and **it is NVIDIA's characterisation of OCI, not an Oracle statement**. Quoting it back to an OCI engineer as an Oracle position would be a mistake, and a visible one.

What Oracle *has* published is a power-procurement record, and it is unusually rich. **Project Jupiter** in Doña Ana County, New Mexico will be "fully powered" by up to 2.45 GW of Bloom Energy fuel cells, replacing previously planned gas turbines and diesel generators and consolidating the facility into "one single microgrid campus" — Oracle's single most consequential published electrical decision. [WEB, verified 2026-09-23] At **Abilene**, Oracle states the campus connects directly to the ERCOT grid with GE Vernova natural-gas turbines for backup, across eight buildings on 1,100 acres — and states no voltage, no rack density and no electrical topology. [WEB, verified 2026-09-23] Oracle is also the tenant rather than the owner there: the campus is developed by a Crusoe, Blue Owl and Primary Digital Infrastructure joint venture and leased to Oracle. [WEB, verified 2026-09-23] At **Project Lighthouse** in Ozaukee County, Wisconsin, Oracle names American Transmission Co. on the interconnection application — the only host transmission utility Oracle names by name anywhere in its own material. [WEB, verified 2026-09-23] And Oracle states plainly that it is "funding new onsite transmission lines, battery storage, and dedicated substations" so costs are not passed to ratepayers. [WEB, verified 2026-09-23]

The twelve questions an OCI electrical engineer would ask

The framing is different. They are not checking you against a specification — they are deciding whether MV-direct conversion belongs in their next design at all. Sell the decision, not the compliance.

#	The question	The published basis
1	Where does this fit in a campus we island on fuel cells?	Project Jupiter, up to 2.45 GW of Bloom fuel cells as one microgrid campus. Solid-oxide fuel cells are DC-native and Oracle has not said what it does with that DC — it is the only Oracle site where the published design makes a DC-native conversion conversation obviously relevant. [WEB, verified 2026-09-23]
2	Does your converter replace or supplement capacitance and energy storage?	Position the SST against OCI's own four published levers by name. [WEB, verified 2026-09-23]

#	The question	The published basis
3	Can it act as an MV UPS, and does that let us delete equipment?	The OCP definition frames an MV SST coupled with energy storage as a medium-voltage UPS replacing the line-frequency transformer, rectifier units and storage. For a tenant paying for onsite transmission, storage and dedicated substations, deletion of apparatus is the argument that lands. [WEB, verified 2026-09-23]
4	What does it do to our interconnection application and our behaviour at the point of common coupling?	Interconnection is on Oracle's critical path and in its public statements. An active front end changes the PCC behaviour they have already filed. [WEB, verified 2026-09-23]
5	We are the tenant, not the owner — who do you actually have to satisfy?	At several sites the specifying authority is the developer. Expect to be routed to Crusoe, Vantage, Related Digital or Digital Realty, and expect to be tested on whether you already knew that. [WEB, verified 2026-09-23]
6	What service voltage do you need, and does it match what we already bought?	Oracle publishes no service-voltage class for any site but does fund dedicated substations. The question is whether your class matches substations already designed or ordered. [WEB, verified 2026-09-23]
7	How does this interact with our closed-loop, non-evaporative cooling standard?	Oracle standardises on it, with water use "on par with a typical office building" and a one-time fill at Abilene. A liquid-cooled SST joins that loop on Oracle's terms, or is air-cooled. [WEB, verified 2026-09-23]
8	What is your availability figure and your failure mode, for an inference fleet?	The published ecosystem targets are 99.9% for training and 99.999% for inference, and OCI runs both on the same campuses. Ask which buildings are which. [WEB, verified 2026-09-23]
9	What is your listing status, and what do we tell the AHJ and our insurer?	Oracle is building in Texas, New Mexico and Wisconsin under three regulatory regimes and is publicly sensitive to permitting. The published gap — no UL certification for the 800 V DC bus — is theirs to carry too. [WEB, verified 2026-09-23]
10	What is the lead time, and is the supply chain domestic?	Oracle publishes no procurement posture, standardisation list, domestic-content rule or exclusion list for electrical equipment, so this is asked commercially rather than against a policy — but its community-facing material emphasises local suppliers and US siting. [WEB, verified 2026-09-23]
11	What is the service model and where are the spares?	NVIDIA's architecture claims up to 70% lower maintenance cost; an operator will ask you to evidence that in a service contract, for campuses Oracle says need more than 1,000 permanent employees. [WEB, verified 2026-09-23]
12	Who else has one running, and at what scale?	Nothing in Oracle's record indicates any SST deployment. With no internal reference and no published specification, the first question a cautious operator asks is for someone else's operating hours. Have the honest answer ready. [WEB, verified 2026-09-23]

ANALYSIS — THE TWO CONVERSATIONS, IN ONE SENTENCE EACH

With NVIDIA you are being measured against a published architecture that does not yet have a box for your product, so the winning posture is technical precision plus an offer to help write the category. With OCI you are talking to an organisation that has published its power *sourcing* in detail and its power *conversion* not at all, so the winning posture is to arrive with the decision framed — what deletes, what it costs, who else is running one — rather than with a compliance answer to a specification that does not exist.

PART IV

Megmeet and the competition

What the company has actually disclosed about its own converter, how it compares vendor by vendor, where it stands in each of its six business groups, and what the policy fence does to it.

9 Megmeet's SST and 800 V DC chain — disclosed against undisclosed

This is the chapter to be most careful in, because it is the one where guessing is most tempting and most costly. Everything below is either what Megmeet has published or an explicit statement that it has published nothing.

9.1 The service-voltage class — the answer is no, and the search was thorough

MEGMEET HAS NEVER DISCLOSED A KV CLASS FOR ITS SOLID-STATE TRANSFORMER

Across the thirty-eight sources pinned in the v7 dossier, the string “kV” does not appear once, and the converter is described only as taking *grid high-voltage input* to 800 V DC. [DOSSIER megmeet v7] A dedicated four-filing text scan performed for this briefing found **zero hits** for kV, 千伏 or even 中压 (“medium voltage”) in the FY2025 annual-report summary, the H1 2026 interim, and two investor-relations records. [WEB, verified 2026-09-23]

Two details sharpen it. In the FY2025 summary the SST is described as directly accepting grid high-voltage input and outputting 800 V DC at over 98.5% efficiency, supporting megawatt-class centralised supply. In the H1 2026 interim the same product carries two new words: 研发中, “under development”, and 预期, “expected”, in front of the efficiency figure. [WEB, verified 2026-09-23] Between April and August the company *narrowed* its own claim. And when an analyst asked point-blank in April 2026 for the specifications of the domestic and international versions, the answer named two versions and attached no voltages to either. [WEB, verified 2026-09-23]

THE ONE CONTRARY SOURCE, AND WHY IT IS NOT USABLE

A Chinese industry report is indexed as stating that Megmeet runs a “domestic 10 kV / overseas 35 kV dual version” with Infineon silicon and a prototype in the first half of 2027. The host returned HTTP 403 to every fetch, so the page was never read. [WEB, verified 2026-09-23] Two reasons not to carry it: 10 kV domestic and 35 kV international is exactly the inference an analyst would make from “domestic version / international version” under Chinese grid norms, which is the inference this document forbids; and **35 kV is not a US service class** — the US class is 34.5 kV. Treat it as low confidence and do not repeat it as Megmeet's specification.

9.2 What is published — the chain, box by box

The chain Megmeet publishes runs: AC compensator → BESS and backup shelves → **SST (>98.5%, 800 V DC out)** → 1 MW+ sidecar → power shelves → point of load. [DOSSIER megmeet v7] Everything in it below the converter is specified and shipping; the converter is the one item not built.

Stage	What is disclosed	What is not
Solid-state transformer	Grid high-voltage input → 800 V DC; efficiency above 98.5%, stated as <i>expected</i> in the most recent filing; megawatt-class centralised supply supporting a PUE below 1.1, as a company claim. [DOSSIER megmeet v7]	Everything else. No service-voltage class, no model number, no datasheet, no MVA or MW rating beyond “megawatt-class”, no topology, no MFT frequency, no footprint, no cooling, no protection scheme, no breaker, no availability date, no measurement boundary for the efficiency figure. [WEB, verified 2026-09-23]

Stage	What is disclosed	What is not
800 V HVDC sidecar	480/380 V AC → 800 V DC external power cabinet, over 1 MW per rack, configurable 800 V DC BBU and supercapacitor shelves; announced for NVIDIA Kyber in May 2025 — first mover among mainland vendors. Claims against legacy in-rack AC: up to +5% end-to-end efficiency, −45% copper, up to −30% TCO. [DOSSIER megmeet v7]	Those three claims are NVIDIA's architecture claims for 800 V DC against a legacy AC hall , not measured Megmeet results, and they are not SST claims at all. [PRIMER ch.4.3]
800 V / 570 kW side rack	800 V DC output rated 570 kW, with integrated battery-backup and power-conversion functions. [DOSSIER megmeet v7]	No efficiency figure, no redundancy scheme published.
Power shelves	33 kW (IPU433-AI616): 1U 19-inch, 33 kW N+1 / 16.5 kW N+N, hot-swap, IPMI/Redfish/SNMP. 110 kW liquid-cooled for GB300 and Vera Rubin-class racks at 45 °C warm-liquid inlet. IPU433K-AA1616: 27.5 kW N+1 / 16.5 kW N+N, six 5.5 kW rectifier slots. [DOSSIER megmeet v7]	—
Backup	BBU shelf 1U: 180 s of backup at 16.5 kW, 240 s at 13.2 kW, peak support to 26.4 kW. Supercapacitor shelf 15 kW. [DOSSIER megmeet v7]	—
Server PSUs	CRPS / M-CRPS from 800 W to 5.5 kW (Ruby M-CRPS launched March 2026); 80 PLUS Platinum / Titanium (96%) / Ruby; power density above 100 W/in ³ as a company claim against under 50 traditional; hot-swap N+1, active PFC, PMBus 1.2 with black box; CB, UL, CE and CQC certifications. [DOSSIER megmeet v7]	—
Not sold at all	—	Megmeet sells no busway and no DC breaker . The shipping chain begins at three-phase 380–480 V AC. [WEB, verified 2026-09-23]

The disclosure gap — nobody has all four, and one column is empty for everyone

	Service-voltage class	Unit MW rating	Evidence beyond an announcement	A listed unit
Heron Power	yes	yes	—	—
Siemens + Reinhausen	yes	—	—	—
Delta Electronics	yes	—	yes	—
SolarEdge	yes	yes	—	—
Enphase	yes	yes	—	—
DG Matrix (Cell-MV, roadmap)	yes	yes	—	—
Novos Power	yes	yes	—	—
Eaton (Resilient Power)	yes	yes	—	—
Sungrow (EnerNeo)	yes	yes	yes	—
Hyosung	yes	yes	partial	—
Alderbuck Energy	yes	—	partial	—
Zhonhen (Panama)	yes	yes	yes	—
JST Power Equipment	yes	yes	partial	—
ETH Zurich (research)	yes	yes	partial	—
EPFL + Navitas	yes	yes	partial	—
ABB (HiPerGuard, MV UPS)	yes	yes	yes	—
NC State / NYPA / EPRI (1 MVA research unit)	yes	yes	yes	—
Megmeet	—	—	—	—
Amperesand	—	—	—	—
GE Vernova	—	—	—	—
Hitachi Energy	—	—	—	—
Vertiv	—	—	—	—
Schneider Electric	—	—	—	—
Sinexcel	—	—	—	—

Figure M12 — The disclosure gap. Boundary: four disclosure fields per vendor scored as published, partial or absent, from the pinned sources only. “Evidence beyond an announcement” is scored from the evidence tier; “a listed unit” means a named certification for a data-centre MV converter. **The last column is empty for every vendor in the market.** Composed from megmeet-sst-briefing-data.json → classLedger.

9.3 Orders, evidence and the US footprint

For the SST specifically: nothing. No order, pilot, shipment, demonstration, tender or named customer anywhere, and the SST has never been exhibited — at the OCP Global Summit in 2025 Megmeet’s own investor record lists it as a roadmap item to match rather than a product shown. [\[WEB, verified 2026-09-23\]](#) No Megmeet paper or white paper on SSTs exists, and no Megmeet SST or MV-converter patent family was located; the closest filing is a high-voltage energy-harvesting winding structure of July 2026, which is not an SST stage. [\[WEB, verified 2026-09-23\]](#)

For the shipping business: real. GB300 volume orders won and contributing from the first quarter of 2026 via a US-headquartered ODM to a North American cloud provider, neither named; Vera Rubin samples under test at several North American clouds; an AIDC agreement with Pace Digitek announced 21 July 2026 with no value or scope disclosed. [\[WEB, verified 2026-09-23\]](#)

The Dallas lab is real, and it is a shelf lab. Megmeet announced a new data-centre power testing laboratory in Dallas, Texas on 26 June 2026 with 360 kW of active capacity, a roadmap to 1.5 MW, and 800 V DC support, testing to the 33 kW shelf and the 110 kW capacitor shelf; the address, square footage and headcount are undisclosed and **the SST is**

not mentioned. [WEB, verified 2026-09-23] The established US entity is Megmeet USA in San Jose, California, self-described as sales, engineering, logistics and technical support with no manufacturing claimed. [WEB, verified 2026-09-23]

ANALYSIS — TWO OBSERVATIONS THE RECORD SUPPORTS BUT DOES NOT STATE

At 360 kW on an 800 V bus, the Dallas laboratory as described **cannot energise a medium-voltage SST**. It is sized for shelves and sidecars, which is exactly what it says it tests. If the SST is to be proved on US soil, the question of *where* has not been answered publicly — and it is question 5 on the week-one list for that reason.

Megmeet appears on NVIDIA's power-system-components tier, and that blog post **does not discuss solid-state transformers at all** — the naming is for rack and sidecar components. [WEB, verified 2026-09-23] No OCP membership listing for Megmeet could be confirmed, though the OCP member directory also returned HTTP 403, so that is an unverified negative rather than a confirmed one. [WEB, verified 2026-09-23]

9.4 What the dossier now contradicts — v7 in the older prep documents, v8 in this briefing

The interview brief of 14 August 2026 was written from dossier **v2** and is five dossier versions stale; the study guide dates from 7 August. [DOSSIER megmeet v7] Neither file has been edited — they are dated documents. This is the correction list.

What the older document says	What the current record says	Kind
"The verdict lands soon — H1 report ~27 Aug 2026 ... the first hard test."	It landed on 27 August and it split : power products +60.92% to RMB 1.841bn at an expanding 25.06% margin, but ex-non-recurring net profit -16.13% and operating cash flow -84.62%. [DOSSIER megmeet v7]	superseded by event
"Power products +67% YoY in Q1 2026 to ¥817M" as the bull datapoint.	The current figure is the half-year rate: +60.92% to RMB 1.841bn. [DOSSIER megmeet v7]	stale quarter
"Consensus embeds a ~6× FY2026 profit rebound premised on AI power."	Post-interim broker consensus is RMB 832M attributable net profit; H1 delivered 252M, about 30% of it, so brokers now model a materially H2-weighted year. No published beat-or-miss verdict on the interim was located. [DOSSIER megmeet v7]	refined, with a caution added
Segment names: smart home appliance controls · power products · NEV & rail · industrial automation · intelligent equipment · precision connection.	Current naming is 电源产品 power products · 智慧生活 smart living · 新能源交通 new-energy transport · 工业自动化 industrial automation · 智能装备 intelligent equipment · 磁电精造 magnetics and precision. And explicitly: there is no separate industrial-power reporting segment — industrial power sits inside power products. [DOSSIER megmeet v7]	renamed and clarified
"~40% of revenue earned overseas."	Two different numbers with two different meanings: H1 2026 <i>direct</i> overseas revenue was RMB 2.180bn, 35.97% of the total and +45.49%; direct plus indirect is above 43%. [DOSSIER megmeet v7]	refined
"8,894 employees at end-2025 (~3,100 in R&D)."	8,894 is still the filed figure at 31 December 2025, but R&D is now about 3,400 engineers, 34.6% of headcount, of whom more than 1,000 are AIDC-power technical staff . [DOSSIER megmeet v7]	moved
"Lite-On — #2, ~30%"; "Delta ... ~70% of Blackwell-platform PSUs."	Both are labelled trade and brokerage estimates in the current record, and Delta's is stated as brokerage-estimated 60–70%. [REPORT 2026-09-08]	tier of the claim
"Megmeet's expected first hyperscaler volume delivery is domestic — a Chinese hyperscaler, in 2026 ... North America is genuinely greenfield."	The current strategy read states that volume delivery to North American majors began in Q1 2026 . [DOSSIER megmeet v7] Both statements are in the corpus and they are not obviously reconcilable.	direct tension — week-one question 6

What the older document says	What the current record says	Kind
The brief carries no market-cap or balance-sheet context.	About ¥24 in early 2024 → ¥190 intraday and about ¥94.7bn market cap on 3 June 2026 → three limit-downs in July 2026 → ¥108.71 on 29 July. Guarantee quota above ¥6bn, more than 50% of net assets, mostly to leveraged subsidiaries, flagged by sell-side as contingent risk. Controlling group stake 28.68% in 2020 → 23.20% in February 2026, mostly passive dilution. [DOSSIER megmeet v7]	new material
"The Huawei dossier still records this as a draft, China-specific rule ... and is stale." (the brief's own flag on the FCC action)	Discharged. The Huawei dossier now records the enacted FCC Covered List prohibition of 28 July 2026, the EU phase-out to April 2027 and the Lithuania remote-access bar. [DOSSIER huawei-digital-power v8]	fixed
The FCC two-prong read — bidirectional <i>and</i> wireless — "is inference from the definition, not a sourced ruling".	Still true, and now better grounded: the definition does have two elements and both must be met, and an August 2026 update removed pure AC-to-DC rectifiers and off-grid inverters. [WEB, verified 2026-09-23] The hedge stands.	still open — carry the hedge
The study guide: "SST — <i>the endgame product</i> ... >98.5% efficient ... this is what enables PUE below 1.1."	The SST is recorded as still in development , with no class disclosed anywhere and the PUE figure a company claim. [DOSSIER megmeet v7]	overclaim
The study guide: "~5% better end-to-end efficiency, and ~45% less copper" as a property of 800 V DC.	Those are NVIDIA's claims for the architecture against a legacy AC hall, and NVIDIA rates the SST and the MV rectifier <i>identically</i> . [PRIMER ch.4.3]	tier error
The study guide: "full 800 VDC facilities with 1 MW racks are projected from ~2027."	Option A is production Q3 2026, Option B deployment as soon as Q3 2027, Option C blocks <i>now</i> as TRUs, and MV-direct SST "toward 2029". [GUIDANCE nvidia-800vdc p10–23]	date conflation
The study guide and lesson plan have no entry for TRU, ISOP, CHB, DAB, MFT, BIL, SSCB, HRMG, IMD/RCM or the 4.8 MW block.	That vocabulary is the whole 2026–27 engineering conversation, and chapter 1 of this document exists to fill it.	gap

Dossier **v8** was cut on the afternoon of 23 September, after this briefing was finished. It re-read every Megmeet filing and investor-relations record from September 2024 to August 2026 and both company websites, and six of its findings contradict statements in this briefing itself; a seventh bounds a lead that Appendix D.2 set aside. [DOSSIER megmeet v8] The body of the document has not been changed to match — these rows are the correction list, and the superseding edition of the companion report, *sst-hall-edge-block-rev2--competitive--2026-09-23*, carries the same corrections.

What this briefing says	What dossier v8 records	Kind
Week-one question 6 and chapter 16.3, and the "direct tension" row above: "volume delivery to North American majors began in Q1 2026."	The interim dates the start of AIDC batch delivery to Q1 2026 across the whole customer chain — technology owners, system integrators and cloud operators — not to North America. The North America sentence is separate: batch delivery to 「部分北美大客户」 began in H1 2026, and the FY2025 annual report already recorded it by its 29 April publication date. [DOSSIER megmeet v8]	premise corrected — week-one question 6
The same row and chapter 16.3: "North America is greenfield" against the Q1 delivery — "not obviously reconcilable."	They reconcile once volume is separated from references. The company says it was late on GB200 and won limited orders there, then won GB300 batch orders contributing from Q1 2026, and names no customer, citing confidentiality. [DOSSIER megmeet v8] A Goldman Sachs note relayed by Sina says the first GB300 order ran through a US-headquartered ODM to a North American cloud provider, neither named. [DOSSIER megmeet v8] North America has volume but no named reference win — "greenfield" is right about references and wrong about volume.	tension resolved
The consensus row above: "Post-interim broker consensus is RMB 832M attributable net profit; H1 delivered 252M, about 30% of it."	On 23 September, 15 institutions over six months carried a FY2026 mean of RMB 787M , in a range of 515–937M; H1's RMB 252.1M is about 32% of it, and RMB 133.9M of that H1 figure was non-recurring. Guosen cut its own FY2026 forecast to RMB 652M on 9 September. [DOSSIER megmeet v8] The conclusion — a heavily second-half-weighted year — stands, and is sharper.	figure moved

What this briefing says	What dossier v8 records	Kind
Chapter 13, the first objection: “give the footprint, and give it precisely: manufacturing across China, Thailand, India and Germany.” The data file’s version of the same answer, which the companion shows, adds “the US”.	The filings name six manufacturing bases: five in China and one in Thailand. India is served through local contract manufacturers, and Germany is an R&D centre in Bielefeld, not a plant. [DOSSIER megmeet v8] The precise footprint is manufacturing in China and Thailand, contract manufacturing in India, R&D in Germany , and a US base that only the company’s website describes (next row).	overclaim — the objection’s own word “precisely”
Chapter 9.3 and chapter 13: the US entity is Megmeet USA in San Jose, “no manufacturing claimed”; the US plant confirmed in November 2024 has a location and status “the company has never published”.	The company’s own About pages place a 35,000 sq ft “U.S. manufacturing base” in the Fujitsu Industrial Park in Richardson, Texas, north of Dallas. [DOSSIER megmeet v8] A Texas licensing record shows a 39,200 sq ft Megmeet renovation at 2821 Telecom Parkway, Richardson, registered in May 2024. [DOSSIER megmeet v8] The HKEX application proof lists no US manufacturing base and describes the US subsidiary as sales and international operations, and nothing states whether the Dallas laboratory sits in the Richardson building. [DOSSIER megmeet v8] San Jose remains the address of Megmeet USA, Inc.	located, but not in the filings
Chapter 14 and week-one question 10: the displaced-LITEON story is “unsupported” and left open as a question of internal position.	Now closed rather than open. The story traces to two early-2025 pieces that attribute it to market rumour, and the company deflected the question when asked directly in December 2024. Soochow (April 2026) expects Megmeet to become NVIDIA’s third NVL72 PSU source, after Delta and LITEON, and the “~41% Delta” figure attached to the story appears in no source. [DOSSIER megmeet v8] The Delta and LITEON dossiers were corrected on the same day. [DOSSIER liteon v7]	closed — carry as settled
Appendix D.2: the 10 kV domestic / 35 kV international SST split “was left out of every table because the page could not be read.”	The page — ifeng, 1 July 2026 — has now been read. It attributes the split to the 20 May earnings call, and the exchange-filed record of that call calls the SST pre-research and contains no kV figure. [DOSSIER megmeet v8] It is carried in the dossier as a bounded lead and stays out of every table here, now for a better reason: the filed record does not support it.	bounded, still not carried

10 The SST competitor matrix — Megmeet against each

Chapter 6 gives the ledger by class. This chapter gives the head-to-head: what each competitor has that Megmeet does not, what Megmeet has that it does not, and the one sentence to use about each. The first three columns are sourced; *the one sentence* is scripted language and is **analysis throughout**.

Competitor	What they have that Megmeet does not	What Megmeet has that they do not	The one sentence
Heron Power 34.5 kV · 4.2 MW · pre-production	The most completely specified hall-edge SST in the public record, at the class US campuses take: thirty 140 kW cells with ten in series per phase — a third of the thirty-five the primer's 1.7 kV assumptions give for this class, which is what a higher-voltage device and a lighter derating buy, and a useful reminder that chapter 6.2's counts are a ratio argument rather than a bill of materials — tolerance of two cell failures per phase with sub-cycle bypass, UL-listed current-limiting fuses and a disconnect rated 40 kA at 34.5 kV, a 40-channel DC panel, a per-rack SSCB clearing under 5 μ s below 2 kA, and a battery on the 800 V bus. Plus the only startup place on NVIDIA's facility-tier roster. [DOSSIER heron-power v1]	Revenue, a filed financial record, a manufacturing base across five countries, named Western OEM customers of long standing, and a shipping rack business. Heron has shipped nothing and does not reach mass production until late 2027. [DOSSIER heron-power v1]	"They have the specification and we have the factory. Neither of us has a listed unit."
Amperesand 3.3 kV devices · 4–10 MW · no unit energised	A ten-year NTU validation lineage with 500 kW and 1.5 MW units tested on the Singapore grid, full-power overload and bidirectional testing of a later generation, eighteen patent families, and a 3.3 kV silicon-carbide front end that roughly halves the cell count of a 1.7 kV design. [DOSSIER amperesand v2]	Published datasheets. Amperesand publishes none, so its unit rating and input class rest on a supplier case study and trade-show reporting — and it publishes <i>no kV figure at all</i> . Also three chief executives in a year. [DOSSIER amperesand v2]	"Strong device choice, long lab record, and you cannot get a datasheet."
DG Matrix shipping LV · MV dated 2028+	The only venture SST vendor shipping a commercial product, an Infineon silicon-carbide supply agreement, a yearlong GPU-load-response test programme with NVIDIA, and ABB and Mitsubishi Heavy Industries on the cap table. [DOSSIER dg-matrix v1]	Nothing at the hall edge — but neither does DG Matrix. Its shipping unit is a 480 V AC skid at 96–97% peak on its own datasheet, and its own CFO deck dates the MV unit to "2028+". [DOSSIER dg-matrix v1]	"They ship a low-voltage skid and market a medium-voltage one. Check which product is in the room."
Novos Power pre-product	Nothing that can be evidenced. [DOSSIER novos-power v1]	Everything. Novos has 2–10 employees, no disclosed round, no Form D, no patent, no paper and no customer, and its claimed 48 kV input is above the 38 kV ceiling of the governing standard. [DOSSIER novos-power v1]	"Know the name; do not price it."
Sungrow (EnerNeo) 10–13.8 kV · 3 MW · supplying	The only SST any filing describes as launched and supplying the market, with framework agreements for 30 MW and 100 MW, 312 kW/m ² , a 99.999% availability claim and a stated roadmap of field trials in 2026, batch orders 2027 and scale 2028. [DOSSIER sungrow v9]	A class that can serve a US hyperscale campus — neither has one, but Sungrow's ceiling is <i>published</i> at 13.8 kV, while Megmeet's is unpublished. Also: asked directly about North American sampling and UL progress, Sungrow left both unanswered. [DOSSIER sungrow v9]	"The most advanced SST in the world tops out at 13.8 kV. That is the state of the market, not a knock on them."
Zhonhen (Panama) 10 kV TRU · supplying in China	A six-year head start on one-stage MV-to-DC conversion at scale, the largest HVDC field record anywhere, and NVIDIA naming the Panama Architecture as one of three TRU families in its own execution paper. [DOSSIER zhonhen v8]	A published US-facing rack business and named Western OEM customers. Zhonhen's exports are 4.1% of revenue, no US entity is on record, and it appears on none of NVIDIA's three rosters. [DOSSIER zhonhen v8]	"Panama is a transformer-rectifier lineup, not an SST — and NVIDIA named the architecture, not the company."

Competitor	What they have that Megmeet does not	What Megmeet has that they do not	The one sentence
Sinexcel samples, no formal sales	The only independent beat verdict among the Chinese interims this cycle, and a high-margin power-quality business with a genuine AIDC hook in its active voltage conditioner. [DOSSIER sinexcel v8]	A published 800 V DC product. The string “800V” does not appear in Sinexcel's 212-page interim, and management states four separate times that the AIDC products are samples with formal sales not yet realised. [DOSSIER sinexcel v8]	“They have said, four times, that it is not selling yet. Take them at their word.”
Delta Electronics 6.6–35 kV claimed · widest range	The deepest published 800 V DC catalogue in the industry, the widest disclosed MV range of any incumbent, and the only vendor claiming shipped SST units — “dozens in Asia, customer pilots underway in the US”, which is the vendor's own claim relayed by trade press. [WEB, verified 2026-09-23]	Nothing at the block. At the shelf, Megmeet competes directly and is growing faster off a smaller base. [DOSSIER megmeet v7]	“Never attack Delta. Sell the second source during a re-qualification the buyer has to do anyway.”
LITEON no SST, no MV product	The battery-backup franchise — BBUs move from optional on GB200 to standard on GB300 — plus USD 919M committed to two McKinney, Texas plants and a target of 70% of capacity outside China by 2027. [DOSSIER liteon v6]	An SST in the published chain at all, and a first-mover position on the mainland sidecar. [DOSSIER megmeet v7]	“US capacity is announced, not producing — and they have no MV product.”
ABB 4.16–34.5 kV MV UPS · orderable	The only orderable solid-state medium-voltage product anywhere: 2.5 MW per unit to 25 MW paralleled at 98%, UL 1741 and the first of its kind to achieve UL 9540, with a 34.5 kV direct-grid-connect class and an installed base at Applied Digital's North Dakota campuses. Plus the first IEC-certified solid-state circuit breaker. [DOSSIER abb v7]	An 800 V DC output. HiPerGuard's output is medium-voltage AC — it is a UPS, and ABB sold its grid business to Hitachi before the supercycle, so it cannot supply the transformers its own campuses wait on. [DOSSIER abb v7]	“The most credible MV solid-state product in the market is a UPS, not a transformer.”
GE Vernova 13.8 kV reference designs	Three jointly developed NVIDIA AI-factory reference designs, an MV-UPS and rectifier catalogue, Prolec GE wholly owned, and the single most important commercial datapoint in the tier — a hyperscaler cost-share carrying a commitment to buy 1,000 SSTs from 2027 if a specification is met. [DOSSIER ge-vernova v6]	A published product. GE Vernova has no dated SST product, no class and no rating on the record, and its stated first hyperscaler delivery of autumn 2026 has not been reported. [WEB, verified 2026-09-23]	“The only disclosed hyperscaler SST purchase commitment in the market belongs to a company with no dated product.”
Eaton owns SST IP	The best-hedged incumbent: it bought Resilient Power outright, published an 800 V DC reference architecture at OCP, owns the prefabricated-enclosure chokepoint and the thermal business, and states on its own earnings call that it is piloting nearly ten SST projects including with hyperscalers, with orders in H2 2026 and shipments from late 2027. [WEB, verified 2026-09-23]	A published specification. Eaton's MVSST rating and class are trade-press only — its own catalogue page and brochure returned HTTP 503 to the research pass, so no vendor spec sheet was obtained. [WEB, verified 2026-09-23]	“If SSTs win, Eaton owns the technology; if not, it owns the skids.”
Schneider Electric no MV SST programme found	The reference designs, the digital twin, the cooling and a 1.2 MW-per-rack 800 V DC sidecar, plus a \$373M capacity-reservation agreement with Digital Realty. [DOSSIER schneider-electric v9]	Nothing at the converter — the sidecar was still a prototype at GTC 2026 with no commercial date, and no Schneider MV SST programme was found. [DOSSIER schneider-electric v9]	“Systems leader, defending the AC facility architecture rather than attacking it.”
Vertiv 34.5 kV stated as a goal	The demand-capture leader by backlog and the highest data-centre revenue concentration among large caps, with a full 800 V DC portfolio releasing in H2 2026 for a 2027 ramp. [DOSSIER vertiv v9]	An SST. Vertiv's own director of engineering states 34.5 kV to 800 V DC at 3–5 MW as a goal, and the shipping sidecar at 400–900 kW is not an SST. [WEB, verified 2026-09-23]	“The incumbent most architecturally exposed if conversion moves upstream — and they have said so in public, in the form of a goal.”
Hitachi Energy no SST product named	The scarcest input in the industry: MV and HV transformer capacity, roughly half of all HVDC projects ever built, and \$1bn committed to US manufacturing including a \$457M Virginia plant. [DOSSIER hitachi-energy v5]	An 800 V DC converter product of any kind. Hitachi Energy has none announced — and is separately quoted urging caution on compact 34.5 kV designs. [WEB, verified 2026-09-23]	“Moat and motive at once: their own transformer waits of 30–40 months are what created this market.”

Competitor	What they have that Megmeet does not	What Megmeet has that they do not	The one sentence
Siemens + Reinhausen up to 36 kV · undated	A credible MV-magnetics partner and a class that covers 34.5 kV. [WEB, verified 2026-09-23]	Anything dated. No MW rating, no efficiency, no date and no NVIDIA mention — a follower posture, announced in August 2026. [WEB, verified 2026-09-23]	"Announced in August 2026 with no rating and no date."

11 The six business units, and the position in each

Megmeet is not an AI company and not an appliance company — it is a power-conversion platform pointed at six industries, and the six do not behave alike. The table below ranks them by how defensible the position is on the evidence rather than by revenue, because the biggest unit is the weakest one.

Business group	Share	GM	The evidence	Who it is up against, and what decides the sale
Intelligent equipment — welding power sources 智能装备 · <i>Strongest in the company</i>	4.6%	39.67%	Megmeet claims the top share of China's robotic gas-shielded arc-welding power-source market for ten consecutive years since 2014 and about 30% share in 2020-21 — company claims on the company's own site, not an audited ranking. [WEB, verified 2026-09-23] What is checkable is commercial: repeat framework deals with Honglu for 1,500 and then 1,800 robotic welding machines, integration with FANUC, ABB and KUKA, exports to 60+ countries. Highest gross margin in the company at 39.67% and rising. [DOSSIER megmeet v7]	Jasic, Riland, Shanghai Hugong domestically; Panasonic and Lincoln Electric among global brands present in Chinese rankings
Power products — AI data centre and network power 电源产品 · <i>Real, growing, and unranked</i>	30.4%	25.06%	The only mainland vendor NVIDIA named for GB200 NVL72 power and on both of the rosters NVIDIA published before it stopped naming vendors — the October 2025 ecosystem post and the August 2026 component tier; the company's own H1 2026 wording is "one of NVIDIA's designated recommended data-centre power suppliers" — membership of a set, not a rank. +60.92%, and the only one of the three largest segments whose gross margin expanded. No AI-data-centre revenue line is disclosed at any granularity. [DOSSIER megmeet v7]	Delta (the benchmark), LITEON (the second source), Flex, Bizlink, Lead Wealth at the shelf; Sungrow, Zhonhen, Sinexcel and the incumbents at the block above
Industrial automation 工业自动化 · <i>Credible, small, well behind the leader</i>	7.4%	27.78%	+39.85% in FY2025, the second-highest margin pool in the company at 27.78%. The company's own scope statement leads with servo and VFD drives, not PLC and HMI. The market is genuinely unconcentrated — across ten listed Chinese players, CR3 is reported around 9% and CR10 around 12%. [WEB, verified 2026-09-23]	Inovance is the reference point and is far larger with a much broader portfolio; INVT also named. No source places Megmeet on a ranked share table against Siemens, Schneider, Mitsubishi Electric, Yaskawa or Estun
New-energy transportation (NEV and rail) 新能源交通 · <i>Volume land-grab, margin-dilutive</i>	9.3%	13.46%	Revenue doubled in FY2025 (+108.65%) at the lowest gross margin of the major segments; 13.46% in H1 2026. More than a million combined OBC plus DC-DC units shipped in 2025. [DOSSIER megmeet v7] Named OEMs: Geely, BAIC New Energy, Leapmotor, Changan, Dongfeng, FAW, Seres. [WEB, verified 2026-09-23]	VMAX, Enpower, Shinry, Fute lead the published OBC rankings — Megmeet appears on none of them
Smart living (appliance controls, smart bathroom) 智慧生活产品 · <i>Biggest unit, weakest position, deteriorating</i>	42.2%	18.39%	42.19% of revenue at 18.39% gross margin, down about 4.2 points in a year. It was the only segment that shrank in FY2025 (-4.8%). Named customers: Xiaomi, Haier, Hisense, Changhong, Skyworth. The smart-bathroom line is genuinely distinctive rather than a controller commodity. [WEB, verified 2026-09-23]	Topband at RMB 11.08bn of FY2025 revenue and H&T at roughly RMB 9.7bn are each about three times Megmeet's whole appliance unit, and are the acknowledged duopoly. Megmeet is third tier

Business group	Share	GM	The evidence	Who it is up against, and what decides the sale
Magnetics and precision manufacturing 磁电精造 · Vertical integration rather than a merchant business	5.0%	8.79%	8.79% gross margin, up from 5.1% but still the lowest in the company. The company describes it as an investment-and-integration period. [DOSSIER megmeet v7] Shaped magnet wire, coaxial cable, ultra-fine flat wire, FFC/FPC feeding internal magnetics supply plus external sales. [WEB, verified 2026-09-23]	No ranked competitor set located

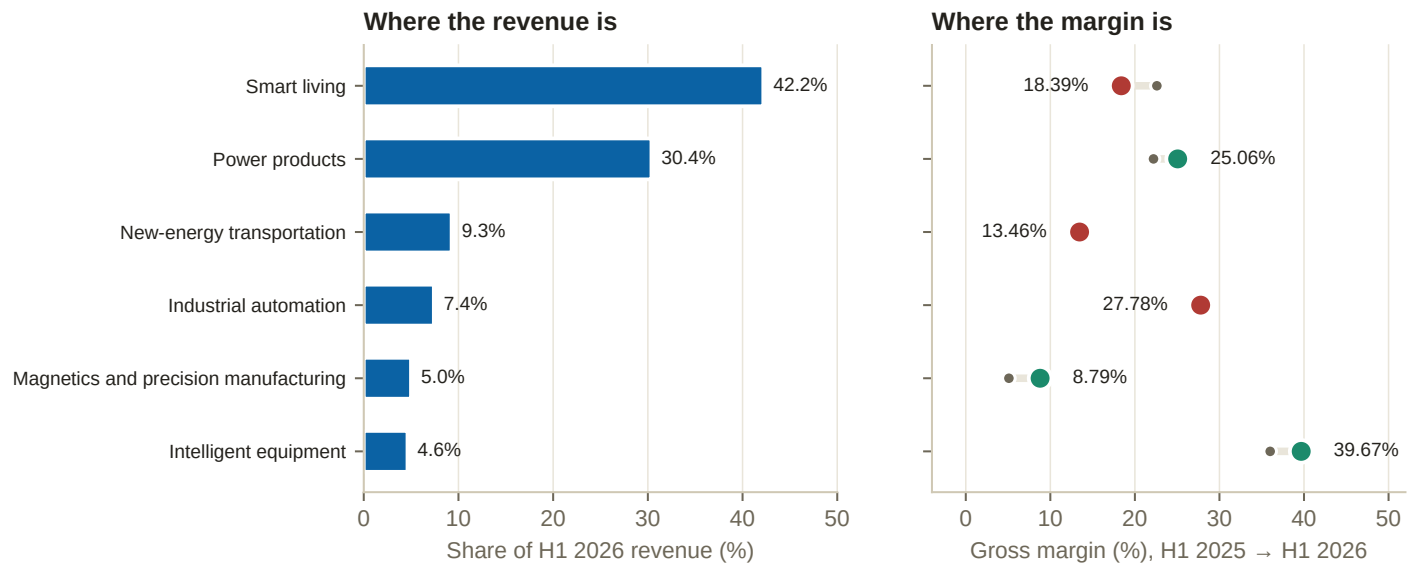


Figure M13 — Where the revenue is, and where the margin is. Boundary: H1 2026 interim, six months to 30 June 2026, RMB; the six named groups are plotted and a 1.14% “other” line completes the reported RMB 6.061bn total, which is why this restatement is the one used. The margin panel shows H1 2025 → H1 2026 as a movement, not a level. **Two panels rather than a dual axis**, which is never correct. Note the unit: 亿元 is one hundred million RMB — FY2025 group revenue is RMB 9.403bn, about USD 1.3bn, *not* USD 13bn. Composed from megmeet - sst - briefing-data.json → megmeetMix.

The six business groups, ranked by how defensible the position is

Intelligent equipment — welding power sources

4.6% of revenue · 39.67% gross margin · Strongest in the company
Megmeet claims the top share of China's robotic gas-shielded arc-welding power-source market for ten consecutive years since 2014 and about 30% share in 2020-21 — company claims on the company's own site, not an audited ranking. What is checkable is commercial: repeat framework deals with Honglu for 1,500 and then 1,800 robotic welding machines, integration with FANUC, ABB and KUKA, exports to 60+ countries. Highest gross margin in the company at 39.67% and rising.
[DOSSIER megmeet v7] [WEB, verified 2026-09-23]

Power products — AI data centre and network power

30.4% of revenue · 25.06% gross margin · Real, growing, and unranked
The only mainland vendor NVIDIA named for GB200 NVL72 power and on both published rosters; the company's own H1 2026 wording is "one of NVIDIA's designated recommended data-centre power suppliers" — membership of a set, not a rank. +60.92% at the only expanding gross margin among the three largest segments. No AI-data-centre revenue line is disclosed at any granularity.
[DOSSIER megmeet v7]

Industrial automation

7.4% of revenue · 27.78% gross margin · Credible, small, well behind the leader
+39.85% in FY2025, the second-highest margin pool in the company at 27.78%. The company's own scope statement leads with servo and VFD drives, not PLC and HMI. The market is genuinely unconcentrated — across ten listed Chinese players, CR3 is reported around 9% and CR10 around 12%.
[WEB, verified 2026-09-23]

New-energy transportation (NEV and rail)

9.3% of revenue · 13.46% gross margin · Volume land-grab, margin-dilutive
Revenue doubled in FY2025 (+108.65%) at the lowest gross margin of the major segments; 13.46% in H1 2026. More than a million combined OBC plus DC-DC units shipped in 2025. Named OEMs: Geely, BAIC New Energy, Leapmotor, Changan, Dongfeng, FAW, Seres.
[DOSSIER megmeet v7] [WEB, verified 2026-09-23]

Smart living (appliance controls, smart bathroom)

42.2% of revenue · 18.39% gross margin · Biggest unit, weakest position, deteriorating
42.19% of revenue at 18.39% gross margin, down about 4.2 points in a year. It was the only segment that shrank in FY2025 (-4.8%). Named customers: Xiaomi, Haier, Hisense, Changhong, Skyworth. The smart-bathroom line is genuinely distinctive rather than a controller commodity.
[WEB, verified 2026-09-23]

Magnetics and precision manufacturing

5.0% of revenue · 8.79% gross margin · Vertical integration rather than a merchant business
8.79% gross margin, up from 5.1% but still the lowest in the company. The company describes it as an investment-and-integration period. Shaped magnet wire, coaxial cable, ultra-fine flat wire, FFC/FPC feeding internal magnetics supply plus external sales.
[DOSSIER megmeet v7] [WEB, verified 2026-09-23]

Figure M14 — The six business groups, ranked by how defensible the position is. Boundary: the ranking is **analysis**; the revenue, share, margin and evidence beside it are sourced and tagged in the data file. Consultancy market-share figures are excluded entirely — two consultancies give the same appliance-controls competitor 13% and 77% of the same market, and neither is quotable. Composed from megmeet-sst-briefing-data.json → buLadder.

ANALYSIS — THREE THINGS THIS TABLE SAYS THAT NOBODY SAYS OUT LOUD

The company's single strongest competitive position is in welding, not in data centres. Intelligent equipment carries the highest gross margin in the company at 39.67%, and the welding line inside it is the only unit with a defensible leadership claim — supported not by a share number but by repeat framework deals for 1,500 and then 1,800 robotic welding machines and by certified integration with FANUC, ABB and KUKA. It is 4.6% of revenue.

The price war is in the biggest unit. Smart living is 42.19% of revenue at an 18.39% gross margin that fell about 4.2 points in a year, against two rivals each roughly three times its size. That is a share-defence position, and it is the wallet funding the AI bet.

The profit pools are not the big units. Intelligent equipment and industrial automation together carry the two best margins in the company and together are 12% of revenue. Any story about Megmeet that starts with the appliance business misreads where the company actually makes money — and any story that starts with AI data centres misreads where it currently earns it.

12 Policy and supply chain

Four regimes bear on a China-headquartered vendor selling medium-voltage power conversion into US AI infrastructure, and they do not compose cleanly. Getting them right is worth more in a procurement conversation than any technical answer, because procurement is where this product line is most likely to be stopped.

12.1 The FCC Covered List — the binding one

On 28 July 2026 the FCC's Public Safety and Homeland Security Bureau added foreign-produced power inverters to the Covered List, implementing national-security determinations issued the previous day. [WEB, verified 2026-09-23] The definition has **two elements and a device must meet both**: it must be “a bidirectional power device or system that converts direct-current electricity to alternating-current electricity, or alternating-current electricity to direct-current electricity”, and it must contain components “enabling remote communication, control, sensing, data collection, or monitoring”.

[WEB, verified 2026-09-23] **There is no voltage or power threshold.** A telemetered MV solid-state transformer is a literal instance of both elements.

Three things are commonly got wrong about it. **It is not a China rule.** “Foreign-produced” is defined by failure of the Buy American standard at 48 CFR §25.101(a) — generally US manufacture with domestic component costs above 65% for 2024–2028 — so it is a content test, not a nationality test, and a US-headquartered vendor building offshore is captured while a China-headquartered vendor building in the US in principle is not. [WEB, verified 2026-09-23] **It is prospective.** Models authorised before 28 July may continue to be imported, marketed, sold and used. [WEB, verified 2026-09-23] **There is a route out**, through conditional approval by the Department of War or DHS, with applications due by 1 January 2028 requiring ownership disclosure to 5% beneficial owners, a component-level bill of materials with country of origin, identification of who is responsible for software updates, and a time-bound US manufacturing plan with committed capital.

[WEB, verified 2026-09-23]

And one thing has changed since the primer: an August 2026 update **removed pure AC-to-DC rectifiers and off-grid inverters** from the list and added a 45X safe harbour — so a unidirectional rectifier-only architecture may sit outside the rule where a bidirectional SST does not. [WEB, verified 2026-09-23]

12.2 Executive Order 14421 — and why it probably misses the hall edge

Executive Order 14421, “Declaring a National Emergency to Secure the United States Bulk-Power System”, was signed on 26 August 2026 under the International Emergency Economic Powers Act. It covers transformers, inverters, circuit breakers, generation turbines, reactors and battery storage, plus associated components, software, firmware, digital services, maintenance services and remote-access capability linked to a covered foreign entity. [WEB, verified 2026-09-23] Crucially for this product line, **it applies to transmission facilities rated 69 kV and above and local distribution is excluded.** DOE must issue implementing regulations within 120 days — by 24 December 2026 — and FAR revision recommendations within 180 days. [WEB, verified 2026-09-23]

ANALYSIS — THE ASYMMETRY TO CARRY INTO A PROCUREMENT CONVERSATION

A 34.5 kV behind-the-meter hall-edge converter is probably *outside* EO 14421, because it sits on local distribution below 69 kV, and *inside* the FCC Covered List, which has no voltage threshold at all. The two regimes were written for different problems and do not line up. Saying this accurately, and saying which one you think binds and why, is a stronger position than either claiming exemption or conceding exposure — and it is a question for counsel and the certification body, which is a strength to say out loud rather than a dodge.

12.3 Tariffs, the EU, and what Megmeet has actually said

A Section 232 tariff of 50% on semi-finished copper and copper-intensive derivatives took effect on 1 August 2025 — copper windings being the second-largest bill-of-materials line in both a line-frequency transformer and an SST.

[PRIMER ch.6.2] Section 301 rates on certain Chinese imports rose from 1 January 2026, and a January 2026 Section 232 proclamation imposed an immediate 25% tariff on certain advanced semiconductor articles; **no transformer-specific or power-electronics-specific line item was located**, with corporate filings describing the exposure generically.

[WEB, verified 2026-09-23] In the European Union, the Commission moved in April–May 2026 to restrict EU funding for projects using inverters from high-risk countries, phasing in from 1 November 2026 with new contracts fully incorporating the restrictions from April 2027. [WEB, verified 2026-09-23]

Megmeet's own language on tariffs is direct: activating Thailand factory capacity can, to a large extent, avoid the US tariff problem. [WEB, verified 2026-09-23] It also states that US sanctions and export-control regimes may involve certain of its customers and suppliers. [WEB, verified 2026-09-23] No FEOC or Section 889 disclosure appears anywhere in its filings.

[WEB, verified 2026-09-23] And on listings: no UL or ETL listing number is disclosed for any Megmeet data-centre product — the company claims UL, TÜV and CNAS *laboratory accreditations*, which are an in-house testing credential and not a listed product. [WEB, verified 2026-09-23]

THE REFRAMING NUMBER

Megmeet stated in April 2025 that direct exports to the United States were **under 3% of total revenue**.

[WEB, verified 2026-09-23] The North American data-centre push is therefore a new build-out, not the scaling of an existing US business — which is the honest way to describe the territory to anyone who asks, and a more defensible position than implying continuity that does not exist.

PART V

The sales layer

Analysis throughout. Nothing in this part is a Megmeet position, a company statement, or anything other than one reading of the evidence in Parts I to IV.

SCOPE NOTE FOR THE WHOLE OF PART V

Every chapter in this part is **analysis**. The objections are constructed from the evidence in Parts I–IV and from the objection set in the August 2026 interview brief, whose format this follows; the answers are one reading of that evidence, not guidance from anyone at Megmeet. Where an answer introduces a fact that appears nowhere else in this document, it carries its tier tag here. Where it restates a fact established in Parts I–IV — which is most of them — the tag is carried at the place the fact is established, and the answer does not repeat it, because a sales script bristling with tags is not a sales script. Nothing in this part is evidence that is not already in the four parts above it.

13 The objections, and the strongest honest answer to each

Seven objections a US data-centre buyer raises about a China-headquartered SST vendor. The strongest move with every one of them is to raise it yourself, before they do.

What they say	What they are really asking	The strongest honest answer	What never to say
"You are a Chinese supplier — will procurement or legal block this after I have spent a qualification cycle?"	Am I going to waste six months?	Answer with the actual regime, not with reassurance. The binding US rule for this product class is the FCC Covered List of 28 July 2026, and it has two prongs that must both be met: a bidirectional AC-DC converter, and components enabling remote communication. It is a content test, not a nationality test — "foreign-produced" means failing the Buy American standard at 48 CFR 25.101(a), so a US-headquartered vendor building offshore is captured too. Then give the footprint, and give it precisely: manufacturing across China, Thailand, India and Germany; a Dallas-area test laboratory on US soil; and a US plant confirmed in November 2024 whose location and status the company has never published — chapter 9.3 shows the US entity on record is a sales and engineering office that claims no manufacturing. [WEB, verified 2026-09-23] Then the proof that actually lands: Ericsson, Cisco, Juniper, Arista and Accton already buy Megmeet power and have already run their supplier audits. [DOSSIER megmeet v7]	<i>Never promise a regulatory outcome, and never say the rule does not apply. Say which prong you think you fall outside of and why, and say it is a question for counsel and the certification body.</i>
"What is your input voltage class?"	Can you even serve my campus?	This is the one you cannot talk your way around, and pretending otherwise is the fastest way to lose an engineer. The published record carries no class for Megmeet's SST. The honest answer on day one is: the sidecar and the rack products are specified and shipping, the converter is under development and the class is not published yet — and then ask what class their campus takes, because that is the answer you need to bring back internally.	<i>Never infer a class from Chinese grid norms. A third-party report circulating a 10 kV domestic / 35 kV international split is exactly that inference, it is unread behind a 403, and 35 kV is not a US class.</i>
"We have never deployed you at this scale."	Am I the guinea pig?	Concede the specific and redirect to the general: new to hyperscale MV conversion, not new to high-volume power manufacturing. Then make the structural point — standardised form factors let a buyer take a limited, low-risk slot first. At the rack that is literally true: a CRPS module fits any compliant chassis.	<i>Never claim an installed base at the hall edge. Nobody in the industry has one — no covered vendor has an energised, named-site, hyperscaler MV SST, and saying so is more credible than implying otherwise.</i>

What they say	What they are really asking	The strongest honest answer	What never to say
"Delta already has this socket."	Why re-open a qualification that works?	Never attack Delta — they are good and the buyer knows it. Sell the second source: the concentration is a risk they already manage elsewhere in the bill of materials. Then sell timing: the architecture change forces re-qualification anyway, so the switching cost you are asking them to absorb is one they are already absorbing.	<i>Never use a share number. Every server-power share figure in circulation is a trade or brokerage estimate, and the buyer will know which ones are contested.</i>
"Your financials look shaky."	Will you still be here mid-programme?	Do not flinch — it is public and they have read it. Profit fell because the company chose to spend ahead of revenue; H1 2026 power products grew 60.92% and was the only one of the three largest segments whose gross margin expanded; the diversified base across six business groups is why the AI bet does not have to work on the first try. Then give them the number that is genuinely uncomfortable and let them see you know it: ex-non-recurring profit fell 16.13% in the same half, and operating cash flow fell 84.62%.	<i>Never quote the headline +45.21% without the ex-non-recurring -16.13% beside it. An analyst will have both, and offering only the first one costs more credibility than the second one costs goodwill.</i>
"What if 800 V DC slips?"	Am I buying a roadmap?	Point at the bridge. The sidecar exists because buildings on 380-480 VAC need megawatt racks now, and the near-term catalogue sells into today's architecture regardless. NVIDIA's own staging has the power rack in production in H2 2026 and the row power center in 2027 — neither needs an SST. The roadmap is upside, not the thesis.	<i>Never cite the 2029 SST date as if it were a commitment to buy one. It is a date attached to a next-generation architecture in a paper that names no vendors.</i>
"Who else has one running?"	Show me operating hours.	The honest answer is that at megawatt scale on a US feeder there is essentially one: a 1 MW research unit from NC State with the New York Power Authority and EPRI, validated on a real distribution feeder, announced August 2026. Everything else is a pilot, a framework agreement or a claim. Say that, and then say what your own first article would look like and who would witness it.	<i>Never let a framework purchase agreement be described as a deployment. Sungrow's 30 MW and 100 MW frameworks are the largest committed volumes anywhere and no unit has been evidenced energised.</i>

14 What not to say in the first month

Ten things that sound knowledgeable and are wrong. Each one is a sentence someone who has read one article says, and each is correctable by someone who has read two.

Do not say	Because
"An SST is about 5% more efficient."	That is NVIDIA's claim for the 800 V DC architecture against a legacy AC hall it puts at under 90% — and NVIDIA rates the SST and the MV rectifier identically at 98.5% or better. The SST's case is stage elimination, footprint and control. [PRIMER ch.1, ch.4.3]
"The bus is plus-minus 400 volts."	NVIDIA specified single-ended 800 V — positive, return, protective earth — and rejected bipolar for lack of three-pole DC breakers. [PRIMER ch.2.2] Plus-minus 400 V is the OCP variant the rack can also accept. [WEB, verified 2026-09-23] The published architectures genuinely diverge, and a vendor without a side fails the first five minutes.
"Zhonhen's Panama / ABB's HiPerGuard / Vertiv's PowerDirect is a solid-state transformer."	They are a transformer-rectifier lineup, a solid-state medium-voltage UPS and a rack DC shelf respectively. [PRIMER ch.1] NVIDIA's own paper names the Panama Architecture as a TRU implementation — as an architecture, not as a company. [GUIDANCE nvidia-800vdc p22]
"Abilene is served at 34.5 kV."	Abilene's Oncor interconnection is 345 kV transmission and the campus distribution class is undisclosed. One decimal point turns a transmission fact into a fabricated service-class finding, and it is the single most likely error to introduce. [WEB, verified 2026-09-23]
"Oracle is designing for 800 V DC."	That is NVIDIA's characterisation of OCI, not an Oracle statement. Oracle has published nothing on 800 V DC, DC distribution, MV conversion or solid-state transformers. Quoting it back to an OCI engineer as an Oracle position would be a mistake. [WEB, verified 2026-09-23]
"We are NVIDIA's named SST partner." (or anyone is)	NVIDIA's ecosystem page names no vendor as an SST vendor and does not break the roster into rectifier or SST sub-categories. [WEB, verified 2026-09-23] Megmeet's own filing says "one of NVIDIA's designated recommended data-centre power suppliers" — membership of a set. [DOSSIER megmeet v7]
"Megmeet displaced LITEON as the number-two NVIDIA power-shelf source."	No supporting source has been located. The company has never claimed it and the H1 interim names no competitor across 238 pages. [DOSSIER megmeet v7] The one contemporaneous supplier account names LITEON second, and two research houses covering the same market in mid-2026 name Delta and LITEON without mentioning Megmeet. [REPORT 2026-09-08]
"Megmeet's revenue is about thirteen billion dollars."	FY2025 revenue is RMB 9.403bn, about USD 1.3bn. Chinese filings quote 亿元 — hundreds of millions — and machine translation renders it as "billion" with depressing regularity. Getting a company's own revenue wrong by 10x in the first month is not recoverable. [DOSSIER megmeet v7]
"The SST relieves the transformer shortage."	Partially at best. It removes the 480 V layer, but not the utility service transformer where the utility owns it, nor the MV switchgear, cable or substation — and it trades grain-oriented-electrical-steel iron for medium-frequency cores and hundreds of SiC modules, on a supply base a fraction of the transformer industry's scale. [PRIMER ch.6.2]
"It is UL listed." (of anything at the 800 V DC bus)	As of the most recent analyst survey no vendor had completed UL certification for data-centre SST deployment, and the most quotable line from PCIM Europe 2026 is that there is no specification and no UL certification for the 800 VDC bus, and that this is a major gap. [WEB, verified 2026-09-23]

15 The ten week-one questions, ranked

Ranking criterion: Ranked by decision leverage: how many of the other answers in this brief would have to be rewritten if the answer came back other than expected. Not by how interesting the question is. They are ordered so that if you only get three answered in the first week, these are the three.

#	The question	What the answer changes
1	What service-voltage class is the SST being designed to, and is it one product or two?	Everything. It decides whether Megmeet is in the hall-edge contest at all, which US campuses are addressable, and whether the cell count is a 15 kV-class or a 35 kV-class bill of materials — a factor of 2.3× to 2.7× in cells, magnetics, gate drivers and control channels, depending on which 15 kV-class voltage the comparison starts from.
2	Is there a target date for a first article, and is there a witness — a customer, a lab, or an NRTL?	It separates a product programme from a roadmap slide. Nobody in the industry has a listed unit; the first vendor with a witnessed first article at a named class takes the layer's narrative.

#	The question	What the answer changes
3	Which listing path are we on — UL 1741 with supplements, UL 347A, UL 2877, IEC 62477-2 — and who is the NRTL?	Certification is 6–18 months and mid-six figures per product on the one published estimate located — a certification body writing about the market it sells into, so read the range as its floor rather than its centre [WEB, verified 2026-09-23] — and each variant recertifies. It sets the earliest possible US revenue date more firmly than any engineering milestone.
4	Where does the FCC Covered List analysis land for the sidecar, the shelves and the SST — and who has written it down?	The rule has no voltage threshold and two prongs. A unidirectional rectifier with wired management plausibly sits outside; a bidirectional, telemetered SST plausibly sits inside. If nobody has written the analysis, that is the first thing to cause to happen.
5	What is the Dallas lab actually for, and what is its ceiling?	It is 360 kW today with a 1.5 MW roadmap — which is a shelf and sidecar validation facility, not somewhere an MV converter can be energised. If the SST is to be proved in the US, the question is where.
6	What did volume delivery to North American majors in Q1 2026 actually consist of, and who were they?	The dossier records it; the older prep documents say North America is greenfield with no reference win. Both cannot be the shape of the territory. The answer decides whether the job is building a first proof case or scaling an existing book.
7	What is the overload specification, the part-load efficiency curve, and the MTBF or FIT number for the converter?	Nobody in the industry publishes any of the three. Whoever publishes first owns the engineering conversation — and if we cannot answer internally, we cannot answer an NVIDIA or OCI engineer either.
8	What is the service model — vendor field engineers or the owner's electricians — and what is the spares strategy and the warranty term?	Only one vendor in the entire research pass publishes a warranty at all. For a first-of-a-kind MV asset the commercial terms are the product, and an owner's engineer will ask before an electrical engineer does.
9	Are we in the OCP LVDC SST specification work, and do we have v0.3?	It is the vocabulary every counterparty will use and more than 80 manufacturers are reported to be building to it. It could not be retrieved in this run because opencompute.org refused every request — so the first practical task is to get the document.
10	What is the internal position on the displaced-LITEON story, and has anyone ever been asked to repeat it?	Least leverage on the market, most on personal credibility. The company has never claimed it and the record points the other way; knowing that the internal answer matches the external one is worth confirming before the first customer meeting.

16 Everything this document could not determine

Named, not smoothed over. Each of these is a real hole, and several of them are holes in the market rather than in the research — which is a different and more useful kind of finding.

16.1 Holes in the record that no amount of research can close

- **Megmeet's SST service-voltage class.** Zero kV mentions across thirty-eight pinned dossier sources and zero hits in a four-filing text scan of the company's own disclosures. This is not a research failure; it is the exact limit of Megmeet's own disclosure. [\[WEB, verified 2026-09-23\]](#)
- **Any audited AI-data-centre power-conversion revenue line, from anyone.** No company in the covered set discloses one, which is why every market-share figure in circulation for this layer is a trade or brokerage estimate. [\[REPORT 2026-09-08\]](#)
- **Any public apples-to-apples \$/MW comparison of an SST against a transformer-plus-rectifier for a US data centre.** None exists; the fragments that circulate are vendor models with vendor-chosen boundaries, none audited on a delivered project. [\[PRIMER ch.6.6\]](#)
- **An MTBF or FIT figure for a complete MV converter, an overload curve, or a part-load efficiency curve.** No vendor publishes any of the three. [\[WEB, verified 2026-09-23\]](#)
- **A warranty term from any hall-edge vendor except one.** Enphase's 10-year limited warranty is the only one located across the entire pass. [\[WEB, verified 2026-09-23\]](#)
- **Any spares list, stocking level or spare-cell price for an MV SST.** None published by anyone. [\[WEB, verified 2026-09-23\]](#)
- **Insurance or bankability treatment of a novel MV asset.** No insurer, reinsurer, broker or owner's-engineer standard naming solid-state transformers was found. [\[WEB, verified 2026-09-23\]](#)
- **Whether NERC CIP attaches to a behind-the-meter hall-edge SST.** Addressed by no source located. [\[WEB, verified 2026-09-23\]](#)
- **Any US authority-having-jurisdiction precedent for MV switching equipment inside an operating data hall.** None located — probably the largest permitting unknown for facility-level SSTs. [\[PRIMER ch.6.4\]](#)
- **Any field-failure, forced-outage or availability data from an operating MV data-centre SST.** None exists to find, because there is no installed base. [\[WEB, verified 2026-09-23\]](#)

16.2 Specific questions asked and not answered

- **Amperesand's voltage class.** No kV figure anywhere on its own site — “configurable MV AC and LV DC” only. A real hole, since it is one of the four pure-plays. [\[DOSSIER amperesand v2\]](#)
- **GE Vernova's class, rating, and the identity of its hyperscaler.** Only the autumn-2026 delivery date is public, and that delivery has not been reported. [\[WEB, verified 2026-09-23\]](#)
- **The feeder voltage of the NC State / NYPA / EPRI 1 MW live-feeder test.** Not stated in any coverage located — for the only independently verified megawatt-class live-feeder SST test in existence. [\[WEB, verified 2026-09-23\]](#)
- **The MV service class at New Albany, Mt Pleasant, Richland Parish, Memphis or the Abilene campus itself.** Nothing disclosed for any of them. [\[WEB, verified 2026-09-23\]](#)
- **Whether any SST has been energised at a named hyperscaler site.** None found. Every hyperscaler-adjacent datapoint is intent, ecosystem membership, or an unnamed counterparty. [\[WEB, verified 2026-09-23\]](#)

- **Eaton’s MVSST specification.** Its own catalogue page and brochure returned HTTP 503 to the research pass, so its class and rating remain trade-press only. [WEB, verified 2026-09-23]
- **Whether Megmeet is in OCP membership or the LVDC SST specification work.** Not found, but the OCP member directory also returned HTTP 403, so this is an unverified negative rather than a confirmed one. [WEB, verified 2026-09-23]
- **Megmeet’s FY2025 gross margin by segment beyond the appliance line, its absolute R&D spend for FY2025 and H1 2026, and any named US customer for any product line.** None located. [WEB, verified 2026-09-23]
- **Whether the US plant Megmeet confirmed in November 2024 is the Dallas facility.** The company confirmed a US factory and never named the location, capacity or timeline; the Dallas laboratory is separately and firmly evidenced by the company’s own June 2026 release, but the two are not confirmed to be the same thing. [WEB, verified 2026-09-23]

16.3 Contradictions flagged and deliberately not resolved

- **“North America is greenfield” against “volume delivery to North American majors began in Q1 2026”.** Both statements are in the corpus, from the August interview brief and the September dossier respectively, and they are not obviously reconcilable. This is week-one question 6. [DOSSIER megmeet v7]
- **DG Matrix’s own 12–34.5 kV banner against an analyst account that it ships LV-input skids first.** Its own datasheet settles it in the analyst’s favour, but both claims are live in the market. [DOSSIER dg-matrix v1]
- **Who acquired Resilient Power.** Eaton’s own newsroom states Eaton completed it in August 2025; a paid market-research source asserts Vertiv acquired it in July 2025. Eaton’s release settles it, and the wrong version is circulating in paid output. [WEB, verified 2026-09-23]
- **Hitachi Energy as simultaneous SST investor and 34.5 kV sceptic.** Its own release announces support for NVIDIA’s architecture and names no SST product or class; trade press then reports it urging caution on compact 34.5 kV designs. Only the release is a primary source. [WEB, verified 2026-09-23]
- **Megmeet’s FY2024 appliance revenue does not reconcile with FY2025’s.** The segments were almost certainly restated when the business groups were renamed; the inconsistency is flagged rather than resolved. [WEB, verified 2026-09-23]

16.4 What was blocked, and what it cost

Four hosts refused this research pass and each cost something specific. **opencompute.org** returned HTTP 403 to every attempt by two tools, which blocked the OCP LVDC Solid-State Transformer Specification v0.3 — the single most important primary document for this subject. Everything this briefing says about its contents is second-hand and flagged as such. [WEB, verified 2026-09-23] **eaton.com** returned HTTP 503 on both the MVSST catalogue page and its brochure, so no Eaton specification was obtained at all. [WEB, verified 2026-09-23] **intertek.com** returned 403, so the UL 347A and UL 2877 scope statements rest on a search-index summary rather than on the certifier’s own page. [WEB, verified 2026-09-23] **ferc.gov** returned 403, so every FERC fact in this document is law-firm-mediated.

[WEB, verified 2026-09-23]

ONE METHODOLOGICAL CAVEAT WORTH KNOWING

Several standards-mapping claims in chapter 7 rest on a niche newsletter that has itself published a self-correction about conflating announcements with deployments. Those are marked low confidence at each use, and the NEC Article 706 “100 V DC default” figure in particular is uncorroborated — the inspectors’-association article that was reachable makes no mention of 800 V DC at all, which is consistent with a gap but is not positive confirmation of it.

[WEB, verified 2026-09-23]

APPENDIX

Glossary, flashcards, sources and colophon

The term system alphabetically, the printed deck, every source by tier, and the record of how this document was made.

A Glossary — the term system, alphabetical

AFE — active front end · Control

A rectifier built from controlled switches rather than diodes. Draws sinusoidal current at unity power factor, can supply reactive power, and can be programmed to a ride-through curve. [\[PRIMER ch.5.5, ch.6.3\]](#) — **IEEE 519 sets 5% total voltage distortion at the point of common coupling with no individual harmonic above 3%** [\[PRIMER ch.6.3\]](#)

BIL — basic insulation level · Magnetism

The lightning or switching impulse a piece of equipment must withstand. Set by the ANSI C84.1 equipment class, not by the vendor. [\[PRIMER ch.6.1, ch.9\]](#) — **15 kV class = 95 kV BIL; 25 kV class = 125 kV; 35 kV class = 150 kV** [\[PRIMER ch.6.1\]](#)

CHB — cascaded H-bridge · Topology

A multilevel converter built from series-connected single-phase full bridges; the standard way to face medium voltage with low-voltage semiconductors. [\[PRIMER ch.3.2, fig.5\]](#) — **13.8 kV needs 14-15 cells per phase with 1.7 kV devices; 34.5 kV needs about 2.5x as many** [\[PRIMER ch.3.2, ch.6.1\]](#)

CLLC / LLC · Topology

Resonant isolated DC-DC topologies whose tank circuit lets the switches turn on and off at zero voltage or current; CLLC is the bidirectional form. [\[PRIMER ch.3.3\]](#) — **NVIDIA specifies a 64:1 LLC with a matrix transformer for 800 V straight to 12 V in-rack** [\[PRIMER ch.2.2\]](#)

Cosmic-ray derating · Devices

Atmospheric neutrons cause single-event burnout in high-field power devices, so the cell DC link is set well below the device rating. [\[PRIMER fig.5\]](#) — **The customary derating is about 0.55 x device rating — the factor that drives every cell-count figure in this brief** [\[WEB, verified 2026-09-23\]](#)

DAB — dual active bridge · Topology

An isolated DC-DC converter with an active full bridge on each side of the transformer; power flow is set by the phase shift between the two square waves, so it is bidirectional by construction. [\[PRIMER ch.3.3, fig.6\]](#) — **5-50 kHz switching** [\[PRIMER ch.3.3\]](#)

DC arc flash · Standards

The hazard NFPA 70E 2027 names and no standard can yet calculate. IEEE 1584-2018 is AC only. [\[PRIMER ch.6.4\]](#) — **70E 2027 adds Article 310 with DC thresholds; the NFPA research project aims its model at 70E 2030 — a three-year gap** [\[WEB, verified 2026-09-23\]](#)

Fail-short · Devices

The property a series-stacked die needs: when it fails it must keep conducting so the stack stays up until the cell can be bypassed. [\[PRIMER ch.5.4\]](#) — **ORNL calls SiC fail-short in series stacks "an open research challenge"** [\[PRIMER ch.5.4\]](#)

Grid-forming · Control

A converter that establishes voltage and frequency rather than following them; includes black start. [\[DOSSIER heron-power v1\]](#) — **Heron claims IEEE 2800 and ERCOT NOGRR 282 compliance with under 1% ripple injected at MV** [\[DOSSIER heron-power v1\]](#)

HRMG / HRRG · Protection

High-resistance midpoint grounding and high-resistance return grounding — two of the four schemes under evaluation for a two-wire 800 V DC bus, alongside floating and solid. [\[GUIDANCE nvidia-800vdc p24-26\]](#) — **Four candidates; the paper leans HRMG** [\[GUIDANCE nvidia-800vdc p24-26\]](#)

IEC 62477-2 · Standards

The international envelope for MV converters: 1 kV AC / 1.5 kV DC up to 36 kV AC / 54 kV DC. Edition 2 went to CDV with voting closing 13 March 2026. [\[WEB, verified 2026-09-23\]](#) — **36 kV AC ceiling** [\[WEB, verified 2026-09-23\]](#)

IMD / RCM · Protection

Insulation monitoring device and residual current monitor. The facility zone uses IMD/IRM and alarms without shutting down; the rack interface zone prefers RCM for fast leakage detection. [\[GUIDANCE nvidia-800vdc p26-27\]](#) — **Two protection zones**

[GUIDANCE [nvidia-800vdc p26–27](#)]

ISOP — input-series, output-parallel · Topology

The connection pattern of an SST: cells stacked in series on the medium-voltage input, tied in parallel on the low-voltage output.

[PRIMER [ch.3.1, fig.4](#)]

MFT — medium-frequency transformer · Magnetics

The kilohertz transformer inside each SST cell. Small because volt-seconds per cycle fall with frequency; demanding because it must hold medium-voltage insulation in that small volume under PWM stress. [PRIMER [ch.3.1, ch.3.3, ch.5.5](#)] — **Raising f from 60 Hz to 10 kHz shrinks core area by roughly 170x; a 4 MW unit weighs about a fortieth of its 60 Hz equivalent — Infineon's comparison is 20 tonnes to about 500 kg** [PRIMER [ch.3.1, ch.5.5](#)]

MMC — modular multilevel converter · Topology

A half-bridge-cell multilevel converter used in HVDC; an alternative front end to the CHB at higher voltages. [PRIMER [ch.9](#)] — **N+1 raises a modular multilevel drive's MTBF by about 1.7x — the one independent redundancy number available** [WEB, verified 2026-09-23]

OCP LVDC SST Specification v0.3 · Standards

The specification the layer is being built to — Microsoft, Google and NVIDIA, July 2026. More than 80 manufacturers reported building to it.

[WEB, verified 2026-09-23] — **v0.3, July 2026** [WEB, verified 2026-09-23]

PCC — point of common coupling · Grid interface

Where the customer's installation meets the utility's system, and where IEEE 519 harmonic limits are assessed. [PRIMER [ch.6.5](#)] — **Limits tighten at higher voltage and lower short-circuit ratio** [PRIMER [ch.6.5](#)]

PD — partial discharge · Magnetics

Localised breakdown inside insulation that does not bridge the electrodes but erodes them over time. Fast dv/dt from wide-bandgap devices drives it. [PRIMER [ch.5.3](#)] — **A 35 kV interface needs roughly 100 kV of insulation — "three times", in TDK's assessment** [PRIMER [ch.5.3](#)]

Ride-through · Control

A load's ability to stay connected through a voltage or frequency disturbance instead of tripping. [PRIMER [ch.6.3](#)] — **NOGRR 282: 2 s at 0.85 pu, 150 ms below 0.35 pu, no more than 150% current, 90% recovery within 2 s, six sags in 90 s** [WEB, verified 2026-09-23]

Service-voltage class · Grid interface

The ANSI C84.1 equipment class of the utility feed. It sets insulation, BIL and cell count — and therefore the product. [PRIMER [ch.6.1, fig.11](#)] — **12.47/13.2/13.8 kV = 15 kV class; 24.9 kV = 25 kV class; 34.5 kV = 35 kV class; gigawatt campuses tie in at 69-345 kV** [PRIMER [ch.6.1](#)]

SiC — silicon carbide · Devices

The wide-bandgap semiconductor that makes 1.2-3.3 kV MOSFETs practical at kilohertz switching. The enabling device of the SST.

[PRIMER [ch.3.2, ch.6.6](#)] — **Production classes 1.2, 1.7 and (from few suppliers) 3.3 kV; 6.5 and 10 kV exist as samples. SiC fell from over \$10/A in 2018 to \$2-3/A in 2025** [PRIMER [ch.3.2, ch.6.6](#)]

SSCB — solid-state circuit breaker · Protection

A semiconductor switch that interrupts DC fault current in microseconds, because DC has no natural current zero for a mechanical contact to exploit. [PRIMER [ch.5.2](#)] — **NVIDIA targets 125 A air-cooled and 1250 A liquid-cooled solid-state breakers, with MCCBs as the Day-1 branch device** [GUIDANCE [nvidia-800vdc p27–30](#)]

SST — solid-state transformer · Topology

A power-electronic converter providing a transformer's voltage change and galvanic isolation through medium-frequency magnetics, with controllable AC-DC and DC-DC stages either side. [PRIMER [ch.3, ch.9](#)] — **97-98.5% for MV-to-DC units; 99.0-99.5% for a good 60 Hz distribution transformer alone** [PRIMER [ch.3.5](#)]

The 4.8 MW block · Grid interface

NVIDIA's standardised data-hall DC power block: MV in, a 6000 A-class DC switchboard, parallel 1250 A busways, 4+1 catcher redundancy through a DC static transfer switch. [GUIDANCE [nvidia-800vdc p17–21](#)] — **4.8 MW per block, 20 MW deployment units of four scalable units** [GUIDANCE [nvidia-800vdc p17–21](#)]

TRU / MV rectifier · Topology

A conventional MV/LV transformer feeding a rectifier — diode, thyristor or active — to produce 800 V DC. [PRIMER [ch.4.1](#)] — **NVIDIA's named near-term path, mature from electrolysis, rail and mining; it rates the MV rectifier and the SST identically at 98.5% or better, as large as 7.5 MVA** [GUIDANCE [nvidia-800vdc p21–23](#)]

UL 1741 / 347A / 2877 · Standards

The US listing routes. UL 1741 with supplements SA/SB is the dominant inverter/converter path; UL 347A and 2877 cover MV power conversion from 1 kV up to 38 kV. **[WEB, verified 2026-09-23] — 1 kV to 38 kV is the listed medium-voltage range; 38 kV is the ceiling**
[WEB, verified 2026-09-23]

B The flashcard deck

Twenty-six cards, one per term. Cover the right-hand side. The same deck drives the spaced-repetition widget in the study companion, from the same data file.

Front — the term	Back — what it is, the number, and the sentence to say it in
SST — solid-state transformer Topology	A power-electronic converter providing a transformer's voltage change and galvanic isolation through medium-frequency magnetics, with controllable AC-DC and DC-DC stages either side. [PRIMER ch.3, ch.9] Number: 97-98.5% for MV-to-DC units; 99.0-99.5% for a good 60 Hz distribution transformer alone [PRIMER ch.3.5] <i>"It is a converter, not a transformer — which is why it can also rectify, regulate and ride through, and why it has the overload manners of a computer rather than of a block of iron."</i>
ISOP — input-series, output-parallel Topology	The connection pattern of an SST: cells stacked in series on the medium-voltage input, tied in parallel on the low-voltage output. [PRIMER ch.3.1, fig.4] <i>"Series on the MV side so each cell sees a fraction of the line; parallel on the LV side so they make one stiff bus."</i>
CHB — cascaded H-bridge Topology	A multilevel converter built from series-connected single-phase full bridges; the standard way to face medium voltage with low-voltage semiconductors. [PRIMER ch.3.2, fig.5] Number: 13.8 kV needs 14-15 cells per phase with 1.7 kV devices; 34.5 kV needs about 2.5x as many [PRIMER ch.3.2, ch.6.1] <i>"The cell count is the cost driver — each cell is a complete converter with its own capacitor, transformer, gate drivers and sensors."</i>
DAB — dual active bridge Topology	An isolated DC-DC converter with an active full bridge on each side of the transformer; power flow is set by the phase shift between the two square waves, so it is bidirectional by construction. [PRIMER ch.3.3, fig.6] Number: 5-50 kHz switching [PRIMER ch.3.3] <i>"Both bridges are active, so power can go back to the grid — a passive rectifier can never do that."</i>
CLLC / LLC Topology	Resonant isolated DC-DC topologies whose tank circuit lets the switches turn on and off at zero voltage or current; CLLC is the bidirectional form. [PRIMER ch.3.3] Number: NVIDIA specifies a 64:1 LLC with a matrix transformer for 800 V straight to 12 V in-rack [PRIMER ch.2.2] <i>"Resonant variants trade control range for soft switching and lower loss — and tend to lose soft switching at light load."</i>
MMC — modular multilevel converter Topology	A half-bridge-cell multilevel converter used in HVDC; an alternative front end to the CHB at higher voltages. [PRIMER ch.9] Number: N+1 raises a modular multilevel drive's MTBF by about 1.7x — the one independent redundancy number available [WEB, verified 2026-09-23] <i>"It is also where the SST's service model comes from: MV drives have been swapping cells in minutes for two decades."</i>
TRU / MV rectifier Topology	A conventional MV/LV transformer feeding a rectifier — diode, thyristor or active — to produce 800 V DC. [PRIMER ch.4.1] Number: NVIDIA's named near-term path, mature from electrolysis, rail and mining; it rates the MV rectifier and the SST identically at 98.5% or better, as large as 7.5 MVA [GUIDANCE nvidia-800vdc p21-23] <i>"NVIDIA's own paper calls the TRU a faster and proven path and reserves the SST for the next generation. Do not sell against the TRU on efficiency — they are rated the same."</i>
MFT — medium-frequency transformer Magnetics	The kilohertz transformer inside each SST cell. Small because volt-seconds per cycle fall with frequency; demanding because it must hold medium-voltage insulation in that small volume under PWM stress. [PRIMER ch.3.1, ch.3.3, ch.5.5] Number: Raising f from 60 Hz to 10 kHz shrinks core area by roughly 170x; a 4 MW unit weighs about a fortieth of its 60 Hz equivalent — Infineon's comparison is 20 tonnes to about 500 kg [PRIMER ch.3.1, ch.5.5] <i>"The magnetics are the part with the fewest suppliers — a low-loss nanocrystalline or ferrite core, MV insulation between windings, and a cooling path for losses in a small volume."</i>
BIL — basic insulation level Magnetics	The lightning or switching impulse a piece of equipment must withstand. Set by the ANSI C84.1 equipment class, not by the vendor. [PRIMER ch.6.1, ch.9] Number: 15 kV class = 95 kV BIL; 25 kV class = 125 kV; 35 kV class = 150 kV [PRIMER ch.6.1] <i>"The class sets the insulation, the BIL and the cell count — which is why a 13.8 kV SST is a different product from a 34.5 kV one."</i>

Front — the term	Back — what it is, the number, and the sentence to say it in
PD — partial discharge Magnetics	Localised breakdown inside insulation that does not bridge the electrodes but erodes them over time. Fast dv/dt from wide-bandgap devices drives it. [PRIMER ch.5.3] Number: A 35 kV interface needs roughly 100 kV of insulation — "three times", in TDK's assessment [PRIMER ch.5.3] <i>"ORNL's point is that IEEE C62.82.1 was written for iron: localised overstress can occur even when terminal-level clamping looks adequate, and impulse tests may not reflect internal stress."</i>
SiC — silicon carbide Devices	The wide-bandgap semiconductor that makes 1.2-3.3 kV MOSFETs practical at kilohertz switching. The enabling device of the SST. [PRIMER ch.3.2, ch.6.6] Number: Production classes 1.2, 1.7 and (from few suppliers) 3.3 kV; 6.5 and 10 kV exist as samples. SiC fell from over \$10/A in 2018 to \$2-3/A in 2025 [PRIMER ch.3.2, ch.6.6] <i>"Moving from 1.7 kV to 3.3 kV devices roughly halves the cell count — which is why Amperesand builds on 3.3 kV parts and why Wolfspeed's wafer capacity is a supply question for the whole layer."</i>
Cosmic-ray derating Devices	Atmospheric neutrons cause single-event burnout in high-field power devices, so the cell DC link is set well below the device rating. [PRIMER fig.5] Number: The customary derating is about 0.55 x device rating — the factor that drives every cell-count figure in this brief [WEB, verified 2026-09-23] <i>"Burnout can occur at 50% or less of rated breakdown, so the derating is not conservatism, it is the design rule."</i>
Fail-short Devices	The property a series-stacked die needs: when it fails it must keep conducting so the stack stays up until the cell can be bypassed. [PRIMER ch.5.4] Number: ORNL calls SiC fail-short in series stacks "an open research challenge" [PRIMER ch.5.4] <i>"Press-pack silicon fails short and keeps conducting for months. SiC's thermal stability, small die area and short short-circuit withstand make that hard — and it is the reliability question a careful buyer will actually ask."</i>
SSCB — solid-state circuit breaker Protection	A semiconductor switch that interrupts DC fault current in microseconds, because DC has no natural current zero for a mechanical contact to exploit. [PRIMER ch.5.2] Number: NVIDIA targets 125 A air-cooled and 1250 A liquid-cooled solid-state breakers, with MCCBs as the Day-1 branch device [GUIDANCE nvidia-800vdc p27-30] <i>"Siemens measured it: at a 200 A/us rise rate, capacitor-discharge interruption was only possible with solid-state breakers. Fuses peaked near 9.4 kA at about 3 ms."</i>
HRMG / HRRG Protection	High-resistance midpoint grounding and high-resistance return grounding — two of the four schemes under evaluation for a two-wire 800 V DC bus, alongside floating and solid. [GUIDANCE nvidia-800vdc p24-26] Number: Four candidates; the paper leans HRMG [GUIDANCE nvidia-800vdc p24-26] <i>"HRMG grounds both conductors symmetrically through a midpoint resistor network: balanced conductor-to-ground voltages, uniform fault detection on both conductors, and it keeps operating after the first fault."</i>
IMD / RCM Protection	Insulation monitoring device and residual current monitor. The facility zone uses IMD/IRM and alarms without shutting down; the rack interface zone prefers RCM for fast leakage detection. [GUIDANCE nvidia-800vdc p26-27] Number: Two protection zones [GUIDANCE nvidia-800vdc p26-27] <i>"The 125 A whip stays de-energised during install and removal — the mechanical lock must engage before an enable signal closes the contactor. That is EV-charging lineage, and they will expect you to know it."</i>
AFE — active front end Control	A rectifier built from controlled switches rather than diodes. Draws sinusoidal current at unity power factor, can supply reactive power, and can be programmed to a ride-through curve. [PRIMER ch.5.5, ch.6.3] Number: IEEE 519 sets 5% total voltage distortion at the point of common coupling with no individual harmonic above 3% [PRIMER ch.6.3] <i>"This is the SST's real US argument: the converter is the model. A transformer cannot be programmed to a ride-through curve; an active front end is the curve."</i>
Grid-forming Control	A converter that establishes voltage and frequency rather than following them; includes black start. [DOSSIER heron-power v1] Number: Heron claims IEEE 2800 and ERCOT NOGRR 282 compliance with under 1% ripple injected at MV [DOSSIER heron-power v1] <i>"Grid-forming is what turns a gigawatt load into a grid citizen rather than a grid problem."</i>
Ride-through Control	A load's ability to stay connected through a voltage or frequency disturbance instead of tripping. [PRIMER ch.6.3] Number: NOGRR 282: 2 s at 0.85 pu, 150 ms below 0.35 pu, no more than 150% current, 90% recovery within 2 s, six sags in 90 s [WEB, verified 2026-09-23] <i>"The July 2024 Virginia root cause was not the transformer — it was a disturbance-counting scheme and manual retransfer. An SST does not override downstream customer logic, and cooling plant is usually not behind the critical bus."</i>
Service-voltage class Grid interface	The ANSI C84.1 equipment class of the utility feed. It sets insulation, BIL and cell count — and therefore the product. [PRIMER ch.6.1, fig.11] Number: 12.47/13.2/13.8 kV = 15 kV class; 24.9 kV = 25 kV class; 34.5 kV = 35 kV class; gigawatt campuses tie in at 69-345 kV [PRIMER ch.6.1] <i>"Large greenfield campuses in Texas and the Southeast trend to 34.5 kV because it moves more power on fewer, smaller feeders. Northern Virginia runs 34.5 kV distribution behind 230 kV ties."</i>

Front — the term	Back — what it is, the number, and the sentence to say it in
PCC — point of common coupling Grid interface	Where the customer's installation meets the utility's system, and where IEEE 519 harmonic limits are assessed. [PRIMER ch.6.5] Number: Limits tighten at higher voltage and lower short-circuit ratio [PRIMER ch.6.5] <i>"For an SST-fed hall the PCC is usually the utility's MV side — so your harmonic signature is measured at 34.5 kV, not at the rack."</i>
The 4.8 MW block Grid interface	NVIDIA's standardised data-hall DC power block: MV in, a 6000 A-class DC switchboard, parallel 1250 A busways, 4+1 catcher redundancy through a DC static transfer switch. [GUIDANCE nvidia-800vdc p17–21] Number: 4.8 MW per block, 20 MW deployment units of four scalable units [GUIDANCE nvidia-800vdc p17–21] <i>"Every vendor now has to answer how it composes to 4.8 MW blocks and 1250 A busways with the tap-can interface — paralleling, busway compatibility, and catcher participation are the three concrete questions."</i>
UL 1741 / 347A / 2877 Standards	The US listing routes. UL 1741 with supplements SA/SB is the dominant inverter/converter path; UL 347A and 2877 cover MV power conversion from 1 kV up to 38 kV. [WEB, verified 2026-09-23] Number: 1 kV to 38 kV is the listed medium-voltage range; 38 kV is the ceiling [WEB, verified 2026-09-23] <i>"Know your listing status precisely. The most quotable line from PCIM Europe 2026 is that there is no UL certification for the 800 VDC bus and that this is a major gap."</i>
IEC 62477-2 Standards	The international envelope for MV converters: 1 kV AC / 1.5 kV DC up to 36 kV AC / 54 kV DC. Edition 2 went to CDV with voting closing 13 March 2026. [WEB, verified 2026-09-23] Number: 36 kV AC ceiling [WEB, verified 2026-09-23] <i>"The envelope exists internationally. What does not exist anywhere is a listed data-centre SST."</i>
DC arc flash Standards	The hazard NFPA 70E 2027 names and no standard can yet calculate. IEEE 1584-2018 is AC only. [PRIMER ch.6.4] Number: 70E 2027 adds Article 310 with DC thresholds; the NFPA research project aims its model at 70E 2030 — a three-year gap [WEB, verified 2026-09-23] <i>"Operators will energise 800 V DC busway for years with no validated way to compute incident energy. The interim answer is conservative table-based PPE, de-energised-work rules and per-project AHJ negotiation."</i>
OCP LVDC SST Specification v0.3 Standards	The specification the layer is being built to — Microsoft, Google and NVIDIA, July 2026. More than 80 manufacturers reported building to it. [WEB, verified 2026-09-23] Number: v0.3, July 2026 [WEB, verified 2026-09-23] <i>"This is the vocabulary your counterparties will use. Read it before the first meeting — and note that it could not be retrieved in this run, because opencompute.org refused every request."</i>

C Sources by tier

Corpus tiers

- **PRIMER** — *Solid-State Transformers for AI Data Centers — Educational Primer*, 12 September 2026. Source repository-information/sst-primer-print.html, PDF SOLID-STATE-TRANSFORMERS-PRIMER.pdf, figures sst-primer-figures/. Ten chapters and fourteen figures, each carrying its own numbered first-party sources.
- **GUIDANCE nvidia-800vdc** — the page-referenced claims ledger and guardrail list for NVIDIA's 800 VDC *Architecture: Industry Alignment & Execution* (August 2026, 36 pp.), at repository-information/industry-guidance/nvidia-800vdc-analysis.md. Page numbers in this document refer to that paper.
- **REPORT 2026-09-08** — Profiler competitive report *AIDC Power Conversion — The 800 VDC Race*, fifteen dossiers across four layers, at live-site-pages/profiler-data/reports/aidc-power-conversion--competitive--2026-09-08.report.json. A companion report scoped to this document's subject was generated alongside it: sst-hall-edge-block--competitive--2026-09-23.report.json, eighteen dossiers and forty-six citations.
- **DOSSIER** — Profiler company dossiers at live-site-pages/profiler-data/<slug>.profile.json, each cited at the version pinned in the tag. **Cited in this document:** abb v7 · amperesand v2 · dg-matrix v1 · ge-vernova v6 · heron-power v1 · hitachi-energy v5 · huawei-digital-power v8 · infineon v1 · liteon v6 · megmeet v7 · novos-power v1 · schneider-electric v9 · sinexcel v8 · sungrow v9 · vertiv v9 · vicor v2 · zhonhen v8. **Read for the companion competitive report but not cited here** — the claims they support are carried in this document at the web tier instead, because the web source is the one that was checked on 23 September: delta-electronics v5 · eaton v8 · oracle v5 · siemens-energy v7.

WEB — research performed for this document, all verified 2026-09-23 (47 sources)

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D Colophon

This document was produced by an unattended overnight run on 23 September 2026, working from a plan written the previous session. Nobody was in the room. That is the reason for this appendix: every judgement made in the developer's absence is recorded here, so that a reader can see where the run chose rather than found.

D.1 Pre-flight, as run

Check	Result on 2026-09-23
Repository identity and branch state	LightAISolutions/Sales, clean tree at origin/main, working branch claude/fervent-lovelace-sk1c5s, no stale remote branch.
Clone depth	Deepened with <code>git fetch --unshallow origin main</code> before any date or SHA lookup — 1,602 commits.
Toolchain	Node 22 and Chromium at /opt/pw-browsers present. matplotlib and playwright installed with <code>pip install</code> ; playwright <code>install</code> was never run, per the environment's rule.
Dossier freshness	Not re-run. The plan's 80 pre-flight of the same morning recorded <code>dueCount: 0</code> across 85 cadence rows, and the run date is inside that window.
Newer AIDC report?	<code>aidc-power-conversion--competitive--2026-09-08</code> was still current in the index. The monthly drift check does not fire until 1 October.
CHANGELOG capacity	Read live: Sections: 99/100 before the first push. Archive rotation was evaluated and is not due — see D.2, assumption 1.

D.2 Every assumption made in the developer's absence

- Archive rotation was not performed, and the plan said it would be.** The plan stated that rotation falls due on the push that takes the CHANGELOG counter to 100. The rotation procedure itself tests the *non-exempt* count, with the current day exempt: at the first push the raw count reached 100 but eight of those sections carried 23 September, leaving **92 non-exempt against a trigger of 100**. That is Scenario A in the rotation examples — a total at or above 100 does not by itself rotate. The run followed the procedure rather than the plan's summary of it, and deepened the clone anyway so that a rotation on any later push would resolve its SHAs.
- The Profiler report was scoped to eighteen slugs, as the plan specified, and no more.** vicor, flex, huawei-digital-power, oracle and nvidia are covered but were left out of scope, with the reason recorded in the report's own scope rationale and gaps block.
- The report builds on the 8 September edition and does not supersede it.** No supersedes field was set and no existing index entry was flipped to superseded, because the two reports answer different questions on different cuts.
- The reader's hearsay about NVIDIA and Oracle engineering contact steered chapter 8's depth and nothing else.** It is stated once, in a box, on the citation-contract page, labelled unverified. No sentence anywhere in this document is sourced to it, and chapter 8 is built entirely from published material.
- Where the plan and a repository rule disagreed, the rule won.** That happened once, on rotation (assumption 1). Everywhere else the plan and the rules agreed.
- The five Phase-B research subagents ran on the same model as the main thread,** as the plan recommended, rather than on the cheaper option the plan offered as a cost lever. The run was already inside its time budget and the pilot ledger depended on reading filings carefully.
- Two figures were redesigned away from the form the plan implied.** The plan's minimum figure set names "NVIDIA + Oracle programme timelines side by side" and a "first-90-days calendar". Both were first drawn on a shared horizontal time axis, rendered, inspected, and found illegible — NVIDIA's milestones cluster inside eleven months and overprint at page width. Both were rebuilt as dated lists, and the reason is recorded in the figure script's own docstrings.
- The competitor positioning map became a grid rather than a scatter** for the same reason: seven of the seventeen vendors sit in just two cells, and no amount of marker jitter separates them. The grid carries the same two axes and loses nothing.

9. **“Undisclosed” is used as a finding, not as a hedge.** Where a vendor has published nothing, the cell says so and the row stays in the table. This applies to Megmeet exactly as it applies to Amperesand, GE Vernova, Hitachi Energy, Vertiv, Schneider and Sinexcel.
10. **Two pieces of third-party information were deliberately not carried.** A report indexed as giving Megmeet a 10 kV domestic and 35 kV international SST split was left out of every table because the page could not be read and the figures are exactly the inference this document forbids. And two consultancies giving the same appliance-controls competitor 13% and 77% of the same market were both excluded rather than one being chosen.
11. **The efficiency figures in the conversion-chain widget are indicative and are labelled so.** The published chain losses are not like-for-like — each author uses a different boundary and load point — and the data file says this in the boundary field that the widget displays.
12. **The older Megmeet prep documents were not edited.** The interview brief, the lesson plan and the study guide are dated documents; chapter 9.4 lists what the current record contradicts in them and changes nothing.
13. **A tier tag is written once per sentence, and the scripted columns are labelled analysis.** Where a source URL had been carried inside a tag it was moved to Appendix C, so every tag in the document and every tag in the data file is one of the five fixed forms or ANALYSIS. Two columns are scripted language rather than record — *the sentence to say it in* in chapter 1 and *the one sentence* in chapter 10 — and each chapter says so rather than carrying a tag per cell. The convention for the wide reference tables is stated on the citation-contract page.

D.3 How this was built

Artefact	Built by
The data file	repository-information/study-prep/megmeet/megmeet-sst-briefing-data.json — the single source for every number in a figure or a widget, 215 values each carrying one of six tier tags. Written before the figures and before the companion.
The fourteen figures	python3 scripts/build-megmeet-sst-briefing-figures.py — a copy of the primer's figure script with the paths changed, the mmsst-fig- prefix applied, and one structural difference: it reads the data file rather than carrying numbers inline. Palette re-validated against the dataviz skill's six checks on the white print surface — all six PASS — and not re-tinted.
The primer's figures	Referenced by number in the text, never copied. The primer remains the source of its own fourteen.
This PDF	node scripts/build-megmeet-sst-briefing-pdf.mjs — a copy of the primer's PDF script with the paths and the DevTools port changed. Rendered through the pre-installed Chromium over the DevTools Protocol so the running header and page numbers survive. The --png proof pages were rendered and read, page by page, before the PDF was called finished. They are 49 slices of the whole document at a 740 px body width with page breaks disabled — the entire content, at a different pagination from the 69 printed pages, which is what the mode is for.
The study companion	megmeet-sst-briefing-companion.html — one self-contained file with the same data file inlined byte-identically, no CDN, works offline. Tested with Playwright from file://.
The Profiler report	live-site-pages/profiler-data/reports/sst-hall-edge-block--competitive--2026-09-23.report.json, dossiers-only, forty-six citations copied verbatim from the cited dossiers' own source lists, python3 scripts/check-profiler-reports.py clean.

The audit pass. A fresh subagent, given no part of the drafting context, audited the finished PDF against the plan's twelve-line rubric and returned thirty-five findings. All of them were worked. The five that mattered: the claim that power products carried “the only expanding gross margin in the company” was false on this document's own data — three of the six segments expanded, and pages in Part IV said so — and is now stated as the only expansion *among the three largest segments*; three figures asserted per-row sourcing they did not print, and now print it; Part V's scope note promised a tag on every fact inside an answer, which two chapters did not do, and the note now describes the convention actually used; the cell-count multiplier appeared as 2.5×, 2.3× and 2.7× on one page without saying which was which; and four dossier versions listed in Appendix C were never cited, and are now separated from the seventeen that are. The full findings list is in the session record, not in this document.

Amendment, evening of 23 September 2026. After dossier v8 landed, chapter 9.4 gained a second correction table — the six places where v8 contradicts this briefing, plus the bounding of the 10 kV / 35 kV lead — and the companion report was superseded by sst-hall-edge-block-rev2--competitive--2026-09-23. Nothing else in the document changed, and the build statistics below describe the original run.

Effort: one unattended session at maximum reasoning effort, with five bounded web-research subagents. **Run:** started 07:36 EST on 23 September 2026, PDF finished 09:24 EST and rebuilt after the audit pass at 10:25 EST — 2 h 49 min in total. **This build:** seventy-one pages, fourteen new figures, 215 tagged records in the data file, forty-seven web sources, forty-six report citations.

Developed by: LightAISolutions